

The Messenger Matters*

Alena Wabitsch

Vienna University of Economics and Business & CEPR

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Abstract

Does the messenger matter in communicating with the public? To be impactful, policy communication must successfully reach people and influence their beliefs. This paper combines causal experimental evidence, data from newspapers, social media, and surveys with theory to study the role of messenger identity in policy communication. When audiences share identity traits with messengers, information becomes more available and more influential. A generalized coordination model demonstrates that diverse messengers maximize welfare when public information is precise, whereas centralized communication is optimal when it is imprecise. This identifies strategic delegation through messenger selection as an additional policy tool, distinct from disclosure.

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JEL: E52, E58, D82, D83, D84

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1 Introduction

Communicating a policy effectively to the public is critical for its impact: people will only respond if they receive the information and believe it. This includes health campaigns promoting vaccine uptake during pandemics, subsidies and taxes incentivizing climate-friendly behavior, and monetary policy aimed at influencing consumption and savings decisions through inflation expectations. How can such policies be communicated effectively to the public?

One important factor that can be varied is the messenger. Communicating with the public often means the audience is diverse in gender, ethnicity, nationality, and other dimensions. This may make the identity of the communicating policymaker more important, given so-called ‘*ingroup effects*’: in interpersonal settings, individuals respond more strongly to information provided by someone who shares their characteristics (i.e., someone from an ‘*ingroup*’).¹ Policy communication, however, differs from these interpersonal settings: information is typically received through the media rather than directly from a messenger, and institutional reputation may overshadow the importance of the individual messenger.

This paper combines novel empirical evidence and theory to address two questions. First, how does messenger identity impact policy communication with the public? Second, how should policy be communicated to the public in light of any messenger effects? To answer these questions, I identify messenger effects on the influence and reach of policy communication with a new inflation forecasting experiment and newspaper data, establish their external validity using social media and representative survey evidence, and assess their welfare consequences through a workhorse coordination model that allows for messenger identity. While the model remains general about the policy domain and dimension(s) of identity, empirically, I focus on nationality-based ingroup effects in communication by the European Central Bank (ECB). I do so because central bank communication is at the frontier of policy communication: carefully designed, anticipated by financial markets and the media, and a core tool of modern monetary policy (Blinder et al., 2024); because the ECB as an international institution provides a prime example of communication with diverse audiences; and because nationality is a salient yet understudied characteristic that individuals feel strongly about. Overall, my results document nationality-based messenger effects along two dimensions: the influence and reach of policy communication both rise for ingroup messengers, as they significantly increase *information use* in belief formation as well as *information availability* in news. Accounting for these effects in optimal communication, my model identifies strategic messenger selection as a new policy tool when policymakers and audiences are diverse.

¹This has been found across various contexts such as education (Carrell et al., 2010; Gershenson et al., 2022), health (Alsan et al., 2019), or finance (Stolper and Walter, 2019). Malmendier and Veldkamp (2022) formalize ingroup effects in economic decision-making and show their relevance for occupational choice.

To identify messenger effects on information use, my first dimension of messenger effects, I design an inflation forecasting experiment. In the pre-registered experiment, participants from Germany, Spain, France, and Italy forecast inflation in six scenarios, each time receiving a signal about future inflation from a different messenger before updating their forecast. The design primarily aims to identify causal nationality-based ingroup effects on information use, with and without institutional context. For this, the within-subject treatments isolate variation in messenger nationality with and without institutional affiliation, while holding other messenger characteristics and the message fixed. This controlled variation is necessary because, in the field, policymakers’ identities combine many observable and unobservable characteristics, making it difficult to isolate a single identity dimension and identify ingroup effects cleanly. Further, varying institutional context is key to understanding whether institutional affiliation moderates ingroup effects, which distinguishes policy communication from interpersonal communication.

Across messenger treatments, participants underweight signals by about 5 percentage points relative to a Bayesian benchmark, consistent with survey evidence of under-reaction to new information (Coibion and Gorodnichenko, 2012). However, signals from ingroup messengers are used 5.9 percentage points more than those from outgroup messengers, a sizable effect that roughly matches the average information underweight. This constitutes the pure nationality-based ingroup effect on information use for inflation expectations. This ingroup effect persists in an institutional context, with signals from ECB ingroup messengers being used 7.1 percentage points more compared to ECB outgroup messengers.

A natural question that arises is whether the increased use of information from ingroup messengers reflects *homophily* (relative overweighting of ingroup signals) or *heterophobia* (relative underweighting of outgroup signals). By including a neutral ECB messenger treatment in the experiment, I can decompose the ingroup effect and show that the ingroup effect is entirely due to homophily: relative to a neutral ECB messenger, participants place greater weight on information from an ingroup ECB messenger. By contrast, I find no support for heterophobia, with information from neutral and outgroup ECB messengers receiving statistically indistinguishable weights.

These ingroup dynamics also extend to institutions: information from neutral messengers affiliated with national central banks (NCBs) is used 6.3 percentage points more than information from otherwise identical neutral ECB messengers. These findings, together with the ingroup effects discussed above, highlight the benefits of two feasible policy actions within the existing institutional framework: delegating communication to diverse policymakers within the same institution, such as different board members, and delegating communication to other institutions, such as the Eurosystem’s NCBs.

Messenger identity may strengthen signal use through two channels. First, it may serve as a shortcut for information processing, making individuals more willing to take the signal at face value. Second, it may increase attention to the information provided. Consistent with the first explanation, I find that ingroup effects are largely associated with self-reported *perceived messenger quality* and *trust*. By contrast, an attention measure incorporated into the experimental interface provides no evidence that messenger identity affects *attention*: participants do not request more information from ingroup messengers than from other messengers. Instead, attention responds causally to the level and uncertainty of inflation across the randomized scenarios, consistent with attention being endogenous to the inflationary environment (Weber et al., 2025). This highlights that audiences are more attentive to policy communication when uncertainty is high, whereas in more stable times, reaching the public is harder and messenger choice cannot offset this inattention. Yet, in a post-experiment survey, participants do report being better informed about ingroup policymakers: they are more likely to know central bankers of their own nationality, and they receive more news about them. Since messenger identity does not raise attention to information, this pattern may instead be explained by greater availability of information about ingroup policymakers through news coverage. Information availability is particularly important for policy communication because, unlike interpersonal communication, it typically reaches the public through the media. Because information availability is purposefully fixed in the experiment, I study it outside the experimental environment.

To identify messenger effects on information availability, my second dimension of messenger effects, I compare newspaper volume across time in the four biggest euro area countries and document that messenger identity is systematically associated with news coverage in national newspapers. To do so, I select articles from two to three major national newspaper outlets per country in LexisNexis between 2016 and 2022 that mention “ECB,” “European Central Bank,” or their local-language equivalents. The 2019 change in ECB presidents, the messengers of major ECB communications, provides quasi-experimental variation in countries’ ingroup status. Controlling for time and country fixed effects, I show that newspaper coverage rises in the ECB president’s home country: an ingroup president increases national ECB-related coverage by around 272 additional articles per year (or roughly 12 additional articles between two monetary policy decisions). By affecting news coverage, messenger identity influences how much policy information is easily available to individuals. Beyond directly improving the reach of policy communication, this may also have important implications for political participation and business-cycle dynamics (Gentzkow et al., 2011; Chahrour et al., 2021).

Finally, to establish the external validity of messenger effects on information use and availability, I provide evidence from social media and a representative inflation expectations

survey. First, I create a new dataset of over 8 million multilingual tweets from 2016 to 2022 and again exploit the transition between ECB presidents of different nationalities in 2019. Consistent with greater information use, Twitter users who share the president’s nationality update their beliefs (proxied by text sentiment) more strongly and in line with the signal from ECB press conferences. Consistent with information availability, tweet volume throughout the sample rises with ingroup status. Second, data from the ECB’s cross-country Consumer Expectations Survey (CES) show that during five years under a French ECB president, French households report more precise inflation expectations that are also more closely aligned with ECB forecasts, again consistent with greater information use and availability.

Does this imply that public policy should always be communicated through multiple messengers to raise diversity? To address this question, I develop a model with strategic complementarities to evaluate welfare-maximizing communication policies. The model is in the spirit of a ‘beauty contest’ (Morris and Shin, 2002), which captures the dual role of disclosing public information: to inform and to coordinate. Agents care about aligning their actions with both an unknown fundamental and the actions of others, and their in- or outgroup status with the messenger shapes how they receive and process public information. While calibrated to the empirical findings based on nationality, the model remains abstract, extending to public communication with diverse messengers and audiences across various dimensions (such as gender, ethnicity, age, socioeconomic status, or expertise), and applicable to policy communication beyond monetary issues such as climate, health or fiscal policy. The model evaluates two communication policies: a disclosure policy and a novel *delegation policy*. The disclosure policy involves setting the precision of public information, while the delegation policy entails choosing the messenger(s) to deliver public information, thereby determining the share of agents classified as ingroup.

The model characterizes optimal communication strategies: when coordination is weak, delegation is always beneficial, especially when public information is precise relative to private information. By contrast, when coordination is strong and the public signal is noisy, both disclosure and delegation should be restrained. As the signal improves, disclosure should rise while communication should remain centralized, so delegation remains restricted. Only once the public signal is sufficiently precise, both disclosure and delegation raise welfare. Thus, the *strategic* delegation of communication is an additional policy tool to maximize welfare: diversifying messengers is optimal when the relative quality of public information is high, while centralizing communication can be preferable when it is low, including certain cases where disclosure would increase welfare, making it *distinct* from disclosure.

The messenger matters. For effective policy communication, the identities of messengers and audiences are important. By strategically delegating communication to messengers with

diverse characteristics, policymakers can influence the reach and influence of their messages, and ultimately maximize social welfare through their disclosure of public information.

Related Literature. This paper contributes to four strands of the literature.

First, the literature on social identity, as this paper studies the role of social identity in policy communication, showing that interpersonal ingroup effects persist in communication from institutions. A well-established literature shows that identity-driven group membership can shape preferences, beliefs, and behavior (e.g., [Akerlof and Kranton, 2000](#); [Chen and Li, 2009](#); [Bursztyn et al., 2020](#); [Malmendier and Veldkamp, 2022](#)). In interpersonal settings, shared identity affects information use and economic behavior in education, healthcare, and finance.² Evidence on similar ingroup effects in public communication is more limited, and most existing evidence comes from political settings, where the identity of interest is partisanship. However, partisanship comes with ideology, making estimated messenger effects also depend on how expected a message is from a given source or messenger.³ This paper instead studies nationality, an identity dimension that is less ideologically loaded, especially in the context of monetary policy, yet strong enough to potentially affect behavior.⁴

Second, methodologically, my experimental design contributes to the growing literature on macroeconomic experiments (see for overviews [Duffy, 2016](#); [Haaland et al., 2023](#); [Fuster and Zafar, 2023](#)), where forecasting inflation is often central (e.g., [Marimon and Sunder, 1993](#); [Kryvtsov and Petersen, 2021](#); [Coibion et al., 2022](#); [McMahon and Rholes, 2023](#)). The experiment introduces several new design features, including an interactive experimental interface to provide a new measure of attention to macroeconomic information, and messenger treatments that vary one characteristic at a time while randomizing both the messenger’s forecasting performance and the inflationary environment. This treatment design limits confounds from other traits of real-world policymakers and from contemporaneous macroeco-

²In education, Black teachers improve Black students’ outcomes, with stronger effects when gender also matches ([Gershenson et al., 2022](#); [Price, 2010](#); [Carrell et al., 2010](#)). In healthcare, [Alsan et al. \(2019\)](#) find that Black men are more likely to seek preventive care when interacting with Black male doctors, and [Alsan and Eichmeyer \(2024\)](#) find that social proximity is effective in promoting vaccination, even when the sender is perceived as less qualified and trustworthy. In financial advising, [Stolper and Walter \(2019\)](#) show that ingroup bias increases clients’ likelihood of following advice.

³[Chiang and Knight \(2011\)](#) show that newspaper endorsements persuade most when unexpected given the outlet, and [Afrouzi et al. \(2024\)](#) find that leaders move their own supporters’ beliefs only when speaking against their party line. [Nicholson \(2012\)](#) finds that out-party cues polarize more than in-party cues persuade.

⁴[Murdock et al. \(2014\)](#) show that nationality features in individuals’ spontaneous self-conceptions, [Shayo \(2009\)](#) links national identification to support for redistribution, and [Shayo \(2020\)](#) reviews evidence that social identities shape economic behavior through ingroup bias and conformity to group norms. In the monetary policy context, existing work suggests that policymakers’ gender and ethnicity affect perceptions, trust, and influence ([Bodea et al., 2021](#); [D’Acunto et al., 2022](#); [Bodea and Kerner, 2022](#)). Note that because these studies use real policymakers who differ along many characteristics, the evidence cannot cleanly identify ingroup effects.

conomic conditions, both of which can shape information use (D’Acunto et al., 2022; Weber et al., 2025). It thereby not only identifies ingroup effects cleanly, but also tests whether institutional affiliation moderates them, and distinguishes homophily from heterophobia as competing mechanisms. These distinctions inform policy by showing whether delegating communication across messengers within or even across institutions carries benefits, and if it may impose outgroup costs.

Third, by showing how the messenger can improve the reach and influence of central bank communication, raising information availability and use, this paper directly contributes to the literature on central bank communication with the public. Blinder et al. (2024) summarize this literature, building on Blinder et al. (2008), Ehrmann et al. (2013), Binder (2017), Blinder (2018), Haldane and McMahon (2018), Coibion et al. (2020), and others. Although communication with the non-expert public is a fairly recent endeavor with contested success, when successful, it can influence institutional trust (Ehrmann, 2026) and inflation expectations (Coibion et al., 2022). While existing work shows that central bank communication can affect media coverage (Lamla and Vinogradov, 2019; Ter Ellen et al., 2022), and reach the non-expert public (Ehrmann and Wabitsch, 2022), this paper highlights the role of the messenger in this.

Finally, by introducing messenger effects into a model of strategic complementarities, this paper contributes to the literature on optimal disclosure in coordination models based on the social value of public information (Morris and Shin, 2002).⁵ My model deviates from existing ones through the use and availability of public signals, reflecting my empirical findings on messenger effects. While Cornand and Heinemann (2008) discuss partial information availability, my model additionally introduces a bias to information use in the form of a *resonance weight* to expectation formation (Malmendier and Veldkamp, 2022), complementing previous behavioral distortions such as myopia, or extra discounting of the future (Gabaix, 2020; Angeletos and Huo, 2021). It also endogenizes signal availability to signal size, extending an idea by Nimark and Pitschner (2019) whereby larger signals are more likely reported by the media. With this, I can evaluate the delegation of communication to other messengers, a new communication policy distinct from but complementary to the widely explored disclosure policy (e.g., James and Lawler 2011, Angeletos and Lian 2018, Bassetto 2019, Kohlhas 2020, Kohlhas 2022).

⁵Public information can disproportionately influence agents’ actions, potentially leading to welfare-reducing overreactions to noisy public signals. Developments since Morris and Shin (2002) include Svensson (2006) arguing that increased precision of public information is likely beneficial for reasonable parameter values, and Amador and Weill (2010) examining a dynamic setting, highlighting varying welfare outcomes.

2 Messenger Effects on Information Use

This section presents causal evidence from a new individual-choice inflation forecasting experiment on how messenger identity and institutional context shape information use, the mechanisms behind these effects, and attention to information. I estimate messenger effects on inflation expectations because they are central to monetary policy and provide a numerical belief outcome that allows for precise measurement of information use. Because the ingroup criterion is nationality, an identity dimension that individuals tend to feel strongly about and that is typically salient for policymakers, I field the experiment in Germany, Spain, France, and Italy, the four largest economies in the euro area, which make up the vast majority of its economic activity.

2.1 Experimental Design

The pre-registered experiment (AEA RCT Registry ID: AEARCTR-0010727) is designed to allow for a controlled assessment of information use, where signal availability is ensured and measures of priors and their precision are of high quality. In the experiment, participants are incentivized to forecast inflation after receiving forecasts attributed to different messengers. The messengers are hypothetical experts who differ in one characteristic at a time, while both messenger order and inflation conditions are randomized to ensure that observed differences can be attributed to a single messenger characteristic. This design thereby isolates the causal effects of messenger identity on information use, limiting confounds such as the inflationary environment or the combination of multiple traits in real-world policymakers. Moreover, the within-subject design increases precision by comparing information use across messenger treatments within the same participant, thereby differencing out individual heterogeneity in beliefs about inflation dynamics, forecasting ability, and general responsiveness to expert forecasts.⁶ For credible macroeconomic dynamics, the design incorporates real-world data from historical euro area inflation and corresponding professional forecasts. Additionally, the design introduces new features to measure attention to information.

⁶While the within-subject design could in principle allow participants to infer that the experiment studies messenger identity, experimenter demand effects are of limited concern here for several reasons. Participants are incentivized for forecast accuracy, so demand-driven responses are costly. In addition, open-ended responses provide little evidence that participants inferred the hypotheses (Table A8), and estimates using only each participant’s first forecasting task, which approximates a between-subjects design, are directionally consistent with the main results, although less precise due to the smaller sample size. This is in line with Mummolo and Peterson (2019), who show that telling participants the study’s hypothesis has no impact on treatment effects in online survey experiments, and De Quidt et al. (2018), who find that typical demand effects are likely small when participants know what researchers *expect* to find and emerge mainly when participants are told that acting in line with the hypothesis would do the researchers a *favor*.

Figure 1 outlines the sequence of steps in the experiment, which the remainder of this subsection describes in detail.⁷

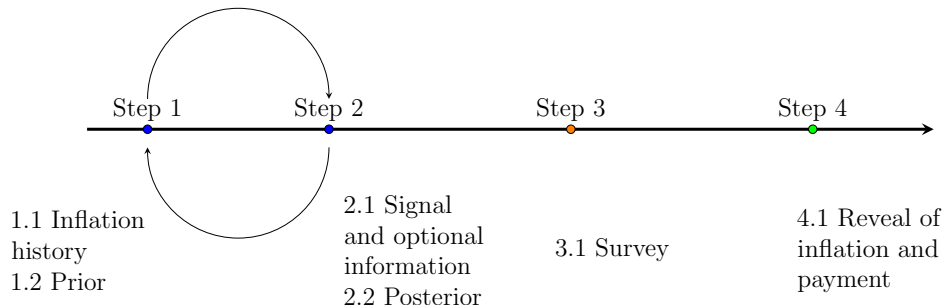


Figure 1: Timeline of Experiment

Notes: The timeline outlines the sequence of steps in the experiment. Steps 1–2 repeat for all six forecasting tasks, followed by step 3 and 4.

Forecasting Tasks. Participants complete six forecasting tasks that elicit prior and posterior beliefs about inflation. In each task, participants first observe ten periods of realized inflation before submitting a point and a range forecast for period 11, capturing their prior beliefs and associated precision. They then receive a forecast for period 11 (the signal), attributed to a randomly assigned messenger and shown together with the messenger’s forecast history over the preceding ten periods. This history conveys signal precision through past forecast errors, mirroring how forecast credibility is evaluated in practice. Historical inflation and forecasts are displayed visually, with exact values available on hover (Figure A17). Finally, participants resubmit their forecasts for period 11, capturing their posterior beliefs. Each sequence of inflation realizations with corresponding forecasts constitutes an inflation scenario; the six scenarios are randomized across messenger treatments at the participant level.

The data for these forecasting tasks consist of six randomly drawn sequences of year-on-year euro area HICP inflation observed quarterly between 1999 and 2023, paired with corresponding one-year-ahead forecasts from the ECB’s Survey of Professional Forecasters (SPF), which serve as the signals. Participants are explicitly informed that both the inflation dynamics and the forecasts are real historical euro area data. Messengers are introduced hypothetically (*“Imagine an expert...”*).⁸

⁷The experimental interface is coded in oTree (Chen et al., 2016), using interactive tools and visuals to facilitate task comprehension. Appendix A.3 shows the entire experimental interface.

⁸Participants do not know the exact dates of the sequences, and with 97 underlying quarters, recognition of specific historical episodes is unlikely. Appendix A.1 displays the full quarterly data series (Figure A9), the forecast-realization pairs for the six selected sequences (Figure A10), and summary statistics per sequence (Table A5).

Incentives. All prior and posterior forecasts are incentivized to reduce forecast error, ensuring high-quality data (Charness et al., 2016). Following common practices in experimental economics, I use a scoring rule to reward the accuracy of point forecasts, with more accurate forecasts earning higher bonuses.⁹ Range forecasts are incentivized to reflect uncertainty: they pay more when they are narrower but zero if inflation realizes outside the range. To avoid hedging, one forecast is randomly chosen for final payment (Appendix A.1 for details).

Messenger Treatments. Each forecasting task features one of the six messenger treatments. The treatments are designed to allow four causal tests of information use: first, whether individuals place greater weight on information from nationality-matched messengers (*pure ingroup effect*); second, whether institutional context affects these ingroup effects (*institutional context*); third, whether ingroup effects stem from overweighting ingroup signals (*homophily*) or underweighting outgroup signals (*heterophobia*); and finally, whether these ingroup dynamics extend to the institutional level by comparing communication attributed to national central banks and the ECB (*institutional affiliation*). The treatments used to implement these tests, summarized in Table 1, fall into three broad categories.

(i) *Nationality-Only Experts*: These treatments identify the pure ingroup effect by comparing

Table 1: Overview of Messenger Treatments

Category	Treatment	Messenger	Hypothesis
Nationality-Only Experts	1–2	Expert from Germany/Spain/France/Italy	H1
Nationality-ECB Experts	3–4	Expert from Germany/Spain/France/Italy who represents the ECB	H2, H3
Institutional Experts	5	Expert who represents the ECB	H3, H4
	6	Expert who represents the participant’s NCB	H4

Notes: Messenger treatments and their mapping to hypotheses H1–H4. The *Nationality-Only* and *Nationality-ECB Experts* treatments consist of one expert with ingroup nationality and one with a randomly selected outgroup nationality. Colors here are for visualization; in the experiment, all text appeared in black.

information use from experts distinguished *only* by nationality, with no other identifying features. The four nationalities used correspond to participants’ nationalities. Each participant faces one expert of their own nationality (the ingroup expert) and one expert of one other nationality, randomly selected from the three remaining options to limit treatment burden and fatigue (the outgroup expert).

(ii) *Nationality-ECB Experts*: These add institutional context where messengers are ECB experts with specified nationalities. One matches the participant’s nationality (the ECB ingroup expert), while the other does not (the ECB outgroup expert). For consistency,

⁹Similar approaches have been adopted by, e.g., Pfajfar and Žakelj 2016, Rholes and Petersen (2021), Mokhtarzadeh and Petersen 2021, Petersen and Rholes 2022, and McMahon and Rholes 2023.

the outgroup nationality is held constant within each participant and matches that of the nationality-only outgroup expert. This design allows direct comparison of ingroup and outgroup effects across institutional and non-institutional contexts, and provides the basis for assessing benefits of delegating communication to different ECB policymakers.

(iii) *Institutional Experts*: These present experts without specified nationalities, attributed either to the ECB (the ECB expert) or to the participant’s national central bank (the NCB expert).¹⁰ These treatments test whether delegating communication to NCBs generates effects similar to nationality-based ingroup dynamics. In combination with the ECB in- and outgroup expert treatments in (ii), they also allow me to distinguish whether institutional ingroup effects are due to homophily (greater use of ingroup signals) or heterophobia (reduced use of outgroup signals).

Each participant faces six treatments, two from each category. Treatment order and the allocation of inflation scenarios to messengers are randomized at the participant level, ensuring valid comparisons across messenger treatments.

Attention to Information. In each forecasting task, participants can request additional qualitative information underlying the messenger’s forecast, providing a measure of attention to information. Three “Read More” buttons reveal qualitative summaries of historical central bank communications by the ECB that correspond to the quantitative SPF forecasts, ensuring consistency between the qualitative information and the numerical signal.¹¹ Attention is measured by the number of buttons clicked: this mimics introducing information as a flow where participants decide when to stop, thereby revealing their chosen level of attention. Similar approaches have been used to measure attention by tracking clicks on résumés (Bartoš et al., 2016). It offers a more reliable measure compared to common proxies for attention such as completion time, where long decision times may reflect slow thinking or unrelated interruptions rather than genuine attention.

Post-Experimental Survey. After completing all forecasting tasks, participants answer a short survey designed to test mechanisms behind the observed effects. All survey questions are asked after the forecasting tasks to avoid priming, but before realized values are revealed, preventing the size of the randomly realized bonus pay from influencing responses. Participants rate each messenger’s *perceived ability*, and report both *trust* in and news *exposure* to institutions (the ECB and their NCB) and to real-world policymakers that are representative

¹⁰This corresponds to the Deutsche Bundesbank for Germans, Banque de France for the French, Banca d’Italia for Italians, and Banco de España for Spaniards.

¹¹Table A6 in Appendix A.1 lists all information pieces and their corresponding inflation scenarios.

of the messenger treatments on 5- and 7-point Likert scales.¹² Participants further complete attention checks and respond to open-text questions about their associations with the experts, their perceived purpose of the study, and their forecasting strategy. These responses allow for ex post checks of experimenter demand. The full survey is shown by Figures A19–A24.

Further details on the data for forecasting sequences, incentive schemes, and provided information are described in Appendix A.1, and screenshots of the entire experimental interface are shown in Appendix A.3.

2.2 Experimental Sample

A total of 400 participants – 100 each in Germany, Spain, France, and Italy – complete the experiment online via Prolific after passing a comprehension quiz. Most data are collected on October 31, 2023, with the remaining responses arriving within 48 hours. One participant with implausibly high forecasts is excluded, leaving a final sample of 399 participants.¹³ The median completion time is 30 minutes, and the average pay is £7.59 (a £4 participation fee plus £3.59 in performance-based bonuses for accurate inflation forecasts, with a £6 maximum). Each participant’s nationality and country of residence are required to coincide, ensuring strong national identification and excluding individuals living abroad. While not designed to be nationally representative, the sample broadly matches national demographics in employment, education, and income; participants are slightly younger than the population average but considerably more diverse than typical student-based laboratory samples. Appendix A.2 reports more demographic details and comparisons with national benchmarks.

2.3 Estimating Information Use

I benchmark information use against a standard linear-Gaussian framework in which posterior beliefs are a weighted average of priors and signals, with weights given by their relative precisions (e.g., Veldkamp 2011). Appendix B.1 provides details on this common framework.¹⁴

¹²These are the ECB board members and NCB governors of the four nationalities in office at the time of the experiment: Isabel Schnabel and Joachim Nagel (Germany), Luis de Guindos and Pablo Hernández de Cos (Spain), Christine Lagarde and François Villeroy de Galhau (France), and Fabio Panetta and Ignazio Visco (Italy), where the first-named is the ECB board member. A policymaker counts as ingroup if their nationality matches the participant’s, and as outgroup otherwise.

¹³All forecasts of this Spanish participant are more than four standard deviations above the remaining sample’s mean, detached from the displayed inflation histories and suggesting systematic misunderstanding of the task and incentive structure. Overall, of 553 participants who began, 153 failed the quiz, withdrew, or timed out.

¹⁴The Bayesian benchmark assumes normally distributed inflation and i.i.d. signals. In reality, euro area inflation is non-normal and forecasts show mild autocorrelation, so this benchmark is imperfect, and deviations from it should not be interpreted as proof of non-Bayesian behavior. However, what matters is relative

To measure deviations from the Bayesian benchmark, I regress posterior beliefs on weighted priors and weighted signals.

$$Posterior_i = \underbrace{\gamma \left(\frac{\alpha_i}{\alpha_i + \beta} A_i \right)}_{\text{weighted Prior}} + \underbrace{\delta \left(\frac{\beta}{\alpha_i + \beta} B \right)}_{\text{weighted Signal}} + \epsilon_i, \quad (1)$$

where A_i denotes the prior, B the signal, α_i prior precision, and β signal precision. Prior precision, α_i , is the inverse subjective variance implied by treating the bounds of a participant's range forecast as a central 95% normal interval.¹⁵ Signal precision, β , is the inverse sample variance of signed SPF forecast errors over the preceding ten quarters and as such is held fixed within each inflation scenario.¹⁶ This reflects the intuitive assumption that participants interpret historical forecast errors as signal noise, although this is not strictly required for identification.¹⁷

I estimate Equation (1) by OLS without an intercept. If participants behave as the Bayesian benchmark, both coefficients equal one ($\gamma = \delta = 1$). Empirically, I find prior weights that are very close to Bayesian, $\gamma = 0.99$, and lower signal weights of $\delta = 0.95$. In other words, participants place roughly 5 percentage points less weight on signals than the Bayesian benchmark predicts. This pattern is consistent with experimental evidence of under-inference (Benjamin, 2019) and survey evidence of under-reaction to macroeconomic news (Coibion and Gorodnichenko, 2012). The signal underweighting is not driven by a single nationality, and no nationality is overweighting signals (see Figure 2 and Table A7). This highlights that, on average, there is value in raising the signal weight. Can varying messengers achieve this? To test this, I extend Equation (1) by interacting signals with messenger treatment dummies

$$Posterior_{ij} = \gamma \left(\frac{\alpha_{ij}}{\alpha_{ij} + \beta_j} A_{ij} \right) + \sum_{j=1}^J \delta_j T_j \left(\frac{\beta_j}{\alpha_{ij} + \beta_j} B_j \right) + \epsilon_{ij}, \quad (2)$$

where T_j is an indicator for messenger treatment $j \in J = \{1, \dots, 6\}$. Each δ_j captures responsiveness to the signal under treatment j , allowing direct tests of the hypotheses on how different messenger identity and institutional context affect the use of signals.

updating across messenger treatments, which are equally affected by these data features.

¹⁵Formally: $\alpha_{ij} = [(UB_{ij} - LB_{ij}) / (2\Phi^{-1}(0.975))]^{-2}$. Nine observations have zero-width ranges, which result in infinite precision and are removed from the analysis. However, all results are robust to setting these observations equal to the maximum prior precision in the sample, retaining all observations.

¹⁶Formally: $\left[\frac{1}{9} \sum_{t=1}^{10} (e_t - \bar{e})^2 \right]^{-1}$, where $e_t = f_t - r_t$, $\bar{e} = \frac{1}{10} \sum_{t=1}^{10} e_t$, f_t is the SPF forecast, and r_t is realized inflation. Table A5 shows all values.

¹⁷Identification in the within-subject design only requires that participants apply a consistent rule for inferring signal precision from the provided data across treatments.

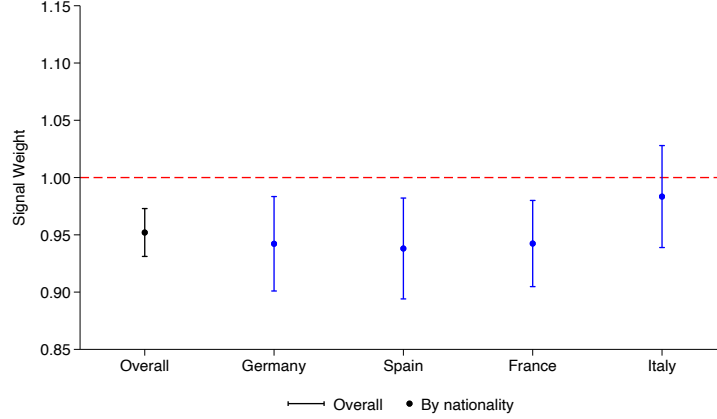


Figure 2: Signal Use Across Treatments by Nationality

Notes: Estimated signal-use coefficients across all messenger treatments, by participant nationality. The benchmark Bayesian weight on the signal is 1 (horizontal dashed line). The full sample is shown in black; nationality-specific estimates in blue. Error bars denote 95% confidence intervals. The complete set of estimates is reported in Table A7.

2.4 Causal Effects of Messenger Identity on Inflation Expectations

Estimating Equation (2) isolates the causal effect of messenger identity by comparing information use across treatments that differ only in identity, with inflation scenarios (including each messenger’s forecast performance) and treatment order randomized. Although not required for unbiasedness under perfect randomization, the preferred specification includes participant, inflation scenario and treatment order fixed effects. Participant fixed effects exploit the within-subject design and absorb individual characteristics, including participant nationality. Scenario and treatment order fixed effects account for differences in information environments and common learning, fatigue, or position effects while addressing any potential imbalance in randomization of treatments. Because adding variables changes the interpretation of the absolute Bayesian coefficients, I focus on relative signal weights across treatments, which capture the treatment effects more precisely under the added controls. Using this specification, I test four hypotheses corresponding to distinct dimensions of messenger identity: nationality, institutional context, the behavioral mechanism (homophily vs. heterophobia), and institutional affiliation. Table 2 reports the estimates for all four hypotheses, where the most conservative specification (column (3)) provides the main effects, including fixed effects for inflation scenarios, treatment order, and individuals.

Hypothesis 1: Pure Ingroup Effect. *Signals from nationality-matched ingroup experts are used more than signals from outgroup experts.*

First, I want to establish the existence of causal nationality-based ingroup effects for inflation expectations. I test this using the *Nationality-Only Expert* treatments (1–2), where experts differ only by nationality. This makes nationality salient while holding all other features constant, allowing a clean identification of the pure ingroup effect.

Echoing the literature on ingroup effects in interpersonal settings, I find support for Hypothesis 1. Participants update significantly more toward ingroup experts, with an estimated effect of 0.059 ($p < 0.10$). Given the average underweighting of signals by five percentage points compared to the Bayesian benchmark (Section 2.3), the finding that ingroup signals are used 5.9 percentage points more is sizable.

Since policy communication occurs within an institutional context, it is vital to know whether ingroup effects persist in that context. To do so, I next estimate ingroup effects within institutional context.

Hypothesis 2: Institutional Context. *Ingroup effects persist within the institutional context of the ECB.*

The *ECB Expert* treatments (3–4), where messengers are ECB-affiliated experts with nationalities attached, allow me to test this hypothesis. This design extends the pure ingroup effect by adding an institutional dimension while keeping the same variation in nationality.

I find support for Hypothesis 2. When forming inflation expectations, participants place significantly more weight on signals from ingroup than outgroup ECB experts. The effect size of 0.071 ($p < 0.05$) is remarkably comparable to the pure ingroup effect outside institutional context and statistically indistinguishable from it. Again, this increase of signal weight of 7.1 percentage points is sizable considering the average gap to the Bayesian benchmark of five percentage points. This shows that nationality still matters for messages from ECB policymakers, despite institutional context. Institutional reputation thus does not dampen the influence of the individual messenger identity.

This is remarkable, because the relevance of messenger identity could weaken when paired with institutions for a few reasons. Participants may be more likely to neglect the precision of signals as provided, if they trust the institution (see Section 2.6), or if they associate expert treatments with real policymakers. Knowledge from outside the experimental environment could then potentially distort perceived signal precision (Esponda et al., 2023), which, across the four nationalities of participants, would likely dampen estimates of ingroup effects. Similarly, other attributes of such associated policymakers may trigger multiple ingroup effects simultaneously (Stolper and Walter, 2019), weakening the identification of isolated nationality-based effects. Given the similarity between the pure ingroup effect and

Table 2: Main Effects from Experiment (H1, H2, H3, H4)

	(1)	(2)	(3)
<i>Pure Ingroup Effect (H1):</i>			
<i>Ingroup Expert – Outgroup Expert</i>	0.006 (0.034)	0.037 (0.041)	0.059* (0.032)
Observations	795	795	795
R-squared	0.982	0.991	0.995
<i>Ingroup Effect within Institutional Context (H2):</i>			
<i>ECB Ingroup Expert – ECB Outgroup Expert</i>	0.025 (0.034)	0.072* (0.038)	0.071** (0.031)
Observations	795	795	795
R-squared	0.982	0.992	0.995
<i>Homophily (H3a):</i>			
<i>Ingroup ECB Expert – Neutral ECB Expert</i>	0.070** (0.035)	0.109*** (0.040)	0.095*** (0.033)
Observations	794	794	794
R-squared	0.980	0.991	0.994
<i>Heterophobia (H3b):</i>			
<i>Outgroup ECB Expert – Neutral ECB Expert</i>	0.045 (0.034)	0.062 (0.038)	0.034 (0.031)
Observations	795	795	795
R-squared	0.982	0.991	0.994
<i>Institutions (H4):</i>			
<i>Neutral NCB Expert – Neutral ECB Expert</i>	0.051 (0.033)	0.071* (0.039)	0.063* (0.032)
Observations	795	795	795
R-squared	0.982	0.991	0.994
Individual fixed effects		✓	✓
Inflation scenario and order controls			✓

Notes: Effects compare coefficients from the same OLS regressions. Observations are for forecasting tasks and vary slightly across treatments due to instances of infinite prior precision. *Inflation scenario and order controls* denotes controls for the underlying data of forecasting tasks and messenger treatment order. Column (3) reports the main specification. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

the institutional ingroup effect, it seems that the experiment, explicitly designed to avoid associations with real policymakers, successfully limited these concerns. For other design choices, however, these considerations remain relevant and may explain why [D’Acunto et al. \(2022\)](#) find no effect of varying real policymakers on inflation expectations in an RCT.

What explains these ingroup effects within institutions? They can arise either from relative overweighting of information from ingroup messengers (*homophily*), relative underweighting of information from outgroup messengers (*heterophobia*), or a combination of both.

Hypothesis 3: Homophily vs. Heterophobia. *Ingroup effects reflect overuse of ingroup signals (homophily), not underuse of outgroup signals (heterophobia).*

To decompose ingroup effects into homophily and heterophobia, I compare ECB ingroup and outgroup experts to a neutral ECB expert without a nationality (treatments 3–5). Homophily corresponds to greater use of signals from ingroup ECB experts relative to the neutral ECB expert, whereas heterophobia corresponds to weaker use of signals from outgroup ECB experts relative to the neutral ECB expert.

I find strong support for homophily and no evidence for heterophobia. Participants update significantly more toward ingroup ECB experts than the neutral ECB expert (0.095, $p < 0.01$), but they do not update less to outgroup ECB experts (0.034, $p = 0.28$). Thus, ingroup effects in the ECB context are entirely due to homophily, a preference for similar messengers, rather than discrimination against outgroup messengers. In the literature, causal evidence that decomposes ingroup effects is rare because it requires a neutral treatment against which ingroup and outgroup treatments can be compared. Here, the neutral ECB expert treatment provides this benchmark, allowing me to provide new systematic evidence decomposing ingroup effects into homophily and heterophobia. These results extend evidence on ingroup favoritism over outgroup hostility in social preferences to the use of information.¹⁸

The neutral institutional messenger is interesting for another test, a test on the precise institutions. Having established the existence and sources of nationality-based ingroup effects within institutional context, the question arises whether similar ingroup dynamics extend to the institutional level.

Hypothesis 4: Comparing Institutions. *Signals from national central banks are used more than those from the ECB.*

To test this hypothesis, I compare signals from national and supranational institutions using the *Institutional Expert* treatments (5–6), in which experts with no nationality are either from the participant’s NCB or from the ECB. This design isolates the effect of different institutional affiliations and, as long as participants associate more with their NCB than the ECB, tests whether ingroup-like dynamics persist across institutions.

¹⁸Halevy et al. (2008) separate these motives and find that contributions benefiting one’s own group exceed those aimed at harming the other; see also Brewer (1999). Chen and Li (2009) show similar ingroup favoritism in social preferences. These studies concern cooperation, allocation, and attitudes, but not information use.

I find support for Hypothesis 4. Participants update significantly more toward NCB signals than ECB signals (0.063, $p < 0.10$). Again, this effect size is substantial, and similar to the pure and the institutional ingroup effect. This effect provides direct evidence of the benefits of delegating communication across institutions (from the ECB to NCBs), while abstracting from potential downsides, such as the risk of a cacophony of voices (Blinder, 2007), which underscores the importance of aligned communication (Tutino, 2016; Do Hwang et al., 2021). Therefore, the benefits of delegating to NCBs should be viewed as an upper bound, achievable only if messages remain identical across institutions.

2.5 Policy Actions: Delegation of Communication

The experimental findings presented above provide causal evidence for nationality-based ingroup effects in information use, which persist within an institutional context. This highlights the importance of policymakers' identity in how information is used, with Section 4 providing observational and survey evidence to support the external validity of these findings.

More concretely, results from Hypotheses 3 and 4 directly support two policy actions: first, they highlight the benefits of delegating communication across policymakers of comparable expertise and standing within the same institution to raise information use, for example by having diverse board members communicate. The positive effect is driven by homophily rather than heterophobia, suggesting that identity-related outgroup costs are limited. In the experiment, the dimension of diversity is nationality, but other characteristics such as age, gender, or ethnicity could be considered to match the target audience. Second, the experiment highlights the benefits of delegating communication across institutions to raise the impact of information. In the ECB context, this means using the existing network of national central banks in the Eurosystem. For the EU and other international organizations, this could mean delegating communication to national governments.

2.6 Mechanisms

Messenger identity may strengthen signal use through two channels. First, it may act as a shortcut for information processing, making individuals more willing to rely on the signal because they perceive the messenger as more capable or trustworthy. Second, it may increase attention to the information provided. This section presents evidence supporting the first but not the second channel.

2.6.1 Information Processing Shortcut: Perceived Capability and Trust

To better understand whether differences in signal use may reflect information processing shortcuts, I focus on the association between ingroup effects and two subjective perceptions: forecasting ability and trust. First, participants may place greater weight on ingroup messengers' signals because they perceive those messengers as better able to forecast inflation, beyond the signal precision conveyed by the messengers' historical forecast performance. To test this, I interact messenger-dependent signal use with self-reported perceived ability of each messenger's forecasting and analytical skill.¹⁹ Results show that controlling for perceived ability reduces the pure ingroup effect by roughly three-quarters (columns (1) and (2) of Table 3). Taking a conservative view, about one-quarter of this ingroup effect (0.016, $p < 0.10$) remains beyond perceived quality.²⁰ Similar or larger reductions occur across all hypotheses: the institutional ingroup effect disappears entirely (0.008, $p = 0.21$), the homophily effect falls to one-fifth of its size to 0.020 ($p < 0.01$), and the institutional effect (NCB vs. ECB) shrinks to slightly less than a third but remains significant (0.020, $p < 0.01$). Thus, perceived ability explains a decisive amount of the increased signal use, although a residual ingroup effect persists. Interestingly, the heterophobia effect even becomes significantly positive after controlling for perceived ability (0.019, $p < 0.01$), reflecting that outgroup messengers are perceived as less capable. This means that, conditional on perceived ability, participants place greater weight on signals from outgroup ECB experts than on signals from neutral ECB experts, further supporting that ingroup effects are not driven by heterophobia.

Second, trust in real-world institutions and policymakers may also influence signal use in the experimental environment. To see whether this is the case, I interact messenger-dependent signal use with ex-post reported trust in institutions and representative policymakers.²¹ Assuming that participants associate the expert treatments with the representative policymakers, I find that for ECB experts, trust in representative policymakers explains two-thirds of the homophily effect but it does not eliminate it (see columns (3) vs. (4) in Table 3).²² By contrast, for institutional experts, the higher signal use for NCBs relative to the ECB disappears

¹⁹Participants are asked '*How confident were you in each expert's ability to accurately predict inflation and provide accurate economic analysis?*' and answer with a 7-point Likert scale rating (Figure A21).

²⁰This estimate assumes participants fully incorporate their stated perceived ability on top of the forecast history, making it a lower bound.

²¹Participants are asked '*How much do you trust the institutions listed below?*' and answer with a 7-point Likert scale rating (Figure A23).

²²Eliciting trust for messenger treatments makes little sense, which is why real-world representative policymakers are chosen instead, resulting in the trust analysis excluding the Nationality-Only treatments. Furthermore, because trust data are only elicited conditional on participants knowing the representative policymakers or institutions, the sample size drops considerably. In this restricted sample, the ECB ingroup effect (H2) is insignificant (see column (3)). Hence the comparison here focuses on the homophily effect (H3), where sufficient observations remain.

Table 3: Mechanisms: Perceived Quality and Trust

	(1)	(2)	(3)	(4)
	Perceived ability		Trust	
<i>Pure Ingroup Effect (H1):</i>				
<i>Ingroup Expert – Outgroup Expert</i>	0.059* (0.032)	0.016* (0.008)		
Observations	795	795		
R-squared	0.995	0.994		
<i>Ingroup Effect within Institutional Context (H2):</i>				
<i>ECB Ingroup Expert – ECB Outgroup Expert</i>	0.071** (0.031)	0.008 (0.007)	0.047 (0.083)	-0.009 (0.022)
Observations	795	795	315	315
R-squared	0.995	0.994	0.998	0.998
<i>Homophily (H3a):</i>				
<i>Ingroup ECB Expert – Neutral ECB Expert</i>	0.095*** (0.033)	0.020*** (0.007)	0.127*** (0.041)	0.040*** (0.011)
Observations	794	794	612	612
R-squared	0.994	0.993	0.996	0.996
<i>Heterophobia (H3b):</i>				
<i>Outgroup ECB Expert – Neutral ECB Expert</i>	0.034 (0.031)	0.019*** (0.006)	0.008 (0.061)	0.008 (0.015)
Observations	795	795	477	477
R-squared	0.994	0.994	0.999	0.998
<i>Institutions (H4):</i>				
<i>Neutral NCB Expert – Neutral ECB Expert</i>	0.063* (0.032)	0.020*** (0.007)	0.060* (0.033)	0.011 (0.007)
Observations	795	795	779	779
R-squared	0.994	0.993	0.994	0.993
Individual fixed effects	✓	✓	✓	✓
Inflation scenario and order controls	✓	✓	✓	✓
Perceived messenger ability		✓		
Trust				✓
Known PM/institution sample			✓	✓

Notes: Effects compare coefficients from the same OLS regressions. Specifications control for inflation scenario and treatment order and include individual fixed effects, as in the main specification. *Perceived Messenger Ability* denotes the interaction with signal use. *Known PM/institution sample* indicates whether the participant has heard of the relevant policymakers or institutions; *Trust*, elicited only conditional on knowing them, captures its interaction with signal use. Observations are for forecasting tasks and vary slightly across treatments due to instances of infinite prior precision or unfamiliarity with PMs/Inst. Column (1) reports the main results. Column (2) adds the interaction with perceived ability. Column (3) repeats the baseline on the restricted sample where policymakers and institutions are known, ensuring comparability with column (4), which adds the interaction with trust. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

entirely once trust is accounted for. Trust therefore renders the institutional ingroup effect statistically indistinguishable from zero (0.011, $p = 0.14$).

Subjective perceptions of forecasting ability and trust thus play an important role in the observed ingroup effects. This evidence is consistent with the view that messenger identity affects signal use by acting as an information processing shortcut: signals from ingroup messengers are more likely to be taken at face value because they are perceived as more accurate and trustworthy. The evidence further underscores the importance of fostering institutional trust for effective policy communication and supports central banks’ efforts to build and maintain public trust (Ehrmann, 2026). One caveat is that both perceived ability and trust are reported after the experiment to avoid priming and may therefore partly reflect ex post rationalizations of choices. Consequently, the results should be interpreted cautiously, as associations rather than causal effects, and the implied explanatory shares should be viewed as upper bounds.

2.6.2 Attention to Information

Besides serving as a shortcut for information processing, a second channel through which messenger identity may strengthen signal use is attention. To test this channel, I use the attention measure described in Section 2.1, defined as the number of “Read More” buttons clicked to access additional qualitative information, and estimate the causal effect of messenger identity on attention.

On average, participants click 1.8 of the 3 buttons, with no statistically significant variation across messenger treatments (Figure 3), indicating that messengers have no causal effect on information demand. This evidence provides no support for an attention-based explanation of the ingroup effects. Instead, quantitative forecasts from ingroup messengers are used more despite no greater attention to qualitative information, which is again consistent with information from ingroup messengers being accepted at face value.

A natural concern, however, is whether button clicks truly capture attention. To validate this measure, I exploit that both messengers and inflation scenarios are randomized. This design choice prevents inflation data from biasing messenger effects, but also allows me to test whether attention varies with the inflationary environment.

While messengers do not affect attention, the inflationary environment does: scenario-level variation in inflation, randomized across tasks and messengers, shifts attention to information (Figure 3). A one percentage point increase in average historical inflation raises the probability of requesting any information by 17.6 percentage points and increases clicks by 0.503 buttons out of a maximum of three ($p < 0.01$; Table A9). Similarly, a one unit increase in the standard deviation of historical inflation over the ten observed periods raises the probability

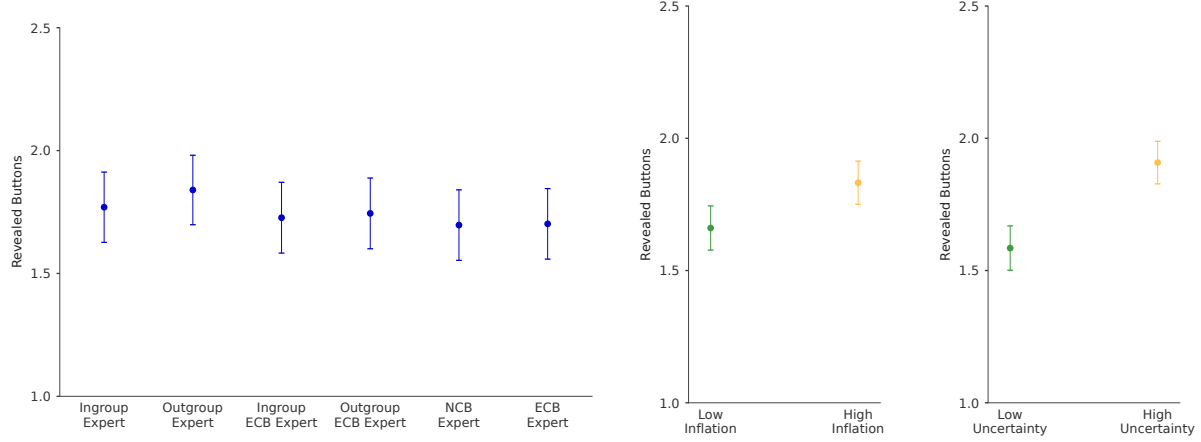


Figure 3: Requests For Additional Information Pieces

Notes: Average number of clicks on buttons by messenger treatment (left panel), by inflation level (middle panel) and by inflation uncertainty (right panel) with 95% confidence intervals. Of the six inflation scenarios, the three scenarios with the lowest (highest) inflation are shown as “Low (High) Inflation.” Similarly, the three with the lowest (highest) standard deviation of inflation history are shown as “Low (High) Uncertainty.”

of requesting any information by 8.4 percentage points and increases clicks by 0.272 ($p < 0.01$). These findings align with Cavallo et al. (2017), who document weaker priors in low-inflation contexts, and Weber et al. (2025), who show that attention is endogenous to inflation levels; my within-subject design adds causal evidence that attention responds strongly not only to inflation levels but also to inflation uncertainty – exactly as theories of rational inattention predict, with attention rising when the returns to information are greatest (Sims, 2003; Maćkowiak et al., 2023).²³

This highlights the importance of the economic environment in reaching the public, as attention to information responds causally to the inflationary environment. When inflation is high or uncertain, individuals are more likely to seek out information, raising attention to policy communication. Conversely, when inflation is low and stable, it is harder for policy communication to get attention, regardless of who delivers it.

At the same time, the absence of messenger effects on attention does not imply that messenger identity is irrelevant for reach. Post-experimental survey responses suggest that ingroup policymakers are more visible to participants: the probability of knowing ingroup policymakers and of receiving news about them is higher by 27 percentage points ($p < 0.01$) and 29 percentage points ($p < 0.01$), respectively (Table A10; Appendix B.1 for more details).

²³Endogenous attention reflects the costliness of being mistaken about inflation. In the experiment, payoffs depend only on forecast accuracy, so expected losses rise with inflation uncertainty but not with its level; the strong response to levels suggests participants carry the real-world salience of high inflation into the experiment.

Given that messenger identity does not influence attention to information, the improved reach of policy communication may instead reflect greater information availability for the ingroup. Information availability cannot be studied in the experimental environment, which by design ensures that every participant receives every signal. Instead, I study the reach of policy communication in the next section using field data on news coverage to document a second messenger effect on the availability of information.

3 Messenger Effects on Information Availability

Messenger effects may operate not only through how individuals use information once they receive it, but also through whether the information reaches them in the first place. Unlike interpersonal communication, policy communication is typically not directly observed by the public but relies on the media to successfully reach its audience. Media coverage therefore shapes how much policy information is readily available to the public. If messenger identity affects this coverage, it increases the reach of policy communication and may also have important implications for political participation and business-cycle dynamics ([Gentzkow et al., 2011](#); [Chahrour et al., 2021](#)).

3.1 Newspaper Data

To assess media coverage, I construct a sample of ECB-related articles from German, Spanish, French, and Italian newspapers obtained through LexisNexis. I restrict the sample to articles containing the terms “ECB,” “European Central Bank,” or their translations into the relevant local language. For each country, I include only articles published by domestic newspapers and written in the local language. The sample spans November 1, 2016, through October 31, 2022 – exactly three years under Mario Draghi and three years under Christine Lagarde. To limit fluctuations arising from changes in LexisNexis availability rather than actual news coverage, I restrict the sample to a balanced set of major newspapers that are continuously available throughout the sample period. The Italian sample includes *Corriere della Sera* and *La Nazione*; the French sample comprises *Le Figaro*, *La Tribune*, and *La Croix*; the German sample comprises *BILD*, including its Sunday and regional editions, *Süddeutsche Zeitung*, including its regional editions, and *Der Tagesspiegel*; and the Spanish sample includes *El Mundo* and *El País*. For the full six-year sample, I observe annual article counts by newspaper (where each year starts in November, in line with the change in presidents). For the two years surrounding the presidential transition, I also observe the content of articles and their exact publication date. Across the full six-year period, newspaper article counts total 25,076:

7,340 in Germany, 6,619 in Spain, 3,339 in France, and 7,778 in Italy.

While the sample covers major mainstream newspapers, it does not span the full universe of news outlets. To mitigate concerns about sample-specific effects, Section 4 presents corroborating evidence from social media, which is not subject to outlet selection and covers a broader information environment.

For the analysis in this section, I define *ingroup* coverage as coverage by newspapers published in the ECB president’s home country: Italy during Draghi’s presidency and France during Lagarde’s. To study whether domestic news coverage shifts with the president’s nationality, I use the transition from Draghi to Lagarde, at which coverage by Italian newspapers loses ingroup status and coverage by French newspapers gains it. Germany and Spain remain in the outgroup throughout the sample period, providing benchmarks for common movements in ECB-related news.

3.2 Information Availability

To study how information availability depends on messenger identity, I compute a measure of relative news coverage and compare how it responds to a change in ingroup status. Relative news coverage is a country’s share of all ECB-related articles across the four countries, which is convenient because it nets out aggregate fluctuations in news volume such as those driven by macroeconomic events affecting the euro area as a whole.

Italy’s share in ECB-related news is on average 35% in my sample under ingroup president Draghi and declines to an average of 27% under Lagarde, when Italy switches to outgroup status (Figure 4a). The opposite pattern arises for France, where relative news coverage rises from 11% under Draghi to 16% under ingroup president Lagarde. These patterns are not driven by a small number of unusual events, but instead reflect a consistent trend, as more granular press-conference-cycle-level data for the two years surrounding the presidential transition suggest (Figure 4b). By contrast, relative coverage in Germany and Spain changes little around the presidential transition, and importantly, changes do not appear systematically related to the presidential transition (Figure A28). Because article shares are compositional – summing to one across countries by construction – an increase in one country’s share mechanically implies a decline elsewhere. However, using absolute article counts returns similar patterns, as corresponding plots show in Figure A29.

To quantify the association between ingroup status and newspaper coverage, I estimate

$$Y_{ct} = \alpha + \beta Ingroup_{ct} + \lambda_c + \tau_t + \varepsilon_{ct}, \quad (3)$$

using both the annual sample and the more granular press conference cycle sample. Y_{ct}

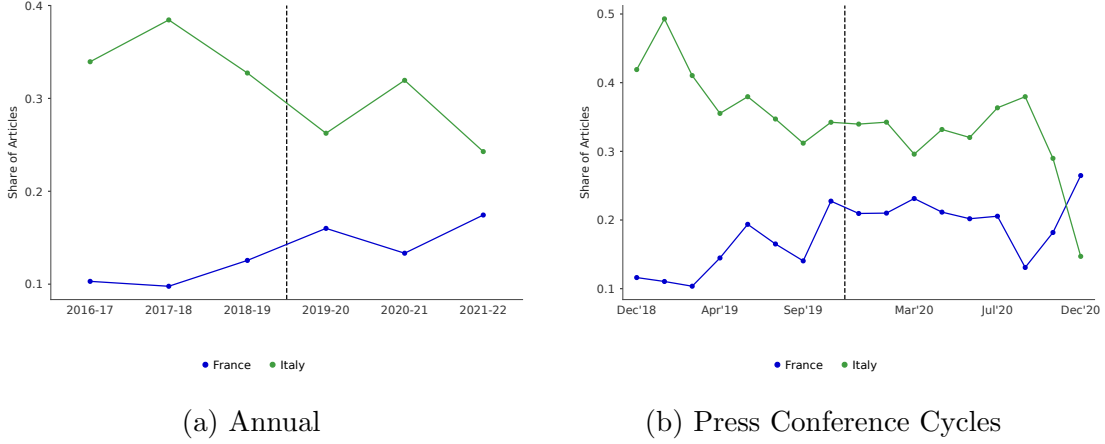


Figure 4: Share of National Newspaper Articles Over Time

Notes: The figure plots countries' share of articles about the ECB over time. The vertical grey dashed line indicates the switch in presidents. Panel (a): each dot represents one year (from November 1 until October 31 of the following year). Three years per president are shown. Panel (b): each dot represents coverage over the preceding six-week press conference cycle, plotted at the cycle-ending press conference date. The first post-transition dot covers a cycle spanning both Draghi and Lagarde. Data are only available for exactly one entire year per president, which means the first and last press conference cycles shown are shorter than six weeks. The newspaper sample is described in Section 3.1.

denotes either country c 's share of ECB-related articles in period t or the corresponding article count. Country fixed effects, λ_c , absorb time-invariant differences in coverage across countries, and period fixed effects, τ_t , absorb shocks to ECB-related coverage common to all countries. In the annual specification, t denotes November-to-October sample years; in the higher-frequency specification, t denotes press conference cycles. The indicator $Ingroup_{ct}$ equals one for coverage by Italian newspapers during Draghi's presidency and for coverage by French newspapers during Lagarde's presidency.

Because ingroup status switches from Italy to France at the presidential transition in November 2019, while remaining zero for Germany and Spain throughout the sample, β is identified by the reversal in coverage between Italy and France. Germany and Spain help discipline common movements in ECB-related coverage through the period fixed effects, but the identifying variation for β comes only from the change in ingroup status for Italian and French coverage. The balanced newspaper panel further implies that time-invariant differences in outlets' ECB coverage are absorbed; what is required is that the sampled outlets do not change their coverage propensities at the presidential transition for reasons unrelated to the president's ingroup status.

Table 4 reports the regression estimates. In the full six-year sample, controlling for country and time fixed effects, ingroup status is associated with a statistically significant 6.1-percentage-point higher share of ECB-related news in the corresponding country (column

(1)). The press-conference-cycle specification, which includes country and press-conference-cycle fixed effects, yields a comparable estimate of 5.1 percentage points (column (2)). To verify that the effects are not merely mechanical consequences of using shares, columns (3) and (4) use absolute article counts as an alternative outcome. The estimates imply approximately 272 additional articles annually and 12 additional articles per press-conference cycle, respectively.²⁴ The results are robust to including newspaper-outlet fixed effects, excluding the November 2019 – October 2020 sample year, which contains the onset of the pandemic, and excluding each individual newspaper outlet in turn.

Table 4: Ingroup Effect on Information Availability: Newspaper Coverage

	(1)	(2)	(3)	(4)
	<i>Share of articles</i>		<i>Number of articles</i>	
	Annual	PC cycle	Annual	PC cycle
Ingroup	0.061*** (0.018)	0.051** (0.023)	271.833*** (68.118)	11.722 (7.592)
Germany	0.191*** (0.013)	0.044** (0.021)	802.750*** (67.044)	9.139 (9.134)
Spain	0.161*** (0.017)	0.117*** (0.021)	682.583*** (80.714)	40.806*** (7.794)
Italy	0.180*** (0.018)	0.157*** (0.023)	739.833*** (68.118)	59.944*** (7.592)
Constant (France)	0.102*** (0.011)	0.158*** (0.028)	601.250*** (76.051)	29.847*** (7.839)
Year FE	✓		✓	
Press Conference Cycle FE		✓		✓
Observations	24	72	24	72
R-squared	0.925	0.674	0.941	0.848

Notes: OLS regressions with robust standard errors in parentheses. Columns (1) and (3) use six years of annual newspaper volume data. Columns (2) and (4) use two years of data at the level of six-week press conference cycles. Columns (1) and (2) report effects on the share of articles for all four euro area countries; columns (3) and (4) on the number of articles. *Ingroup* equals 1 for the ECB president’s country of origin during their term (Italy under Draghi, France under Lagarde). Baseline country: France. The newspaper sample is described in Section 3.1. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The estimates in this section should be interpreted as suggestive associations rather than strictly causal effects, because the presidential transition is anticipated, coincides with changes in the policy and economic environment, and involves two real policymakers whose identities are multidimensional and whose communication styles may differ. At the same time, none of these factors is tailored to produce the specific pattern the ingroup channel predicts – a

²⁴The ingroup effect in column (4) is positive but imprecise, just missing the 10% significance level (11.7, $p = 0.12$). This lower precision likely reflects higher residual variance in article counts at the six-week frequency, while the annual estimate in column (3) has the same sign but is estimated much more precisely.

reversal in relative coverage between Italy and France with little change in Germany and Spain – so that this pattern emerging despite the many other differences between the two presidencies is all the more notable.

The content of the articles corroborates the interpretation that messenger identity drives these coverage shifts. I measure how salient a messenger is by whether an article mentions the ECB president; because articles are collected based on ECB-related keywords, not policymakers’ names, such mentions capture the president’s prominence within ECB news. Italian newspapers mention Draghi 884 times during his presidency but Lagarde only 365 times during hers, while French newspapers mention Lagarde more often than Draghi (358 versus 298 times). These differences go beyond the volume shifts documented above: across press-conference cycles, the probability that an ECB-related article mentions the president is systematically higher in the president’s home country, rising for France and falling for Italy at the transition (Figure A30). Germany and Spain, where neither president is an ingroup messenger, mention the two presidents at similar rates, with only a slight common decline under Lagarde that plausibly reflects broader time trends, and show no shift around the transition. Similar patterns emerge using the number of articles mentioning the president or normalizing by article length.²⁵

These findings constitute a second dimension of messenger effects: besides shaping how agents use policy information, messenger identity is also associated with how much information reaches them in the first place. For ingroup ECB presidents, national media systematically provide more coverage of the central bank’s communication, resulting in agents receiving more information, even when they do not actively follow the ECB.

4 External Validity

This section provides evidence from social media and household inflation expectations to show that the messenger effects identified in the experiment (Section 2) and through newspaper coverage (Section 3) extend beyond a controlled experimental setting and the print media context. Thereby, this section establishes external validity.

4.1 Validating Evidence I: Social Media

High-frequency Twitter data for the euro area confirms that identity shapes responses to policy communication: following major ECB communication events, messenger identity

²⁵Messenger salience matters in its own right, as salient attributes disproportionately capture attention and shape choices (Bordalo et al., 2012, 2013; Mullainathan et al., 2008).

affects both the public’s use of policy-relevant information and the reach of that information, validating the causal experimental findings and corroborating the newspaper evidence.

4.1.1 Twitter Data

To capture how individuals engage with ECB communications, I use textual data from Twitter. Compared to survey or market data, Twitter offers several advantages. It provides unsolicited, high-frequency expressions of public opinion that reflect how individuals from different nationalities engage with policy communication. Its scale allows for the measurement of attention, reach, and belief formation with a granularity unmatched by other data sources. Users can be tracked over time, enabling panel-style analysis of beliefs and their updates among both professionals and non-experts. Although Twitter users are not fully representative of the general population, the platform’s scale provides valuable insights into cross-national variation in responses to policy communication. I construct a multilingual dataset of publicly available tweets posted between 1 November 2016 and 31 October 2022, using the Academic Twitter API. This period covers three full years under Mario Draghi’s presidency (1 November 2016–31 October 2019) and three years under Christine Lagarde (1 November 2019–31 October 2022). Tweets are included if written in German, French, Italian, or Spanish, reflecting the main languages of the four largest euro area countries. Tweet language proxies nationality and does so well (see Appendix B.3 for validations). English tweets are also collected for completeness, but are excluded from nationality-specific analyses because they cannot be reliably attributed to a single nationality. Tweets are selected if they contain “ecb,” “European Central Bank,” or their language-specific equivalents.²⁶ The final dataset contains 8,031,937 tweets after extensive cleaning and translation (Table A12 shows a summary by language). Appendix B.3 documents the cleaning process and provides summary statistics.

For the evidence presented in this section, I define the *ingroup* as users whose tweet language corresponds to the nationality of the ECB president. Thus, Italian-language tweets are ingroup during Draghi’s term and French-language tweets during Lagarde’s.

4.1.2 Stronger Information Use

To show the external validity of the causal results from the experiment (Section 2), I exploit the panel structure of the Twitter data and study how individuals use information, and find that messenger identity affects how strongly beliefs adjust in response to new information.

²⁶Selection terms follow Ehrmann and Wabitsch (2022), with two key modifications. First, tweets are excluded if they mention “Draghi” or “Lagarde” without reference to the ECB, to avoid capturing content unrelated to ECB communication (e.g., Draghi’s role as Italian Prime Minister or Lagarde’s prior tenure at the IMF). Second, tweets must contain keywords in the original text, including tweets, replies, quote tweets, and retweets.

Specifically, I assess how within-user beliefs change following ECB press conferences (the signals). These press conferences are ideal for studying belief updates because, as the institution’s primary event for communicating monetary policy, they are highly anticipated, and their messages are carefully crafted. As press conferences occur every six weeks, the sample includes 48 communication events.²⁷ The beliefs of interest are about the economy and are proxied with sentiment, a widely used measure of shifts in expectations about economic conditions (e.g., Jaimovich and Rebelo 2009; Chahrour et al. 2021; Flynn and Sastry 2025). I measure tweet sentiment using a dictionary-based approach for its simplicity and transparency, which provides a score of how positive or negative a tweet is, normalized to $[0, 1]$.²⁸ Sentiment distributions are highly similar across languages, making it unlikely that language-specific differences drive the results (Appendix B.3).

For each user i who tweets both before and after a press conference t , $Prior_{i,t}$ is the sentiment of i ’s last tweet during the quiet period – the seven days preceding the press conference, during which no major ECB communication occurs – and $Posterior_{i,t}$ is the sentiment of i ’s first tweet within 24 hours after it. Measuring priors within the quiet period allows for a reasonably wide window to collect beliefs, while ensuring that no other ECB communication contaminates them, so that any changes from $Prior_{i,t}$ to $Posterior_{i,t}$ can be attributed to the press conference itself. To measure how strongly users draw on the press conference signal compared to their own prior beliefs, I estimate

$$\begin{aligned} Posterior_{i,t} = & \beta_1 Prior_{i,t} + \beta_2 Prior_{i,t} \times Ingroup_{i,t} + \beta_3 Ingroup_{i,t} \\ & + \beta_4 Signal_t + \beta_5 Signal_t \times Ingroup_{i,t} + \epsilon_{i,t}, \end{aligned} \quad (4)$$

where $Signal_t$ is the monetary policy surprise from the euro area Monetary Policy Event-Study Database (EA-MPD) of Altavilla et al. (2019), measured as the high-frequency change in two-year Overnight Index Swaps (OIS-2Y) around the announcement to capture monetary policy news including any informational effects of forward guidance. Estimating Equation (4) without a constant lets the coefficients be read as the weights a Bayesian agent places on each source of information, and identifies prior reliance (β_1, β_2) and signal use (β_4, β_5) by ingroup status.²⁹ Because ingroup status varies within language across the presidential transition,

²⁷The monetary policy announcement changed from 13:45 to 14:15 CET after July 21, 2022. Similarly, the starting time of the press conference shifted from 14:30 to 14:45 CET (see [here](#)).

²⁸Specifically, I use the polarity measure integrated in the Python package *TextBlob*, which is based on Princeton University’s WordNet lexicon (Loria, 2018). Results are robust to alternative natural language processing (NLP) methods such as *BERT*, a large language model (LLM).

²⁹Unlike in the experiment, where priors are induced and held fixed across messenger conditions by design, priors here are formed endogenously and may differ systematically between in- and outgroup users. Imposing a common coefficient on priors would risk attributing group differences in prior reliance to the signal interaction. I therefore interact both the prior and the signal with ingroup status, allowing the data to determine how

estimated differences in prior and signal use between in- and outgroup do not merely reflect differences in languages.

Both in- and outgroup users incorporate new information, but they weigh the two sources very differently. Outgroup users rely overwhelmingly on their priors, placing a weight of 0.81 on prior beliefs against only 0.20 on the press conference signal – a ratio of four to one. Ingroup users, by contrast, weight the two sources almost equally (0.53 and 0.53), placing nearly three times as much weight on the ECB signal as outgroup users and about a third less on their priors. Another advantage of the Twitter data is that I can distinguish between user types, showing that this ingroup effect is not solely driven by experts such as journalists nor solely by non-experts. Following [Ehrmann and Wabitsch \(2022\)](#), I classify users by their monetary expertise and find that results are remarkably similar across expert and non-expert subsamples, though the shift toward the signal is somewhat larger among non-experts, whose weight on the signal rises from 0.16 to 0.50, compared with 0.19 to 0.48 among experts (Figure 5a, Table A13).³⁰ These estimates rest on a specific proxy for the content of each press conference and on a linear updating rule, but neither is strictly required, as an alternative analysis based on the relative size of belief updates shows in Appendix B.3.

Overall, these patterns on Twitter confirm stronger information use for the ingroup. The size of ingroup effects may seem more pronounced on Twitter than in the experiment, which reflects that outside the controlled experimental environment, measured information use reflects both a greater weight on a given signal – the channel isolated by the experiment, which holds signal availability fixed and better controls priors – and greater exposure to signals, the second dimension of messenger effects (Section 3). To corroborate this dimension of messenger effects directly, I next assess how ECB communication spreads on social media.

4.1.3 Higher Information Availability

To assess information availability on Twitter, I compare tweet volume across languages. For each language, I compute the relative share in tweet volume and compare it across presidencies in Figure 5b. To mitigate concerns about general Twitter adoption or one-off events driving results, I split the sample into six-week intervals between ECB press conferences, which as the primary communication events present a natural basis for structuring the sample over time.³¹ Spanish users are consistently the most active, accounting for nearly 40% of ECB-related

each group weights the two sources of information.

³⁰Users are classified as experts if they comment on at least every second press conference; non-experts if they do so only sporadically and predominantly tweet about ECB-unrelated topics; and neither experts nor non-experts if they fall in between. Users falling in between are included in the full sample but not analyzed separately.

³¹The 48 press conferences in the sample result in 49 distinct press conference cycles. The presidential transition occurred during a cycle, which I split for cleaner results, yielding 50 president-cycle observations.

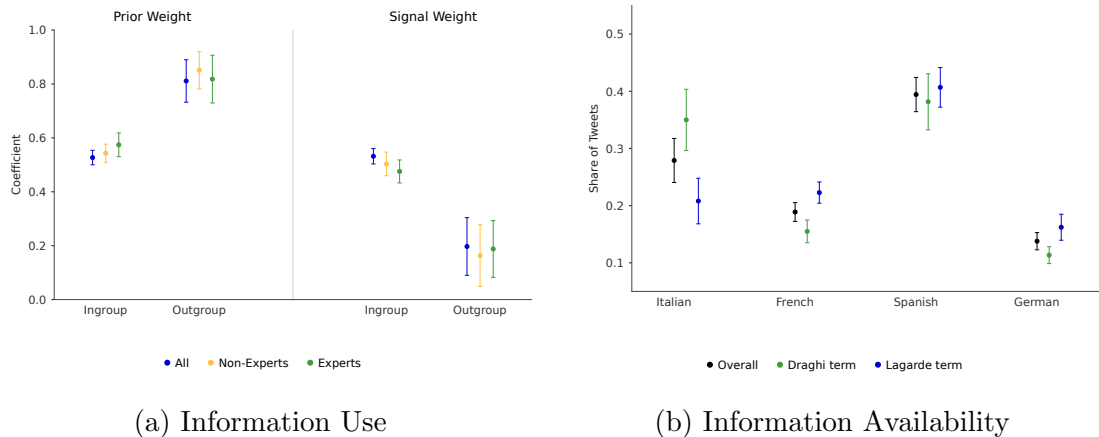


Figure 5: Ingroup Effects on Information Use and Availability

Notes: Panel (a) plots linear combinations of OLS coefficients from columns (3)–(5) of Table A13 with 95% confidence intervals based on two-way clustered standard errors (by press conference and user). The left (right) panel shows the weight placed on prior beliefs (on the press conference signal) by individuals of the in- and outgroup. Blue: entire sample; orange: non-experts; green: experts. The expert classification follows Ehrmann and Wabitsch (2022). Panel (b) shows tweet volume expressed as the shares of tweets by language across each six-week press conference cycle, with 95% confidence intervals. Green and blue bars indicate Draghi’s and Lagarde’s presidencies, respectively. Black indicates the entire sample.

tweets, followed by Italians, who make up about one-third of the sample. Importantly, during Draghi’s presidency, the Italian share is markedly higher than under Lagarde, while French tweet shares increase sharply under Lagarde. By contrast, Spanish and German tweet shares remain relatively stable.³²

A simple regression estimates that ingroup status increases information availability by 10.5 percentage points ($p < 0.01$) (Table A15), a sizable effect given average tweet shares of 19% in France and 28% in Italy. As with newspaper articles, messenger salience corroborates that the identity of the messenger is a driver of the available information.³³

At face value, the effect is about twice the size of its print equivalent, where a similar specification yields 5.1 percentage points (Table 4, column (2)). Scaled by the dispersion of shares within each medium, however, the two effects are comparable: the ingroup effect accounts for 41% of the cross-language spread in tweet shares and 33% of the cross-country spread in article shares. The ingroup effect thus explains a similar share of total variation in

³²The German share rises slightly during Lagarde’s term, which may reflect Isabel Schnabel’s entry to the ECB Executive Board or the coinciding disproportional growth of German Twitter (see Table A11 in Appendix B.3). These factors highlight that, while patterns corroborate ingroup patterns on information availability, they should not be interpreted as causal.

³³Comparing how often ECB presidents are mentioned in ECB-related Twitter samples across languages shows that “Draghi” ranks in the Italian top 10 but around 50th in French, while “Lagarde” is in the French top 10 but much lower in Italian. In English, German, and Spanish samples – where neither president is in the ingroup – both names appear with similarly low frequency. Figure A31 illustrates this with word clouds.

traditional and social media.

The composition of the two samples may further explain the larger raw effect on social media. Newspaper coverage is produced entirely by professionals, whereas the Twitter sample spans a wider range of monetary policy expertise (Ehrmann and Wabitsch, 2022), and the effect is markedly stronger among non-experts (7.8 percentage points) than among experts (2.4 percentage points). This means that professionals are less susceptible – but not immune – to messenger effects, but also that information availability is influenced not just by experts and professionals, but also by non-experts spreading information. Altogether, these findings confirm that messenger identity systematically shapes the availability of policy information.

Overall, the evidence from Twitter confirms that messenger identity shapes not only the availability of policy information but also the way audiences process it. There is more information available for ingroup audiences who also use information more, both suggesting that policy communication is more influential when conveyed by an ingroup messenger.

While tweets highlight modern information supply beyond traditional media, and their high frequency allows changes in experts’ and non-experts’ beliefs alike to be attributed to specific communication events, they come with limitations: Twitter activity is shaped by contemporaneous news, platform algorithms, and changing user composition, and language differences may add measurement noise. The within-language, over-time structure of the analysis mitigates these concerns, but may not fully eliminate them. That messenger effects emerge so clearly despite this noise, mirroring the cleaner evidence from Sections 2 and 3, is reassuring. Still, Twitter users are not fully representative of the population, and tweet sentiment cannot directly speak to inflation expectations. I therefore next complement this evidence with data from a representative survey of household inflation expectations.

4.2 Validating Evidence II: Household Inflation Expectations

According to the messenger effects identified in this paper, the ingroup faces a higher number of signals and uses them more strongly when forming beliefs. On both accounts, the ingroup’s inflation forecasts should therefore – *ceteris paribus* – lie closer to the ECB projections, and be more precise. To assess whether this is indeed the case, I first compare the distance between the ECB’s 12-month-ahead inflation forecasts and forecasts of households in the euro area, and then analyze the precision of households’ forecasts by assessing their provided forecast uncertainty. Values for the ECB projections are taken from the ECB’s Macroeconomic Projection Database (MPD), which is published four times a year.³⁴ Household

³⁴The MPD is published in March, June, September, and December and contains information on the outlook for the euro area and contributes to the ECB Governing Council’s assessment of economic developments and

inflation expectations come from the ECB’s Consumer Expectations Survey (CES), which is a representative survey of consumers in selected euro area countries started in 2020. I use their aggregated monthly mean and median indicators of quantitative household inflation expectations over the next 12 months.³⁵ Similarly, for forecast uncertainty, I use the ECB-provided aggregate of median individual forecast uncertainty, where each household’s uncertainty is measured as the interquartile range of the subjective probability distribution implied by its allocation of probabilities to predefined inflation bins.

Forecasts of one-year-ahead inflation by households in France, the only ingroup in the sample, indeed most closely resemble the ECB’s forecasts (Figure 6a).³⁶ During this period, French median forecasts exceed the ECB’s projections by 0.5 percentage points on average, compared with 1.1 percentage points across countries. Mean forecasts show the same pattern: the average gap is 1.7 percentage points in France versus 2.6 percentage points across countries (Figure A33). Regressing the distance on country indicators with survey-wave fixed effects confirms that, relative to French households, median forecasts in other countries are significantly further from the ECB’s projections, except in Germany, where the difference is statistically insignificant (Table A16, column (2)).

Turning to the second moment of expectations, French and German households report the lowest forecast uncertainty (1.4 and 1.3 percentage points, respectively), compared with up to 2.9 percentage points in Italy (Figure 6b). Regressing uncertainty on country indicators with survey-wave fixed effects confirms that expectations are more precise in France: relative to France, uncertainty is significantly higher in Belgium, Spain, the Netherlands, and Italy (all $p < 0.01$), while only Germany displays marginally lower uncertainty ($p < 0.10$; Table A16, column (3)).

Both the distance and the precision of household forecasts thus move in line with the messenger effects of this paper’s main findings: under a French ECB president, French households’ forecasts are closest to the ECB’s projections and among the most confident. Importantly, these patterns are correlational rather than causal, as cross-country differences in attitudes, institutional trust, or realized inflation over this period may contribute to them. Nevertheless, they are jointly consistent with the cleaner evidence in Sections 2 and 3.

risks to price stability. The June and December projections are conducted by Eurosystem staff, while the March and September projections are conducted by ECB staff.

³⁵The survey provides mean and median values that are population-weighted and winsorized at the 2nd and 98th percentiles of the weighted distribution of responses for each survey wave and country.

³⁶Forecasts are lagged by one month, giving information from ECB forecasts time to reach individuals. Not lagging the ECB’s forecasts yields very similar findings. Available data span from April 2020 to December 2023, when the ECB’s president is French (Christine Lagarde).

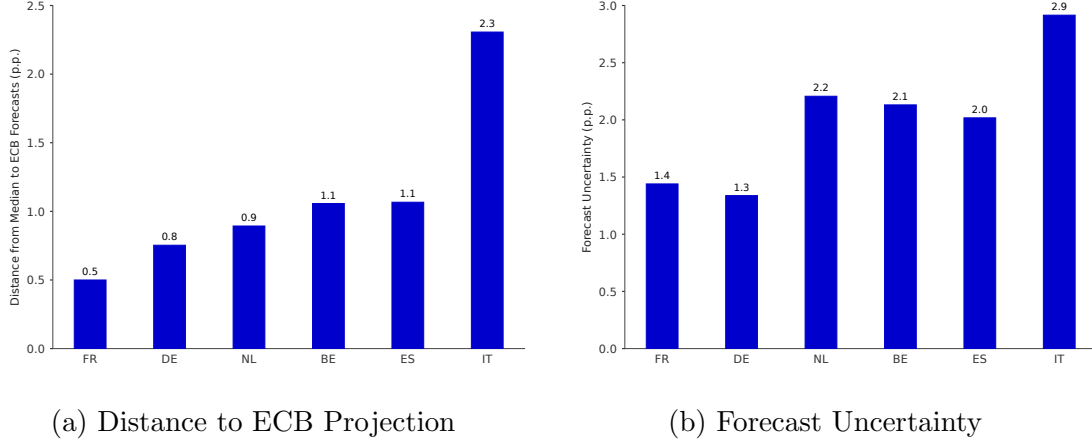


Figure 6: Household Inflation Forecasts

Notes: Panel (a) shows average difference between median household inflation forecasts over the next twelve months and the ECB’s projection (lagged by one month). Panel (b) shows households’ forecast uncertainty (median across households). Point forecasts are winsorized at the 2nd and 98th percentiles of the weighted distribution for each survey wave and country. Panel (a) covers the 15 survey waves coinciding with ECB projection releases between April 2020 and December 2023, whereby data for Belgium are missing for the first wave (April 2020); panel (b) covers all 45 monthly waves over the same period. Countries are ordered throughout by the distance measure in panel (a).

Altogether, the social media and survey evidence validate the dimensions of messenger effects: messenger identity influences how information spreads and how beliefs adjust. In-group messengers therefore improve both the reach and influence of policy communication. Motivated by these facts, Section 5 develops a stylized coordination model to evaluate optimal communication strategies under messenger effects, providing welfare-based guidance.

5 Modeling Optimal Communication

In light of the empirical messenger effects, how should policymakers select messengers to communicate with the public? To derive welfare-maximizing communication policies, I extend the ‘beauty contest’ framework by [Morris and Shin \(2002\)](#), incorporating messenger identity. In this model, agents receive private and public signals, allowing me to capture the dual role of public information disclosure: to inform about an economic fundamental and to coordinate agents with others.³⁷ There are in- and outgroup agents – defined by whether they share characteristics with the messenger – who differ in the availability and use of the public signal. The model is deliberately kept abstract yet general, allowing it to apply to any policy communication with diverse messengers and audiences. While nationality in the euro

³⁷Examples of this coordination motive for inflation are firms setting prices considering their competitors’ prices, or agents negotiating wages similar to others.

area is the motivating example, the model extends to other dimensions of heterogeneity, such as age, gender, ethnicity, socioeconomic background and economic expertise. I study two communication policies: (i) a *disclosure policy* that sets the precision of the public signal, and (ii) a *delegation policy* that selects the messenger(s) and thereby changes the composition of in- and outgroup agents.

5.1 The Model Environment

A continuum of agents $i \in [0, 1]$ choose actions with a beauty contest coordination motive, caring about aligning actions with others' actions in addition to an unknown fundamental. Each agent i chooses an action $a_i \in \mathbb{R}$ to maximize payoff $u_i \in \mathbb{R}$ (that is, minimize the loss):

$$u_i = -(1 - r)(a_i - x)^2 - r(a_i - \bar{a})^2, \quad \bar{a} = \int_0^1 a_i di, \quad r \in [0, 1], \quad (5)$$

where $x \sim \mathcal{N}(0, \tau_x^{-1})$ is an unknown fundamental, such as future inflation, with $\tau_x \rightarrow 0$. The parameter $r \in [0, 1]$ captures the strength of the coordination motive.

Agents are of three types: g , o^I , o^U . Those whose characteristics match the messenger of public information are *ingroup* agents (g); all others are *outgroup* agents (o). Types are identical except in their access to and processing of public information, reflecting the messenger effects documented in Sections 2–4. Outgroup agents may be either *informed* (o^I) or *uninformed* (o^U), depending on whether they observe the public signal. When informed, they downweight its precision relative to ingroup agents.

Information Structure. Depending on agent type, the information structure is as follows: each agent i observes a private signal

$$y_i = x + \epsilon_{iy}, \quad \epsilon_{iy} \sim \mathcal{N}(0, \tau_y^{-1}), \quad (6)$$

where ϵ_{iy} is independent of all other disturbances, $\mathbb{E}[\epsilon_{iy}\epsilon_{jy}] = 0$ for all $j \neq i$, and $\int \epsilon_{iy} di = 0$ almost surely. Besides private information, some agent types also observe a public signal

$$Y = x + \epsilon_Y = x + \epsilon_z + \epsilon_V, \quad \epsilon_z \sim \mathcal{N}(0, \tau_z^{-1}), \quad \epsilon_V \sim \mathcal{N}(0, \tau_V^{-1}), \quad (7)$$

where ϵ_z and ϵ_V are independent of each other and all other disturbances. The two noise components capture (i) the exogenous uncertainty in the policymaker's information about the fundamental (ϵ_z), and (ii) additional noise ϵ_V that is fully controlled by the policymaker and determines the overall precision of the public signal, $\tau_Y = \tau_z + \tau_V$.

Crucially, the public signal is not strictly common knowledge in the standard sense.³⁸ Ingroup agents always observe Y . By contrast, outgroup agents observe Y only if its magnitude exceeds their idiosyncratic threshold d_o (i.e., $|Y| \geq d_o$). For each outgroup agent, $d_o \sim \mathcal{N}_+(0, 1)$ (truncated normal on $[0, \infty)$), i.i.d. across agents and independent of x , ϵ_z , ϵ_V , and ϵ_{iy} . Hence, conditional on Y , the share of informed outgroup agents is

$$A \equiv \Pr(d_o \leq |Y| | Y) = 2\Phi(|Y|) - 1, \quad (8)$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function. The fraction of outgroup agents who do not observe the signal is $1 - A$. As $|Y|$ becomes very large, A approaches 1, meaning nearly all outgroup agents observe large signals, whether positive or negative. This parsimoniously captures newsworthiness thresholds in media coverage (Nimark and Pitschner, 2019). Intuitively, larger signals are more likely to be reported, increasing the probability that outgroup agents receive them. The truncated normal choice ensures non-negativity and closed-form expressions for A , ensuring tractability.

The information set of agent i can thus be stated as

$$\Omega_i = \begin{cases} \{y_i, Y\}, & \text{if } i \text{ is type } g, \\ \{y_i, Y\}, & \text{if } i \text{ is type } o^I \text{ where } (|Y| \geq d_o), \\ \{y_i\}, & \text{if } i \text{ is type } o^U \text{ where } (|Y| < d_o). \end{cases} \quad (9)$$

Finally, to account for the ingroup effects on information use shown in Section 2, I assume that informed outgroup agents apply *information resonance* (Malmendier and Veldkamp, 2022), meaning they downweight the accuracy of public information based on an additional *resonance weight* ρ_{im} . The *perceived accuracy* of the public signal for agent type o^I is

$$\omega_{im} \equiv \rho_{im} \tau_Y, \quad \rho_{im} \in (0, 1], \quad \rho_{im} = 2 - 2\Phi(\chi \|\theta_i - \theta_m\|), \quad (10)$$

where $\theta_i, \theta_m \in \mathbb{R}^k$ are vectors of observable characteristics, and ρ_{im} defines the resonance that a signal coming from messenger m has for receiving agent i . Resonance is strictly decreasing in $\|\theta_i - \theta_m\|$, the Euclidean distance between agent and messenger in characteristic space, and depends on $\chi > 0$, the sensitivity to mismatch.³⁹ The normal cumulative distribution function $\Phi(\cdot)$ equals 0.5 at 0, and hence the expression $2 - 2\Phi(\cdot)$ takes on a value of 1 at zero

³⁸This departs from standard coordination models in the literature. An exception is Cornand and Heinemann (2008), who restrict the availability of public signals but do not allow it to vary by agent type or signal size.

³⁹If $\chi = 0$: agents learn from everyone without discounting anyone's information ($\rho_{im} = 1$). If χ is very large: agents disregard any information from messengers that do not exactly match their characteristics ($\rho_{im} \rightarrow 0$). $0 < \chi < 1$ represents cases in-between.

and then declines quickly towards zero.

Communication Policies. The policymaker makes two choices in the model. First, as part of their *disclosure policy*, they choose the precision of the public signal by setting τ_V , which governs the magnitude of the endogenous noise component ϵ_V in Equation (7). Full disclosure corresponds to $\tau_V \rightarrow \infty$, complete opacity to $\tau_V \rightarrow 0$, and partial disclosure to any interior value $\tau_V \in \mathbb{R}_+$. The noise term ϵ_z represents the *irreducible* component of the policymaker’s information, while ϵ_V is the *reducible* part under their control.

Second, as part of their *delegation policy*, the policymaker chooses the messenger(s) responsible for communicating the public signal, thereby determining the share of ingroup agents α . Delegation means appointing additional or different messenger(s), who differ in characteristics, to convey the *same* public signal. This raises the share of the ingroup α , as agents are assumed to attend to signals from ingroup messengers whenever available. The opposite is *centralization*, where the policymaker limits communication to a few or very homogeneous messengers to reduce α . Delegation affects both the reach and resonance of the message, but does so only via α , leaving τ_V unchanged and A determined by the realized signal magnitude.⁴⁰ Two concrete implementations of delegation in the context of central bank communication are (i) using other board members and (ii) involving national central bank (NCB) governors in the Eurosystem, as suggested by the reduced-form evidence in this paper. For tractability, I treat $\alpha \in [0, 1]$ as the endogenous outcome of the delegation choice, increasing either when a single messenger shares characteristics with a larger share of agents, or when multiple messengers are chosen to increase representational diversity.

The shares of the population in the ingroup (α), outgroup ($1 - \alpha$), and informed outgroup (A) are common knowledge. However, consistent with the literature (e.g., Angeletos and Huo (2021); Gust et al. (2025)), agents assume homogeneous information use among informed agents: agents are unaware of their own or other agents’ biases. Consistent with this assumption of homogeneous information use within each agent type, I treat ρ_{im} as a group-level parameter that is constant for all informed outgroup agents (denoted ρ henceforth).

Timeline. The model consists of two stages. First, the policymaker sets disclosure τ_V (hence τ_Y) and delegation (the messenger(s), hence α), aiming to maximize the ex-ante welfare of agents. Second, agents observe signals, form posteriors, and choose actions, aiming to maximize expected utility using their information sets.

⁴⁰Perceived expertise may vary with the messenger. I relax the assumption that delegation does not affect expertise in Appendix C.5, allowing τ_Y to decrease in α .

5.2 Optimal Actions and Equilibrium

Having specified the information structure and timing, I now characterize how agents optimally weight available signals and how these choices aggregate in equilibrium. This will be important later to characterize the welfare effects of disclosure and delegation policies. In equilibrium, neither policymakers nor agents want to deviate from their strategies. Each agent chooses her action a_i to maximize her own expectation of realized utility $\mathbb{E}[u_i|\Omega_i]$. Given the quadratic payoff, which is strictly concave in a_i for $r \in [0, 1)$, the first-order condition delivers the best-response function

$$a_i = (1 - r)\mathbb{E}[x|\Omega_i] + r\mathbb{E}[\bar{a}|\Omega_i]. \quad (11)$$

Agents do not observe others' actions. Instead, they form expectations about $\bar{a}$ from beliefs about others' posteriors. Suppose that all agents of each type use a linear strategy

$$a_i = \kappa_i y_i + (1 - \kappa_i)Y. \quad (12)$$

In equilibrium, all informed agents of a given type – ingroup (g) or informed outgroup (o^I) – choose the same optimal signal extraction weight κ_i^* . Uninformed outgroup agents (o^U) rely solely on their private signal. The fixed point of these best-response functions yields the equilibrium choices summarized in Proposition 1.

Proposition 1. *There exists a unique linear equilibrium. The equilibrium actions are*

(a) *Ingroup agents*

$$a_g(y_i, Y) = \frac{\tau_Y Y + \tau_y q y_i}{\tau_Y + \tau_y q}, \quad (13)$$

(b) *Informed outgroup agents*

$$a_{o^I}(y_i, Y) = \frac{\rho \tau_Y Y + \tau_y q y_i}{\rho \tau_Y + \tau_y q}, \quad (14)$$

(c) *Uninformed outgroup agents*

$$a_{o^U}(y_i) = y_i. \quad (15)$$

where $q \equiv 1 - r(\alpha + (1 - \alpha)A) = 1 - rs$.

Proof in Appendix C.1, which also derives the signal extraction weights κ_i^* by agent type. Equilibrium actions of informed agents differ only in their perceived precision of Y : ingroup agents use τ_Y , while informed outgroup agents place less weight on Y , using $\rho \tau_Y$. For informed agents, the weight on the private signal is further scaled by $q = 1 - rs$, which captures the coordination pull toward the public signal, and is independent of ρ . A stronger coordination

motive r increases the desire to align with others, raising the relative weight on the public signal, as in [Morris and Shin \(2002\)](#). In addition, q decreases in $s = \alpha + (1 - \alpha)A$, the share of informed agents who observe Y . Conversely, uninformed outgroup agents do not observe Y and solely rely on y_i , which weakens the coordination effect, leading all agents to place greater weight on their private signals.

In equilibrium, the population average action equals

$$\bar{a} = \alpha a_g + (1 - \alpha)A a_{oI} + (1 - \alpha)(1 - A)a_{oU}(y_i), \quad (16)$$

which summarizes how the composition of agent types shapes aggregate behavior.

The model nests [Morris and Shin \(2002\)](#) in two special cases (i) $\alpha = 1$, so that all agents are ingroup (homogeneous), and (ii) $|Y| \rightarrow \infty$ with $\chi = 0$, so that all agents are informed ($A \rightarrow 1$) and update without bias ($\rho = 1$).

5.3 Social Welfare

With equilibrium actions characterized, I now turn to the planner's problem of evaluating how communication policies affect aggregate welfare. When the equilibrium weighting of information coincides with the socially optimal weighting, the policymaker's best strategy is to fully disclose their information. In general, however, because agents optimally weight available information subject to their objectives, greater transparency may amplify coordination distortions, creating a welfare trade-off. To evaluate welfare, I follow the literature and define social welfare as the population average of individual utilities

$$W(\{a_i\}, x) \equiv \frac{1}{1 - r} \int_0^1 u_i(a, x) di = - \int_0^1 (a_i - x)^2 di. \quad (17)$$

The policymaker chooses disclosure and delegation policies to maximize the ex-ante expected value of utilitarian welfare, $\mathbb{E}[W(\{a_i\}, x)]$, evaluated from the perspective of a benevolent social planner who aggregates utilities across the population. This assumption abstracts from any political or strategic incentives.

To characterize and discuss the optimal policy, it is helpful to first decompose utilitarian welfare $\mathbb{E}[W(\{a_i\}, x)]$ into its constituent components.

Lemma 1. *The policymaker maximizes utilitarian welfare $\mathbb{E}[W(\{a_i\}, x)]$ with respect to τ_Y and α , which can be decomposed into*

$$\mathbb{E}[W(\{a_i\}, x)] = -\alpha \frac{\tau_Y + \tau_y q^2}{(\tau_Y + \tau_y q)^2} - (1 - \alpha)A \frac{\rho^2 \tau_Y + \tau_y q^2}{(\rho \tau_Y + \tau_y q)^2} - \frac{(1 - \alpha)(1 - A)}{\tau_y}. \quad (18)$$

Expected welfare decreases with r as stronger strategic complementarities lead agents to over-rely on the public signal at the expense of private information. The effect of information diffusion among outgroup agents (A) is ambiguous and depends on r and the quality of public information: under strong coordination, larger A raises welfare by improving the reach of high-quality public signals, while lower A can preserve welfare if noisy public signals lead to coordination distortions due to strategic complementarities. Next, I assess the specific welfare implications of both disclosure and delegation policies.

5.4 Welfare Implications of Communication Policies

To evaluate the welfare effects of communication policies – both disclosure and delegation policies – I conduct partial comparative statics on expected social welfare. I fix the resonance weight ρ , and take the realization of x and Y as given. Thereby, I condition on realized signal magnitude $|Y|$, holding the fraction of informed outgroup agents A fixed, while varying τ_Y , τ_y , r , and α .

5.4.1 Public Signal Precision – The Disclosure Policy

To examine how the precision of public information affects welfare, recall that disclosure increases τ_Y , the precision of the public signal Y . Under full disclosure ($\tau_Y \rightarrow \infty$), Y is as precise as the policymaker’s underlying information (τ_z); under complete opacity ($\tau_Y \rightarrow 0$), Y is pure noise. When strategic complementarities are strong, disclosure improves welfare only if public information is sufficiently precise relative to private signals. Otherwise, disclosure may reduce welfare, as agents over-coordinate on a noisy public signal, neglecting better private information (Morris and Shin, 2002). Introducing messenger effects alters the conditions under which disclosure enhances welfare, as formalized in the following proposition.

Proposition 2. *Let $c \equiv \tau_y/\tau_Y$, let $s \equiv \alpha + (1 - \alpha)A \in [0, 1]$ be the share of agents who observe Y , and note that $q \equiv 1 - r(\alpha + (1 - \alpha)A) = 1 - rs$. Then*

$$\frac{\partial \mathbb{E}[W]}{\partial \tau_Y} \geq 0 \quad \text{whenever} \quad \begin{cases} q \geq \frac{1}{2}, \\ \frac{\rho}{2} \leq q < \frac{1}{2} \text{ and } c \leq \frac{1}{q(1 - 2q)}, \\ 0 < q < \frac{\rho}{2} \text{ and } c \leq \min \left\{ \frac{1}{q(1 - 2q)}, \frac{\rho^2}{q(\rho - 2q)} \right\}. \end{cases}$$

In particular, the upper bound is $c \leq 8$ for all r, α, A, ρ .

Derivation in Appendix C.2. The sufficient condition depends on *effective exposure* to the public signal (i.e., the extent to which Y is received and coordinated on), as captured by q .

- *Low effective exposure* ($q \geq \frac{1}{2}$): Welfare is non-decreasing in τ_Y for any c , as $q(1 - 2q)$ becomes non-positive at $q = \frac{1}{2}$, eliminating the risk of harmful over-coordination.
- *Intermediate effective exposure* ($\frac{\rho}{2} \leq q < \frac{1}{2}$): Welfare improves with higher τ_Y provided that private information is not too precise relative to public, i.e., $c \leq \frac{1}{q(1-2q)}$.
- *High effective exposure* ($0 < q < \frac{\rho}{2}$): The condition tightens to $c \leq \min \left\{ \frac{1}{q(1-2q)}, \frac{\rho^2}{q(\rho-2q)} \right\}$.

Intuitively, as effective exposure increases (that is, as q falls), the permissible range for c shrinks, requiring relatively better public signal precision for disclosure to improve welfare. Considering disclosure across different levels of effective exposure, low exposure poses no welfare concerns; intermediate exposure requires that private information quality not dominate; and high exposure is the most restrictive. Notably, the constraint $c \leq 8$ suffices to ensure welfare improvements from increased τ_Y across all parameter values (r, α, A, ρ) . Moreover, with weak r (i.e., $rs \leq \min\{\frac{1}{2}, 1 - \frac{\rho}{2}\}$), welfare increases with disclosure for any $c \geq 0$.

This means that in addition to r , τ_Y , and τ_y , two additional parameters now determine the welfare impact of disclosure: signal availability s and resonance ρ in information use. Figure 7 visualizes their impact separately by comparing the Morris and Shin (2002) benchmark in panel (a), with restricted signal availability (panel (b)) and a resonance-based updating bias (panel (c)). Notably, both s and ρ shrink the set of $(r, \tau_y/\tau_Y)$ for which additional disclosure is harmful. Intuitively, disclosure shifts weight toward the public signal in agents' actions; it harms welfare when the public signal is insufficiently precise relative to private information (large τ_y/τ_Y) when coordination motives are strong (large r), so valuable private information gets neglected. Outgroups mitigate such potential welfare losses in two ways: (i) limited reach $A < 1$ lowers the share of informed agents $s = \alpha + (1 - \alpha)A$, raising q , and (ii) the informed outgroup agents' underweighting of the public signal $\rho < 1$ further reduces the effective pull of Y (effective precision $\rho\tau_Y$). Both effects expand the parameter region in which increasing τ_Y is welfare-improving, allowing outgroups to buffer potential welfare losses from disclosure.⁴¹

Corollary 1. *The presence of outgroup agents can mitigate potential welfare losses from increased disclosure when public information is noisy.*

This raises the question of the optimal composition of in- and outgroup agents.

⁴¹For completeness, Appendix C.3 analyzes private signal precision. Heterogeneity in processing biases implies that increasing private signal precision can reduce welfare under strong coordination and sizable bias.

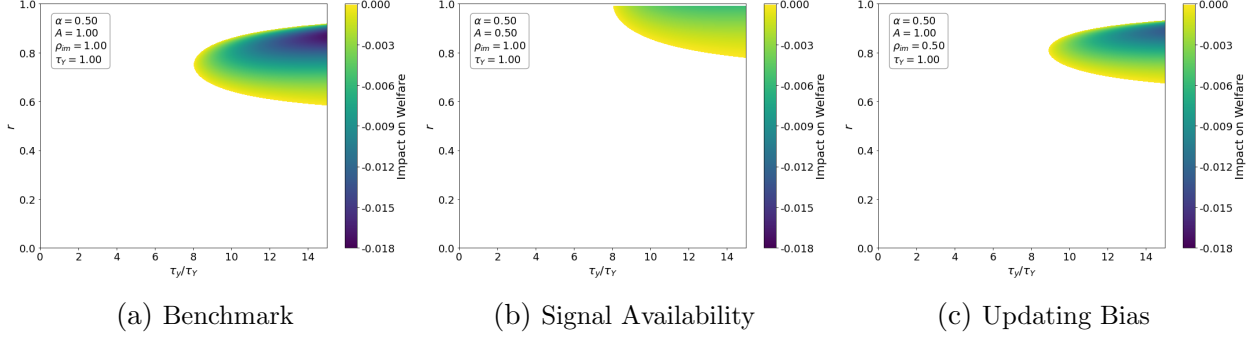


Figure 7: Disclosure's Effect on Social Welfare

Notes: Panels depict the welfare implications of increasing public signal precision ($\partial \mathbb{E}[W]/\partial \tau_Y$) for the two messenger effects. White: non-negative marginal effects, color: negative. Panel (a): all are informed, no bias ($A = 1, \rho = 1$, replicating [Morris and Shin \(2002\)](#)). Panel (b): a fixed fraction of the outgroup are informed ($A = 0.5$), without bias. Panel (c): all are informed but outgroup strongly underweight public information ($\rho = 0.5$). In all panels $\tau_Y = 1$ for scaling; the vertical axis reports r , and the horizontal axis τ_Y/τ_Y .

5.4.2 In- and Outgroup Composition – The Delegation Policy

Public communication can be delegated to messengers whose characteristics raise the ingroup share α . What determines the optimal α^* ?

Proposition 3. *There exists $\alpha^* \in [0, 1]$ that maximizes $\mathbb{E}[W]$, with the optimal value depending on τ_Y/τ_y , r , ρ , and A . Any interior optimum satisfies $\partial \mathbb{E}[W]/\partial \alpha = 0$ and $\partial^2 \mathbb{E}[W]/\partial \alpha^2 \leq 0$. The two corner solutions satisfy*

$$\alpha^* = 0 \Rightarrow \left. \frac{\partial \mathbb{E}[W]}{\partial \alpha} \right|_{\alpha=0} \leq 0, \quad \alpha^* = 1 \Rightarrow \left. \frac{\partial \mathbb{E}[W]}{\partial \alpha} \right|_{\alpha=1} \geq 0.$$

Derivation in Appendix C.4. An explicit closed form for α^* is not feasible because of the updating bias ($\rho < 1$).⁴² Numerical methods show that there exist corner and interior solutions of α^* (Figure A34), which is jointly shaped by four factors: (i) the public–private precision ratio (τ_Y/τ_y), (ii) the strength of coordination r , (iii) the degree of updating bias ρ , and (iv) realized signal size $|Y|$ (availability A). Delegation (higher α^*) is optimal when the public signal is sufficiently precise; centralization (lower α^*) is more likely optimal under strong coordination and noisy public information – especially when many outgroup agents

⁴²With only restricted availability but no updating bias: $\alpha^* = \min \left\{ 1, \frac{\tau_Y + \tau_y}{3r \tau_y} \right\}$. Thus $\alpha^* < 1$ whenever $\frac{\tau_Y}{\tau_y} < 3r - 1$ (requiring $r > 1/3$). $r < \frac{1}{3}$ always results in the corner solution of only ingroup agents ($\alpha^* = 1$), in line with the optimal publicity result of [Cornand and Heinemann \(2008\)](#). Intuitively, when coordination motives are sufficiently strong and the public signal is relatively noisy, keeping some outgroup agents (who miss small signals) can offset over-reaction to the public signal. By contrast, with weak complementarities, broad delegation is optimal, ensuring homogeneous ingroup agents fully use the public signal.

are informed yet biased. Appendix C.4 details these mechanisms.

A Caveat: Expertise Loss under Delegation. Delegating away from the (presumably) most expert messenger may lower public signal precision. If τ_Y declines with α at rate $k \equiv -\frac{d\tau_Y}{d\alpha} > 0$, then delegation is locally welfare-improving only if $\frac{\partial \mathbb{E}[W]}{\partial \alpha} > k(\alpha) \frac{\partial \mathbb{E}[W]}{\partial \tau_Y}$. Two implications follow: (i) When $\frac{\partial \mathbb{E}[W]}{\partial \tau_Y} > 0$ (disclosure is beneficial), delegation must overcome the positive marginal cost of expertise loss $k(\alpha) \frac{\partial \mathbb{E}[W]}{\partial \tau_Y}$. (ii) When $\frac{\partial \mathbb{E}[W]}{\partial \tau_Y} < 0$ (disclosure is harmful), expertise loss helps to reduce over-coordination – yet, so does centralized communication (Section 5.5). The trade-off is tighter under strong coordination and with noisy public signals. Appendix C.5 shows this in detail.

5.5 Policy Implications: When to Delegate?

Since both disclosure and delegation decisions affect welfare, how should optimal policy communication balance delegation and disclosure policies to maximize the social value of public information? To show the optimal policy mix, Figure 8 plots welfare contours in (τ_Y, α) for low and high coordination, fixing $A = 0.5$ (i.e., conditioning on the realized $|Y|$) and a modest underweighting $\rho = 0.95$ (roughly reflecting empirical ingroup effects). White lines indicate welfare contours, lighter shades of color indicate higher welfare, and arrows on the black lines indicate the direction in which welfare rises, suggesting whether to delegate and whether to disclose.

1. Low Coordination: Delegate and Disclose Always

If $\tau_Y > \tau_y$, the corner $\alpha^* = 1$ is optimal with gains from delegation being large and full disclosure also being optimal. If $\tau_Y < \tau_y$, gains from delegation are smaller but still positive with larger rises in α needed for noticeable welfare improvements. Intuitively, the marginal benefit of shifting agents toward a relatively noisy public signal is lower (Figure 8a).

2. High Coordination: Disclose and Delegate Strategically

When strategic complementarities are strong, the policy space can be divided by boundaries, indicating where disclosure and/or delegation are socially desired.

First, reflecting Proposition 2, the red dotted *disclosure boundary* in Figure 8b shows that as delegation increases (higher α), effective exposure to Y rises (i.e., q falls), potentially pushing the economy into a region where disclosure reduces welfare – particularly when τ_Y is low (i.e., large c). A smaller resonance weight ρ shifts this boundary left (lower τ_Y needed for

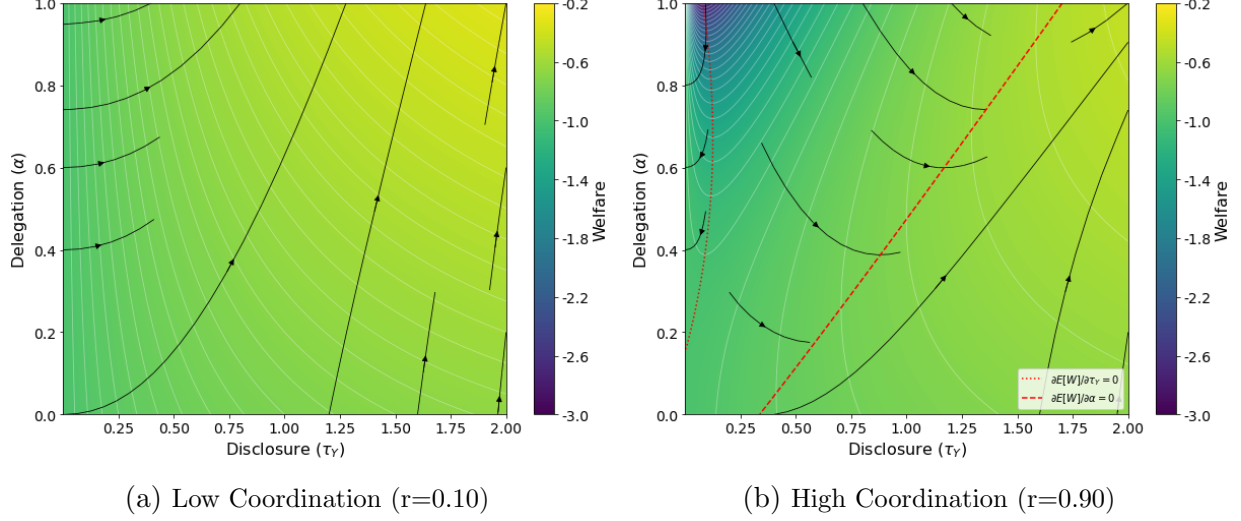


Figure 8: Social Welfare Contours

Notes: Welfare ($E[W]$) over disclosure (τ_Y) and delegation (α), conditional on a “moderate” signal ($A = 0.5$). Parameters: $\tau_y = 1$, $\rho = 0.95$. Shading indicates welfare: lighter = higher (less negative) welfare. Thin white lines are iso-welfare (contour) levels. Black stream arrows show the local welfare-ascent direction. Panel (a): low coordination $r = 0.10$. Panel (b): high coordination $r = 0.90$. In (b): red *dotted* line: $\partial E[W]/\partial \tau_Y = 0$ (disclosure boundary); red *dashed* line: $\partial E[W]/\partial \alpha = 0$ (delegation boundary).

a given α) and up (more α is tolerable for a given τ_Y), thereby expanding the region where disclosure is welfare-improving.

Second, to the right of the red dashed *delegation boundary* in Figure 8b, delegation raises welfare as the composition gain from agents switching weight on Y from ρ to 1 outweighs the higher coordination risk from lower q . A smaller ρ shifts this boundary left and down, expanding the region where delegation improves welfare.

Thus, the policy space can be divided into three regions:

(i) *Restrict and centralize*: When public information is noisy, and disclosure harms welfare, centralizing communication (i.e., reducing α) preserves welfare ($\partial E[W]/\partial \tau_Y < 0$ and $\partial E[W]/\partial \alpha < 0$). Thus, limiting the number or diversity of messengers emerges as an *additional* policy tool, particularly valuable when policymakers cannot limit disclosure (e.g., because of periodic announcements). This reflects the finding discussed in Section 5.4.1, and occurs for parameter combinations left of the red dotted line.

(ii) *Disclose and centralize*: As the quality of public information improves, disclosure improves welfare, while delegation still hurts ($\partial E[W]/\partial \tau_Y > 0$ and $\partial E[W]/\partial \alpha < 0$). The policymaker should increase disclosure but centralize communication. This identifies delegation not just as an additional policy tool, but also a *distinct* one. This result arises because outgroup agents place relatively more weight on private signals (and expect others to do the same), which buffers inefficient coordination on a still-imperfect public signal. This occurs

between the two red lines (dotted and dashed).

(iii) *Disclose and delegate*: When public information is sufficiently precise, disclosure and delegation both raise welfare ($\partial \mathbb{E}[W]/\partial \tau_Y > 0$ and $\partial \mathbb{E}[W]/\partial \alpha > 0$). This occurs to the right of the red dashed line.

Delegation is therefore an additional policy tool, distinct from disclosure. Maximizing the social value of public information requires policymakers to choose disclosure and delegation policies jointly. In practice, this means that optimal communication policy involves *strategic* delegation, accounting for the realized signal size, relative precisions, potential expertise losses, and the strength of strategic complementarities.

6 Conclusion

This paper combines causal experimental evidence with data from newspapers, social media, and surveys to show that messenger identity affects public communication both in reach and in influence. Based on these findings, a coordination model identifies the selection of messenger(s) as an additional policy tool to maximize the social value of policy communication.

Two limitations merit emphasis. First, the experiment purposefully isolates identity along a single dimension, yet policymakers have many traits such as age, gender, ethnicity, socioeconomic background, or expertise. Natural extensions include studying interactions among characteristics and assessing their relative importance. Second, I abstract from ‘cacophony of voices’; accounting for the risk of multiple public messages that differ is an interesting next step.

To conclude, policy communication is not immune to messenger effects, and policymakers can exploit these effects by strategically selecting who communicates policy to the public. Thus, whenever policy is communicated to diverse public audiences, the messenger matters.

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Appendix

A Experiment: Design, Sample, and Interface

This appendix extends Section 2 by detailing the experimental design, the experimental sample, and providing the complete interface of the experiment.

A.1 Details on the Experimental Design

This appendix provides additional details of the experimental design described in Section 2.1.

Data in Forecasting Tasks. Data are quarterly observations of year-on-year euro area HICP inflation between 1999Q1 and 2023Q1 and the corresponding one-year-ahead forecasts from the ECB Survey of Professional Forecasters (SPF).⁴³ Figure A9 shows the full series of realizations, forecasts, and forecast errors. From these 97 quarters, six sequences of 11 periods each are randomly selected – one per forecasting task; Figure A10 illustrates them and Table A5 reports mean inflation, forecast precision, and volatility for each.

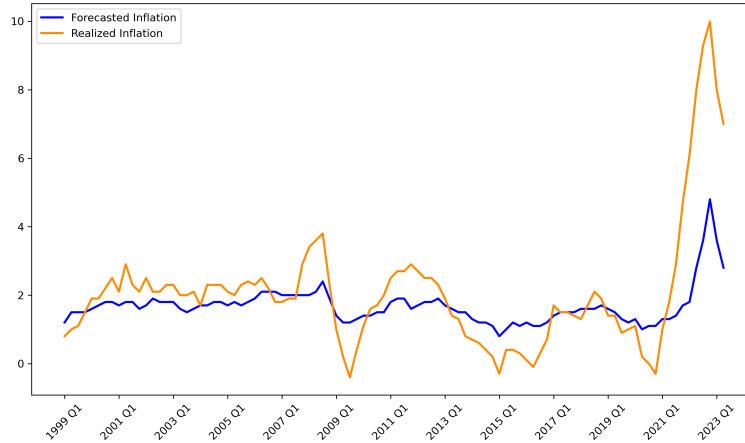


Figure A9: Quarterly Euro Area Inflation, SPF Forecasts, and SPF Forecast Errors

Notes: Realized quarterly euro area HICP inflation and one-year-ahead forecasts from the ECB Survey of Professional Forecasters.

⁴³The SPF is a quarterly survey of expectations for inflation and other macroeconomic variables, answered by experts affiliated with financial and non-financial institutions based in the European Union; see more [here](#).

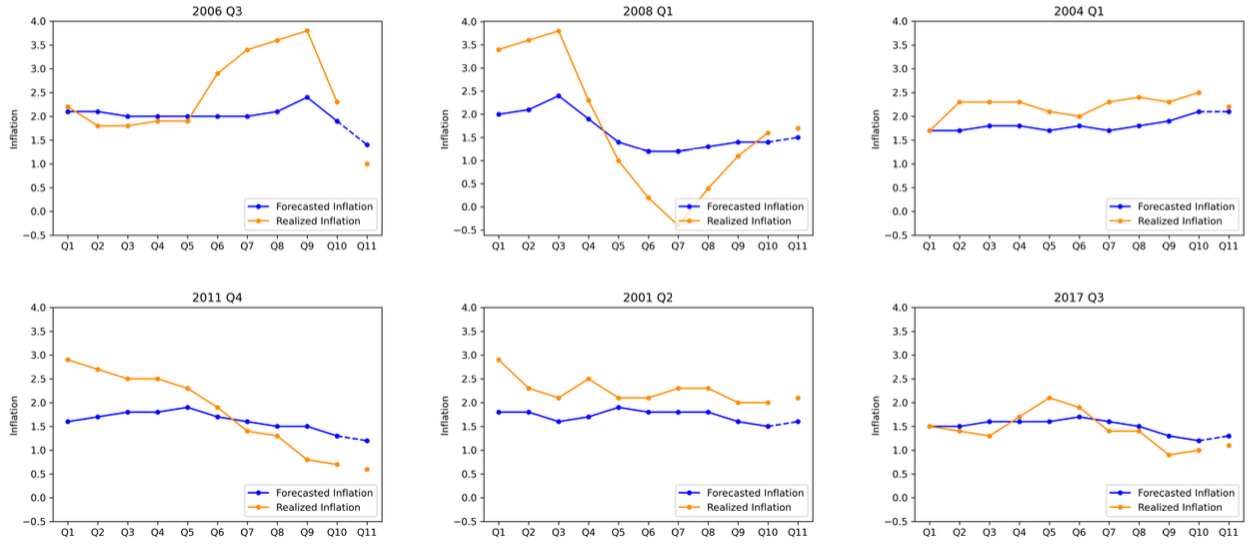


Figure A10: Inflation Sequences

Notes: Sequences are randomly selected from real euro area HICP and SPF one-year-ahead forecast data. The solid line shows the history of realized inflation and forecasts presented to participants. The dashed line shows the forecast leading to the signal, and the unconnected final dot shows the value participants are asked to forecast. Dates are omitted in the experiment to prevent identification of historical episodes. Sequences are randomized across treatments.

Table A5: Overview of Randomly Selected Sequences

10-Period Starting Date	10-Period Average of Inflation	10-Period STD of Inflation	Next-Period Signal	Next-Period Realization	10-Period Signal Precision
2001Q2	2.3	0.3	1.6	2.1	15.5
2004Q1	2.2	0.2	2.1	2.2	26.8
2006Q3	2.6	0.8	1.4	1.0	1.9
2008Q1	1.7	1.5	1.5	1.7	0.8
2011Q4	1.9	0.8	1.2	0.6	2.2
2017Q3	1.5	0.4	1.3	1.1	14.4

Notes: Summary statistics for the six randomly selected inflation sequences. The indicated date marks the beginning of the 10-quarter historical window. The average and standard deviation are calculated from realized inflation during these ten quarters; the standard deviation is the sample standard deviation. “Next-Period Signal” and “Next-Period Realization” denote the SPF forecast and realized inflation, respectively, in the immediately following (11th) period. Signal precision is the inverse sample variance of the signed SPF forecast errors over the ten-quarter historical window. This is the signal-precision measure used in the Bayesian benchmark (Section 2.3). Data on realized euro-area HICP inflation and SPF forecasts are taken from the ECB Statistical Data Warehouse.

Details on Forecast Incentives. Point forecasts and range forecasts are incentivized using scoring rules standard in the experimental literature. For *point forecasts*, participants receive a bonus that decreases with the absolute forecast error. A perfect forecast yields the maximum bonus, which decays geometrically with each percentage point of error. The point forecast score $F_{i,t}$ is given by

$$F_{i,t} = 6 \cdot 3^{-|\mathbb{E}_{i,t-1}\{\pi_t\} - \pi_t|}, \quad (19)$$

where π_t is realized inflation at time t , and $\mathbb{E}_{i,t-1}\{\pi_t\}$ is participant i 's forecast made at $t - 1$. A perfect forecast yields $F_{i,t} = 6$; the score declines by two-thirds for each percentage point of error.

For *range forecasts*, participant i submits at $t - 1$ a range $[\underline{u}_{i,t-1}, \bar{u}_{i,t-1}]$ for π_t . Let $r_{i,t-1} = \bar{u}_{i,t-1} - \underline{u}_{i,t-1} \geq 0$ denote the width. The score realized at t is

$$U_{i,t} = \begin{cases} 0, & \pi_t \notin [\underline{u}_{i,t-1}, \bar{u}_{i,t-1}], \\ \frac{\phi}{1 + r_{i,t-1}}, & \pi_t \in [\underline{u}_{i,t-1}, \bar{u}_{i,t-1}], \end{cases} \quad \text{with } \phi = 6.$$

To avoid learning effects and any impact from forecast performance on survey answers, realized inflation values are not shown until the very end of the experiment. Participants then receive feedback on all 12 forecasts (six priors and six posteriors) and learn which of the 24 decisions (12 point forecasts and 12 ranges) is randomly selected for payment. Randomization ensures equal effort is applied to all forecasts.

Information Treatments. In addition to the numerical forecast (the signal), participants may request up to three qualitative pieces of information linked to the same inflation scenario. These are based on ECB press conferences held at approximately the same time as the corresponding SPF forecast, ensuring consistency between numerical and qualitative information. Table A6 lists all information pieces revealed by clicking “Read More” by inflation scenario.

Table A6: Additional Information Pieces and Corresponding Inflation Scenarios

Inflation Scenario	Information Pieces
2001 Q2	1) Real economic activity in the Euro area was weak recently, but survey data and recent financial market developments indicate a gradual upturn in economic activity. 2) The expected pick-up in activity is supported by both external factors, such as global recovery, and domestic factors, such as ongoing adjustment efforts by companies to enhance competitiveness and profitability. 3) Downside risks to the main scenario for economic growth have declined, but they have not disappeared. Macroeconomic imbalances in some regions and high oil prices are mentioned as potential risks to economic activity in the Euro area.
2004 Q1	1) The inflation rates are forecasted to remain elevated. Factors contributing to this include higher energy prices, indirect effects of past oil price increases, and potential stronger-than-expected wage developments. 2) The strength of monetary and credit growth, coupled with ample liquidity, pose inflationary risks over the medium to longer term. 3) The rapid rate of monetary growth is primarily driven by the stimulative impact of low interest rates.
2006 Q3	1) Overall, the evidence that inflationary pressures are diminishing further has increased and, looking forward, inflation rates are expected to be in line with price stability, supporting the purchasing power of incomes and savings. 2) The recent decline in headline inflation mainly reflects the considerable easing in global commodity prices over the past few months, which more than offsets the impact of the sharp rise in unit labour costs in the first half of this year. Lower commodity prices and weakening demand should lead to inflationary pressures diminishing further. 3) The large declines in commodity prices and the impact of weakening demand on price developments strongly impacted this forecast.
2008 Q1	1) Price developments are expected to remain moderate over the medium-term horizon. Global inflationary pressures may persist, while domestic price pressures are expected to remain low. 2) The still weak annual growth rate of bank loans to the private sector conceals the fact that monthly flows have now been positive for a number of months. At the same time, these aggregate developments continue to reflect mainly an ongoing strengthening in the annual growth of loans to households, while the annual growth of loans to non-financial corporations has remained negative. 3) Overall, price stability is expected to be maintained over the medium term, thereby supporting the purchasing power of Euro area households. Inflation expectations remain firmly anchored.
2011 Q4	1) The risks to the economic outlook for the Euro area are considered to be broadly balanced over the medium term, both for upside and downside risks to price developments. 2) The monetary analysis confirms subdued underlying growth in broad money and credit, and the annual growth rate of loans to the private sector continues to contract. 3) Progress has been made in improving the funding situation of banks, but further steps are needed to strengthen the resilience of banks and reduce the fragmentation of Euro area credit markets.
2017 Q3	1) While there are some initial signs of stabilization in the growth slowdown and a mild increase in underlying inflation, overall inflation remains low. 2) Ongoing employment growth and increasing wages are underpinning the resilience of the Euro area economy. 3) Inflation expectations are at low levels, and measures of underlying inflation have remained generally muted. Inflation is expected to increase over the medium term, supported by monetary policy measures, economic expansion, and solid wage growth.

Notes: Information pieces are linked to inflation scenarios (each scenario's first quarter shown in the first column). Each inflation scenario includes three buttons, each displaying one piece of information. Information pieces are summarized statements based on ECB press conferences, which can be found in full [here](#).

A.2 Experimental Sample

Prolific provides background data on participants before the study, including age, sex, income, marital and employment status, student status, ethnicity, country of birth and residence,

years in current country, childhood residence, financial decision-making, languages, education, migration history, investments, property ownership, and cultural and linguistic background. Figure A11 summarizes key participant characteristics.

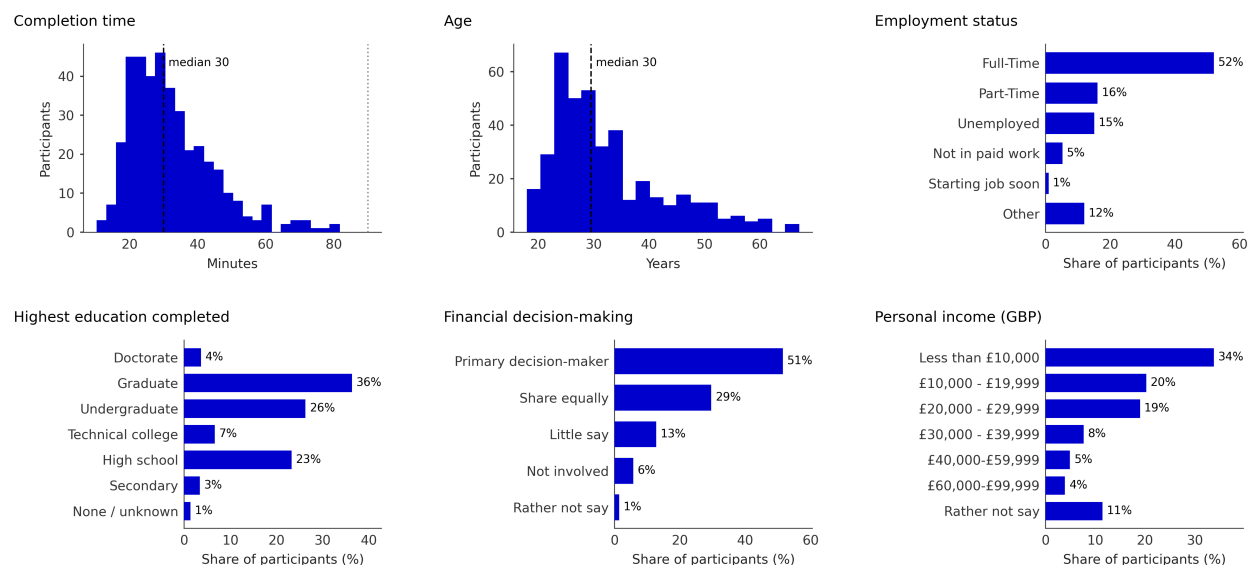


Figure A11: Descriptive Evidence of Experiment Participants

Notes: Key characteristics of participants. Median completion time is around 30 minutes; most finish within an hour. Prolific automatically disqualifies participants who take longer than 90 minutes.

Employment rates align closely with national figures (67.6% in the experiment compared to 68.1% in the population) (Eurostat, 2023a). The unemployment rate is somewhat higher in the experiment (14.8% versus 7.6%) (Eurostat, 2023e), but cross-country differences follow population patterns, with higher unemployment in Spain and Italy than in France and Germany. This discrepancy may be driven by students on Prolific classifying themselves as unemployed, a view supported by the relatively low combined shares of respondents reporting being “not in paid work” (5.0%) or “other” (11.8%) compared to the inactive share of the population (26.3%) (Eurostat, 2023b). It could also partly reflect a lower proportion of retirees, as participants are younger on average: across all countries, the median age in the experiment is 30, compared to a population median of about 45 in 2023 (Eurostat, 2023d). Educational attainment is, on average, somewhat higher than in the population, which is the case across all four countries and might simply reflect the lower age of participants. The French sample is the most educated, in line with population patterns (Eurostat, 2023c). Further, reported personal income is consistent with the population average.⁴⁴

⁴⁴Average personal income after taxes of the 353 participants who disclose it is just over 19k GBP, using the midpoints of each bracket. This method might produce slightly higher values if those in the 60–99k GBP

A.3 Interface of the Experiment

Figures A12–A26 below show screenshots of the entire experiment.

Welcome!

Name of Study: Forecasting Inflation

Description: The aim of this experiment is to better understand macroeconomic expectations. No prior knowledge is necessary to participate in the experiment. Your task will be to forecast inflation in different economic scenarios. After these forecasting tasks, we will ask you a few short questions about your views related to the economy.

How will I get remunerated for my time? We highly value your participation in this 25-minute experiment, which is why we would like to remunerate you fairly. You will get £4 for participating in this survey. In addition, you have the chance to get up to £6 added to your payment. The more accurately you forecast, the more of the additional £6 you will take home. The maximum amount of money you could get is thus £10, and the minimum is £4. We will reward you as quickly as possible. Typically, you can expect to get rewarded within 1-3 business days after submission.

How will my data be used? Your answers will be completely anonymous, and we will keep them confidential. Your data will be stored in a password-protected file and may be used in academic publications. Your IP address will never be stored. Research data will be stored for a minimum of five years after publication or public release. By submitting your personal data, you agree to this transfer, storing or processing.

Who will have access to my data? Data are collected for the sole purpose of research and processed by researchers at the University of Oxford. Any personal information that could identify you will be removed or changed before files are shared with other researchers or results are made public. Responsible members of the University of Oxford and funders may be given access to data for monitoring and/or audit of the study to ensure we are complying with guidelines.

Do I have to take part? No. Please note that your participation is voluntary. You may withdraw at any point during the experiment for any reason, before submitting your answers, by closing the browser. However, we are only able to reimburse participants who complete the full survey.

Who has reviewed this project? This project has been reviewed by, and received ethics clearance through, the University of Oxford's Economics Department's Research Ethics Committee [reference number: ECONCIA21-22-24].

What if there is a problem? If you have a concern about any aspect of this project, please speak to Alena Wabitsch (alena.wabitsch@economics.ox.ac.uk), who will try to answer your query. The researcher should acknowledge your concern within 5 working days and give you an indication of how they intend to deal with it. If you remain unhappy or wish to make a formal complaint, please contact the relevant Chair of the Research Ethics Committee at the University of Oxford:

The Department of Economics Departmental Research Ethics Committee can be contacted at:

Email: ethics@economics.ox.ac.uk

Address: Department of Economics

Manor Road 10

Oxford OX1 3UQ

United Kingdom

The University of Oxford's Data Protection Officer can be contacted at:

Email address: data.protection@admin.ox.ac.uk

Telephone number: + 44 1865 280 199

Address: Planning and Council Secretariat

University Offices

Wellington Square

Oxford OX1 2JD

United Kingdom

If you have read and understood the information above, and agree to participate with the understanding that the data (including any personal data) you submit will be processed accordingly, please check the relevant box below to get started.

☐ Yes, I have understood the above information and agree to take part.

Please provide your Prolific ID:

Click **Next** to start the experiment.

Next

Figure A12: Welcome Page

bracket cluster closer to the lower bound. [OECD \(2023\)](#) reports annual gross wages, which, after accounting for taxation, amount to roughly 31k EUR in France and Germany and around 23k EUR in Italy and Spain.

Instructions

Please read these instructions carefully to make good decisions during the experiment. The quality of your decisions directly impact your bonus payment, which can be up to £6, in addition to the £4 participation payment. There will be a quiz on these instructions on the next page. If you answer at least one question incorrectly three times, the experiment will be ended early.

Throughout the experiment, you can always access these instructions by clicking the "Show Instructions" button at the bottom of any page.

Your Goal in the Experiment:

Your task is to **accurately forecast the annual inflation rate in the Euro area**. This means, you will be predicting the percentage increase or decrease in prices in the entire Euro area between a specific point in time and exactly 1 year earlier. Your goal is to **minimise forecast error**, which is the difference between your forecasted value of annual inflation and the realised annual inflation.

*Note: The more accurate your inflation forecasts, the lower your forecast error, the **more bonus payment** you get!*

Whenever we talk about inflation in this experiment, what we mean is the *annual* inflation rate (i.e., the percentage change in prices of goods and services over the course of exactly 1 year.) Data for inflation reflect real economic scenarios of the Euro area. Sometimes you will be shown professional forecasts, which are based on analyses of real European experts for the given, real economic scenarios.

The Experiment:

You will make two types of inflation forecasts:

- **Point Forecast:** Your 'Point Forecast' of inflation is your best guess of the exact value of inflation for the next unknown period.
- **Range Forecast:** Your 'Range Forecast' of inflation is the range of plausible values that you think will almost certainly contain the realised value of inflation in the next period (i.e. the upper and the lower inflation bounds of inflation).

The closer your point forecast is to the realised inflation, the more money you get. Similarly, the smaller your indicated range of inflation, the more money you get (as long as the realised inflation rate is inside your forecasted range).

What to Expect:

You will face 6 **distinct** forecasting tasks. Each task is independent and unrelated to the others. In each task, you will first make your inflation forecasts ('*Initial Forecasts*'). You will then receive additional information about the next period's inflation rate and have a chance to adjust your forecasts ('*Updated Forecasts*'). After all 6 forecasting tasks, we will ask you a few questions, and then unveil the realised inflation rates and your performance.

What to do in each Forecasting Task:

1. Examine the given 10 periods of inflation history.
2. Make your Initial Forecasts for the next period:
 - A point forecast of inflation
 - The corresponding range forecast of inflation
3. You will be provided professional inflation forecasts, which are based on real historic forecasts of European experts for the respective scenarios. *Note: The provided historic differences between the realised inflation and the forecasted inflation represent the experts' historic forecast errors.*
4. Make your Updated Forecasts for the next period:
 - As before, provide a point forecast, and the corresponding range forecast.
 - *Note: Your Updated Forecasts can be the same as your Initial Forecasts, use some of the same values, or use completely different values.*
 - We will remind you of the values of your Initial Forecasts, when you are making your Updated Forecasts.

Overview of all the steps of a forecasting task:

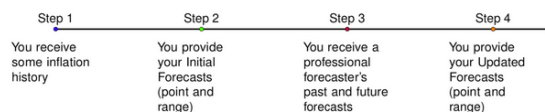


Figure A13: Instructions (1)

Interacting with the data:

Historical data, professional forecasting history, and your point and range forecasts will be visualised in an interactive chart.

- Hover over chart dots for values.
- Type all your forecasts in percentages (up to 1 decimal) into the clearly labelled input text boxes.
- You may submit positive values (prices are going up), negative values (prices are going down), or a value of zero (prices are staying the same).
- Your forecast range can be as big or small as you prefer.
- Your point forecast doesn't have to be the mid-point of your range forecast. Your point forecast may be closer or farther from the upper than the lower bound.
- Your point forecast must always lie within your forecasted range - the software will prevent other inputs.
- Your point inflation forecasts will be visualised with green squares, and the corresponding range forecasts with green triangles.
- Submit your forecasts by clicking the 'Next' button. Note: You won't be able to change your answers after submitting them.

How your decisions get you bonus payment:

You can earn up to £6 in bonus payment for the accuracy of your point and range forecasts. The full £6 bonus requires your point forecast to match realised inflation (down to 1 decimal), and your range forecast to have the upper and lower bound on the value of the realised inflation (also down to 1 decimal).

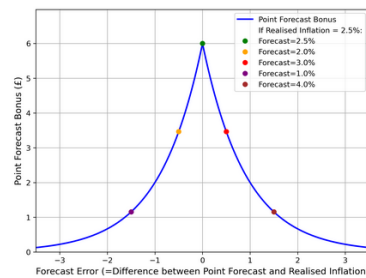
Your bonus will be awarded for **either** your point forecast accuracy **or** your range forecast accuracy in **one** set of forecasts (*Initial or Updated*) in **one of the 6** forecasting tasks. Here's how it works:

- Our software *randomly* selects one of your six forecasting tasks.
- For that forecasting task, the software *randomly* chooses either your Initial Forecasts or your Updated Forecasts.
- Within these selected Forecasts, the software *randomly* picks either your point forecast or your range forecast.

*Note: This means that to get a high bonus payment, you must take **both** the point and range forecasts in **all** Initial and Updated Forecasts across **all** forecasting tasks equally seriously. This is because you will only be paid for **one** randomly selected decision across the entire experiment.*

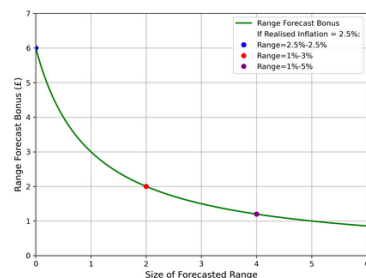
Point forecast bonus:

- Your exact forecast payment is computed like this: $P = 6 \cdot 3^{-|\text{Forecast Error}|}$
- The larger your forecast error (distance of forecasted inflation above or below the realised inflation), the less you earn.
- The figure below shows how your point forecast bonus falls, the further away your forecast is from the realised inflation value.
- The colourful dots in the figure visualise the following example: Suppose that realised inflation turns out to be **2.5%**:
 - If your point forecast of inflation was 2.5%, you would earn $P = 6 \cdot 3^{-0} = £6.00$
 - If your point forecast of inflation was 2.0%, you would earn $P = 6 \cdot 3^{-0.5} = £3.46$
 - If your point forecast of inflation was 3.0%, you would earn $P = 6 \cdot 3^{-0.5} = £3.46$
 - If your point forecast of inflation was 1.0%, you would earn $P = 6 \cdot 3^{-1.5} = £1.15$
 - If your point forecast of inflation was 4.0%, you would earn $P = 6 \cdot 3^{-1.5} = £1.15$



Range forecast bonus:

- If realised inflation is inside your range forecast, you score $R = 6 \cdot [1 / (1 + \text{total range})]$
- The total range of your forecast is given by the gap between the upper bound of the range forecast and the lower bound of the range forecast.
- The larger the range you create, the less money you earn, when inflation is within your range forecast.
- If inflation falls **outside** your forecast range, you earn **nothing**
- The figure below shows how your range forecast bonus falls, the larger your forecasted range is (as long as the realised value of inflation falls within that range).
- The colourful dots in the figure visualise the following example: Suppose that realised inflation turns out to be **2.5%**:
 - If your forecasted range was between 2.5% and 2.5%, then you would earn $R = 6 \cdot [1 / (1 + 0)] = £6.00$
 - If your forecasted range was between 1% and 3%, you would earn $R = 6 \cdot [1 / (1 + 2)] = £2.00$
 - If your forecasted range was between 1% and 5%, you would earn $R = 6 \cdot [1 / (1 + 4)] = £1.20$
 - If your forecasted range was between 0% and 2%, you would earn nothing because realised inflation is not within your range.



Note that the "Next" button may appear with a delay on certain pages. Similarly, parts of certain pages will be displayed only after a few seconds. Go through the experiment at your own pace and don't rush any decision.

Figure A14: Instructions (2)

Comprehension Quiz

Before starting with the first forecasting tasks, we ask that you successfully complete the following comprehension quiz. You are always allowed to consult the instructions whenever you need to (see button at the bottom of the page).

If you get an answer wrong, you will be told so, and will have the opportunity to correct your answer. If you submit the quiz three times with at least one wrong answer, we will end the experiment early.

Your performance on this quiz does not affect your payment in any way, but you will not be able to proceed to the forecasting tasks until you have correctly answered all of the quiz questions.

1. True or False? Your goal in the experiment is to forecast inflation as accurately as possible.

☐ True ☐ False

2. True or False? For your bonus payment the software randomly selects a single forecasting task, of which it further randomly selects the Initial Forecasts or the Updated Forecasts, and finally it randomly selects either your point or your range forecast.

☐ True ☐ False

3. If inflation turns out to be 5.0% in the period selected for bonus payment, and your forecasted range is between -0.2% and 4.7%, how much extra payment will you get for your range?

4. True or False? If inflation turns out to be 2.2%, and your point forecast (that was selected for bonus payment) is 1.0%, your bonus payment is less than if you had forecasted 0.3%.

☐ True ☐ False

5. How many forecasting tasks will you have to do?

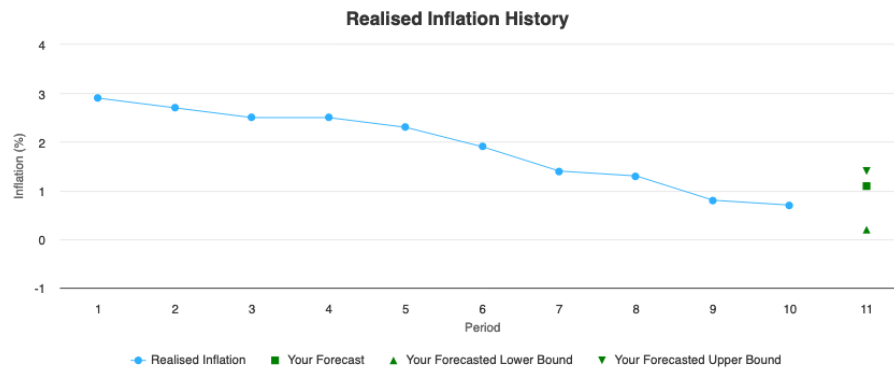
Next

Show instructions

Figure A15: Comprehension Quiz

Notes: Participants must answer all questions correctly within three attempts to proceed. Those who fail are excluded from the experiment.

Forecasting Task 1: Initial Forecasts



Provide your **point forecast** of inflation in period 11:

%

Provide the **highest probable value** of inflation in period 11 (i.e. the **upper bound of the range** you forecast inflation to fall in):

%

Provide the **lowest probable value** of inflation in period 11 (i.e. the **lower bound of the range** you forecast inflation to fall in):

%

Note: Enter any numeric value. You may use up to 1 decimal. Indicate decimals with a dot ("."). Use the "-" sign for negative values. The chart will automatically update its scaling to visualise your inputs.

[Next](#)

[Show instructions](#)

Figure A16: Prior elicitation

Notes: Participants are required to spend at least ten seconds on this page to reduce the risk of rushing without processing the information.

Forecasting Task 1: Updated Forecasts

Now imagine an expert from Italy who represents the European Central Bank (ECB) provides a forecast of 1.2% for inflation in period 11.

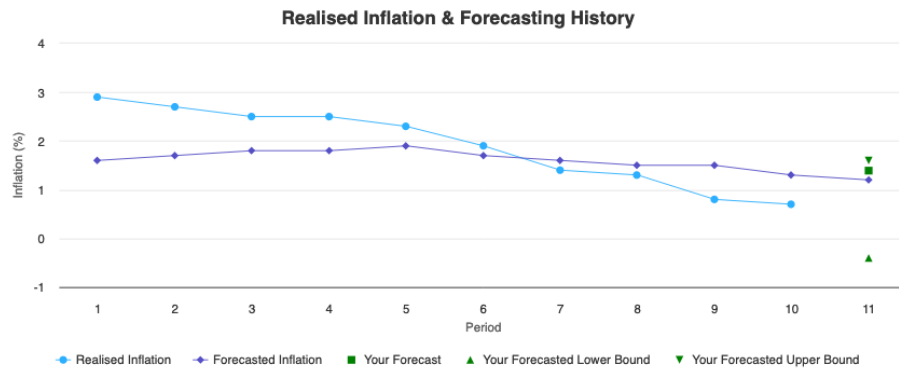
You find this forecast, as well as the expert's corresponding forecasting history, displayed in the graph.

In addition, you may click on any or all of the 3 "Read more" buttons below for detailed explanations by the expert from Italy who represents the European Central Bank (ECB) about the analyses underlying the forecast for period 11.

Read more

Read more

Read more



You forecasted 1.1% inflation for period 11. You may now keep or revise this point forecast:

%

You forecasted the **highest probable value** of inflation in period 11 to be 1.4%. You may now keep or revise this **upper bound** of your range forecast:

%

You forecasted the **lowest probable value** of inflation in period 11 to be 0.2%. You may now keep or revise this lower bound of your range forecast:

%

Note: Enter any numeric value. You may use up to 1 decimal. Indicate decimals with a dot ("."). Use the "-" sign for negative values. The chart will automatically update its scaling to visualise your inputs.

Next

Show instructions

Figure A17: Posterior elicitation

Notes: The first sentence of this page is shown in isolation for 5 seconds before the rest of the page appears to ensure participants attend to the messenger and the signal. Participants are then required to spend at least another 10 seconds on this page to reduce the risk of rushing decisions without paying attention to the provided information. Hovering over the dots in the graph reveals exact values of realized inflation and inflation forecasts.

Attention Check

The following questions assess your attentiveness during the **last (most recent)** of the 6 forecasting tasks you just completed. Please answer to the best of your ability. Your responses will not impact your final payment.

Which expert provided the forecast in the last forecasting task?

What value of inflation for Period 11 did the expert forecast in the last forecasting task?

Next

Show instructions

Figure A18: Attention Check

Notes: The order of answer options for experts is randomized. The options for period 11 values reflect actual signal values of the six forecasting tasks.

Survey

Before we reveal the correct values for inflation and the bonus payment that you have earned, we would like to ask you a few questions.

We are interested in your honest opinions and perceptions. Therefore, there are no right or wrong answers.

Note: You may leave any of the comment boxes below empty, but we highly appreciate hearing your opinion.

What do you think is the purpose of this study?

What - if any - strategy did you apply to your decision-making to maximise the bonus payment from this study?

Next

Show instructions

Figure A19: Survey: Purpose of Experiment & Strategy

Notes: Participants are allowed to skip any or all of these questions entirely.

Survey

What - if anything - did you **think about**, or associate with, the **experts** who provided you the forecasts?

Write down **anything that comes to mind**, in any form you would like. We are also interested to hear if you did not think about an expert at all.

Keywords are sufficient - you do not have to write full sentences. You may leave some boxes empty, but we highly appreciate to hear about your views and perceptions.

Note: There are no right or wrong answers.

The expert from the European Central Bank (ECB):

The Spanish expert from the European Central Bank (ECB):

The French expert:

The Spanish expert:

The French expert from the European Central Bank (ECB):

The expert from the Banque de France:

Next

Show instructions

Figure A20: Survey: Association

Notes: Participants are allowed to skip any or all of these questions entirely. To make it easier for participants to recall, the order of experts represents the occurrence of experts in forecasting tasks (i.e., randomized across individuals).

Survey

How confident were you in each **expert's ability** to accurately predict inflation and provide accurate economic analysis?

Give your answer **on a scale of 1 to 7**, where **1** represents **Complete Lack in Confidence** in their abilities, **4** represents being **Neutral** about their abilities and **7** represents **Complete Confidence** in their abilities.

Note: Again, there are no right or wrong answers.

The expert from the European Central Bank (ECB):

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

The Spanish expert from the European Central Bank (ECB):

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

The French expert:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

The Spanish expert:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

The French expert from the European Central Bank (ECB):

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

The expert from the Banque de France:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

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Figure A21: Survey: Perceived Ability

Notes: Order of experts represents occurrence of experts in forecasting tasks (i.e., randomized across individuals).

Survey

How well are you **informed** about the **institutions** or **policymakers** listed below? Give your answer **on a scale of 0 to 5**, where these values represent the following:

1: *I have **never** heard of them before this experiment.*

2: *I have heard of them, but I **don't know with** certainty what they do.*

3: *I **know** with certainty what/who they are and what they do, but I **don't get any news** about them.*

4: *I **know** with certainty what/who they are and what they do, and I get news **occasionally** about them.*

5: *I **know** with certainty what/who they are and what they do, and I get news **regularly** about them.*

Remember: There are no right or wrong answers.

Pablo Hernández de Cos:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Joachim Nagel:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

The Banque de France (the national central bank of France):

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

François Villeroy de Galhau:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Fabio Panetta:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Ignazio Visco:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Luis de Guindos:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Isabel Schnabel:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Christine Lagarde:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

The European Central Bank (ECB):

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

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Figure A22: Survey: Exposure

Notes: The order in which institutions and policymakers appear is randomized at the participant level.

Survey

How much do you **trust** the institutions or policymakers listed below?

Give your answer **on a scale of 1 to 7**, where **1** represents **Completely Distrusting** them, **4** represents being **Neutral** (neither trusting nor distrusting them) and **7** represents **Completely Trusting** them.

Note: Again, there are no right or wrong answers.

The Banque de France (the national central bank of France):

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Isabel Schnabel:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Christine Lagarde:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Luis de Guindos:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Ignazio Visco:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Joachim Nagel:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

François Villeroy de Galhau:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Fabio Panetta:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

The European Central Bank (ECB):

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

Pablo Hernández de Cos:

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7

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Figure A23: Survey: Trust

Notes: This page is only shown to participants who indicate knowing at least one institution or policymaker. Only institutions and policymakers that the participant indicates knowing appear in this list, since indicated trust is only meaningful if participants know the institutions and policymakers. The order in which institutions and policymakers appear is randomized at the participant level.

Survey

This is our last question. On the next page, we will show you how well you did in the forecasting tasks, and thus how much money you made in this experiment.

Note: Please select a single answer to the question below. Again, there are no right or wrong answers.

Every 6 weeks, the Governing Council of the ECB makes a monetary policy decision for the Euro area, which may include potential changes to key interest rates. Subsequently, the ECB announces this decision during a press conference.

On average, over the past 8 years, how often have you been aware of these monetary policy decisions by the ECB (even if there was no change in interest rates)? This can be through receiving news in any form of media, from the ECB directly, through discussions with others, or any other information sources.

- ☐ every 6 weeks (ca. at least 8-9 times a year)
- ☐ every 6-12 weeks (ca. 5-7 times a year)
- ☐ every 12-18 weeks (ca. 3-4 times a year)
- ☐ every 18-24 weeks (ca. once or twice a year)
- ☐ less than once a year

Next

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Figure A24: Survey: Monetary Policy Expertise

Notes: The question allows for potential comparison of experimental and Twitter results, distinguishing between experts and non-experts. With some caveats, answers to this question can be seen as an imperfect proxy for monetary policy expertise.

Realised Inflation & Your Performance

We now reveal the realised values for inflation in period 11, your corresponding forecasts, and the bonus payment that you will earn, if the software selects a forecast.

Forecasting Task 1: The realised inflation was 2.1%.

	Initial Forecasts		Updated Forecasts	
	Your Forecasts	Bonus Payments	Your Forecasts	Bonus Payments
Point Forecasts	1.0%	£1.79	1.0%	£1.79
Range Forecasts	-0.2%-1.4%	£0.0	1.0%-2.0%	£0.0

Forecasting Task 2: The realised inflation was 2.2%.

	Initial Forecasts		Updated Forecasts	
	Your Forecasts	Bonus Payments	Your Forecasts	Bonus Payments
Point Forecasts	1.1%	£1.79	1.0%	£1.61
Range Forecasts	1.1%-2.9%	£2.14	-124.0%-1.0%	£0.0

Forecasting Task 3: The realised inflation was 1.0%.

	Initial Forecasts		Updated Forecasts	
	Your Forecasts	Bonus Payments	Your Forecasts	Bonus Payments
Point Forecasts	1.1%	£5.38	1.0%	£6.0
Range Forecasts	0.0%-2.9%	£1.54	1.0%-1.0%	£6.0

Forecasting Task 4: The realised inflation was 1.1%.

	Initial Forecasts		Updated Forecasts	
	Your Forecasts	Bonus Payments	Your Forecasts	Bonus Payments
Point Forecasts	1.1%	£6.0	1.0%	£5.38
Range Forecasts	-0.2%-2.2%	£1.76	1.0%-1.0%	£0.0

Forecasting Task 5: The realised inflation was 1.7%.

	Initial Forecasts		Updated Forecasts	
	Your Forecasts	Bonus Payments	Your Forecasts	Bonus Payments
Point Forecasts	1.6%	£5.38	1.6%	£5.38
Range Forecasts	0.0%-2.9%	£1.54	0.4%-2.5%	£1.94

Forecasting Task 6: The realised inflation was 0.6%.

	Initial Forecasts		Updated Forecasts	
	Your Forecasts	Bonus Payments	Your Forecasts	Bonus Payments
Point Forecasts	1.0%	£3.87	1.0%	£3.87
Range Forecasts	1.0%-1.0%	£0.0	1.0%-1.0%	£0.0

Click the "Next" button below for the software to randomly select the forecast for your bonus payment.

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Figure A25: Payoff Overview

Notes: Realized inflation values for period 11 and the corresponding performance of participants are only shown at the very end of the experiment and after all survey questions to avoid learning effects between forecasting tasks and performance-driven biases to survey answers asked after the experiment.

Your payment

The software randomly selected your **Range Forecast** of the **Updated Forecast** of **Forecasting Task 1** for your bonus payment.

Given your performance for this Range forecast, your bonus payment will thus be £0.00.

Together with the participation payment of £4, you will be paid a **total of £4.00** for your participation in this experiment.

Please let us know of any comment you may have in the text box below, and click the "Next" button below to end the experiment.

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Figure A26: Payoff Reveal

Notes: The software reveals the randomly selected decision for each participant's bonus payment.

B Additional Information and Results

This section provides additional information and results for Sections 2–4.

B.1 Experiment

Bayesian Belief Updating Framework. To assess how agents update their inflation expectations following new forecast information, I use a standard Bayesian belief updating framework. This framework estimates how agents balance novel information (the signal) against their existing beliefs (priors). In the Bayesian case, agents weigh information and priors equally when both have the same precision.

Formally, let participant i 's prior belief about inflation x be $x \sim \mathcal{N}(A_i, \alpha_i^{-1})$. The signal is $B = x + e$, with $e \sim \mathcal{N}(0, \beta^{-1})$. Bayes' law implies

$$E_i[x|B] = \frac{\alpha_i A_i + \beta B}{\alpha_i + \beta}, \quad (20)$$

i.e., the posterior is a precision-weighted average of prior and signal. As discussed in the main text, euro area inflation is non-normal and forecast errors show mild autocorrelation, so the Bayesian case serves as a benchmark; deviations are interpreted comparatively across treatments.

Average signal use. Table A7 shows that participants in all countries underweight the signal compared to a Bayesian benchmark across treatments.

Table A7: Average Signal Use by Nationality

	(1) Overall	(2) Germany	(3) Spain	(4) France	(5) Italy
Signal weight	0.952*** (0.011)	0.942*** (0.021)	0.938*** (0.022)	0.942*** (0.019)	0.984*** (0.022)
Prior weight	0.990*** (0.004)	0.997*** (0.007)	0.984*** (0.007)	1.002*** (0.007)	0.976*** (0.008)
Observations	2,385	600	594	593	598
R-squared	0.982	0.982	0.981	0.984	0.979

Notes: OLS regressions without an intercept (estimating Equation (1)). Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Experimenter Demand Effects. To assess the scope for experimenter demand effects, I examine whether participants identified the research questions and adjusted their responses accordingly. I do so using open-text responses to questions about the study’s purpose (*‘What do you think is the purpose of this study?’*) and about participants’ forecasting strategy (*‘What – if any – strategy did you apply to your decision-making to maximise the bonus payment from this study?’*). Figure A19 shows the corresponding screenshot of the experiment. I classify the responses using a large language model (Anthropic’s Claude Sonnet 4.5; temperature 0). The model first derives a category scheme from a random sample of responses and then assigns each response to one category, coding unmatched responses as “Other.” Table A8 reports the five most frequent categories. Participants rarely identify variation in experts’ characteristics and institutional affiliation as the study’s purpose. Even if they did, experimenter demand would be a concern only if participants adjusted their forecasting strategy accordingly. The five most common reported forecasting strategies provide no evidence of this. Experimenter demand effects therefore appear to be of little concern.

Table A8: Experimenter Demand Effects: Perceived Study Purpose and Forecasting Strategy

Rank	Category	Definition	N	Share
<i>Panel A. Perceived Study Purpose</i>				
1	Expert Influence on Judgment	Responses indicating the study examines how expert opinions, forecasts, or advice influence, sway, or change participants’ own predictions or decisions	131	0.328
2	Layperson Forecasting Ability	Responses suggesting the study assesses how well non-experts or ordinary people can predict, forecast, or understand inflation without specialized knowledge	52	0.130
3	Uncertain or No Idea	Responses where participants explicitly state they do not know, are unsure, or have no idea about the study’s purpose	48	0.120
4	Trust in Experts	Responses focusing on measuring trust, credibility, or reliability of experts, forecasters, or institutions, such as the ECB, in participants’ eyes	29	0.073
5	Inflation Perception	Responses suggesting the study examines how people perceive, think about, or understand inflation trends and their real-world impact	29	0.073
<i>Panel B. Forecasting Strategy</i>				
1	No Particular Strategy	No specific approach applied; answering to the best of ability without a defined method	89	0.223
2	Trend/Pattern Analysis	Following the direction, trajectory, or patterns visible in historical inflation data; extrapolating curves or trends from previous periods	59	0.148
3	Combined Information Synthesis	Integrating multiple sources including historical data, expert forecasts, and contextual information; balancing different inputs to form judgment	58	0.145
4	Expert-Guided Adjustment	Starting with own forecast then adjusting or updating based on expert predictions; incorporating expert opinion into final estimate	50	0.125
5	Expert Accuracy Evaluation	Assessing how accurate experts were in past predictions and weighting their current forecast accordingly; identifying systematic biases in expert predictions	36	0.090

Notes: Five most frequent categories from classification of open-text responses to post-experiment questions about participants’ perceived study purpose and their forecasting strategy. Classification is performed with a large language model (Anthropic’s Claude Sonnet 4.5; temperature 0). Shares are computed relative to all 399 participants.

Table A9: Endogenous Attention

Dep.:	OLS Attention (Continuous)	Probit Attention (Binary)	Marginal Effects Attention (Binary)
Inflation Level	0.503*** (0.055)	0.464*** (0.051)	0.176*** (0.019)
Inflation Uncertainty	0.272*** (0.082)	0.223*** (0.073)	0.084*** (0.027)
Signal Precision	-0.004 (0.003)	-0.005* (0.003)	-0.002* (0.001)
<i>Messenger Treatments</i>			
Outgroup Expert	0.060 (0.060)	0.048 (0.055)	
ECB Ingroup Expert	-0.062 (0.061)	-0.071 (0.056)	
ECB Outgroup Expert	-0.020 (0.069)	-0.035 (0.062)	
ECB Expert	-0.066 (0.066)	-0.052 (0.061)	
NCB Expert	-0.062 (0.064)	-0.045 (0.058)	
Treatment Order	✓	✓	✓
Individual-FE	✓		
Observations	2,394	2,394	2,394
R-squared	0.686		
Pseudo R-squared		0.020	

Notes: OLS regression in column (1) with the number of additional information pieces requested (buttons clicked) as dependent variable. Probit regression in column (2) with a binary outcome equal to 1 if at least one information piece is requested and 0 otherwise as dependent variable. Marginal effects corresponding to probit in column (3). Inflation level is the average inflation over the ten periods in a scenario; inflation uncertainty is the standard deviation over those periods. The reference category for messenger treatments is the *Nationality-Only Ingroup Expert*. Standard errors, clustered at the participant level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Exposure to News in the Media. Survey responses indicate greater reach for ingroup policymakers. Table A10 shows higher probabilities of having heard of, knowing, and following news about ingroup policymakers compared to similar outgroup policymakers.⁴⁵ The likelihood of knowing policymakers increases by roughly a third if they are in the ingroup rather than outgroup. Participants are 29 percentage points ($p < 0.01$) more likely to follow news of a given policymaker if they are in the ingroup.

Figure A27 plots this by participant nationality. Effects for ECB ingroup policymakers are driven by the French and Spanish – coinciding with the nationalities of the ECB president and vice president – suggesting nationality strongly matters, especially for (vice) presidents.

⁴⁵As previously mentioned, the policymakers used to proxy for messengers are NCB governors (Villeroy de Galhau, de Cos, Nagel, Visco) and ECB board members (Lagarde, de Guindos, Schnabel, Panetta).

Table A10: Information Reach: Survey Evidence

	(1) Heard of	(2) Knows of	(3) Informed of
Ingroup	0.080** (0.033)	0.271*** (0.029)	0.286*** (0.023)
Constant (Outgroup)	0.489	0.175	0.020
Individual-FE	✓	✓	✓
Observations	798	798	798
R-squared	0.564	0.617	0.626

Notes: OLS regressions with individual fixed effects; each participant contributes one observation for their ingroup policymaker and one for the average of the three outgroup policymakers. Representative in- and outgroup policymakers are ECB board members (Lagarde, de Guindos, Schnabel, Panetta); results for NCB governors (Villeroy de Galhau, de Cos, Nagel, Visco) are similar, as are results excluding Lagarde and French participants. The *Ingroup* coefficient gives the percentage-point difference in the probability of having heard of, knowing, or being informed about ingroup relative to outgroup policymakers; the constant reports the corresponding outgroup baseline. Data come from the survey question shown in Figure A22. Standard errors, clustered at the participant level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

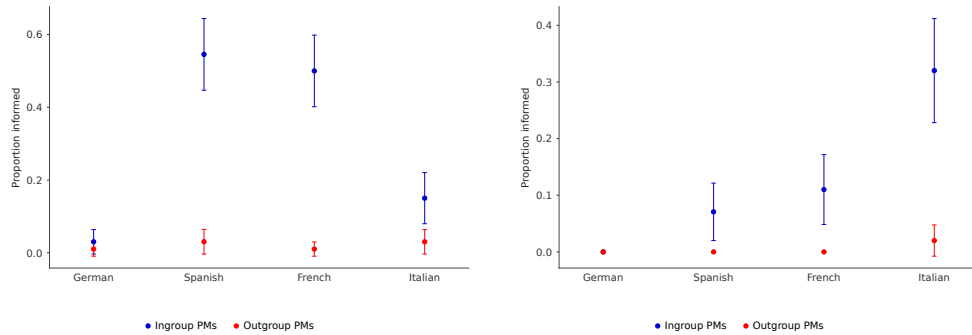


Figure A27: Knowledge of Policymakers and News Exposure

Notes: Shares of participants who regularly receive news about ECB board members (left) and NCB governors (right). Ingroup denotes the respondent's own-country policymaker (in blue); outgroup the average across the three others (in red). Whiskers are 95% confidence intervals; points without whiskers are shares of exactly zero, for which the interval has zero width.

B.2 Information Availability



Figure A28: Share of National Newspaper Articles Over Time: Germany and Spain

Notes: Countries' share of articles about the ECB over time. The vertical grey dashed line indicates the switch in presidents. Left panel: each dot represents one year (from November 1 until October 31 of the following year). Three years per president are shown. Right panel: each dot represents coverage over the preceding six-week press conference cycle, plotted at the cycle-ending press conference date. The first post-transition dot covers a cycle spanning both Draghi and Lagarde. Data are only available for exactly one entire year per president, which means the first and last press conference cycles shown are shorter than six weeks. The newspaper sample is described in Section 3.1.

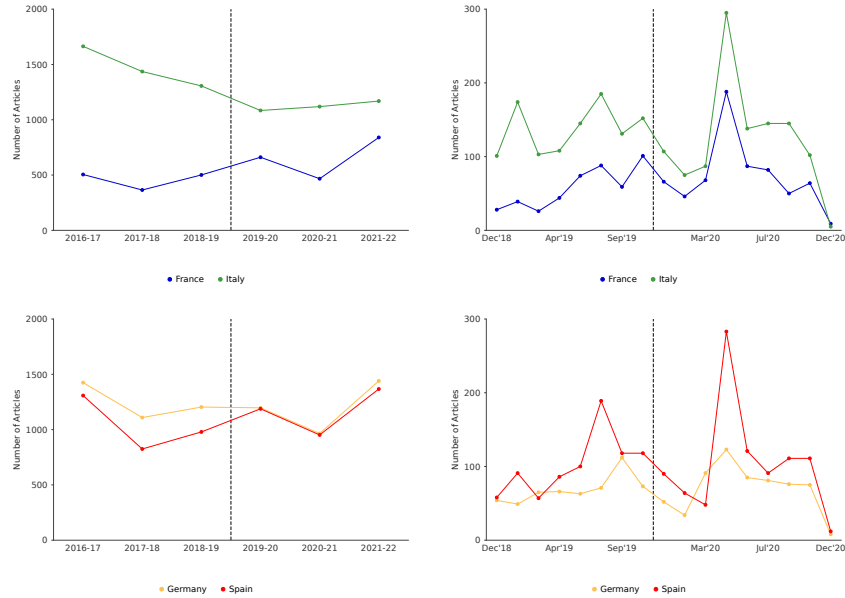


Figure A29: Number of National Newspaper Articles Over Time

Notes: The left panels show the yearly volume of national newspaper articles over six years, while the right panels plot the volume of articles each press conference cycle over two years. Each point is based on the volume of newspaper articles over 12 months – from November 1 until October 31 of the following year – (left), or over 6 weeks (right). Since the selection period is one entire year per president, the first and last press conference cycles shown are shorter than six weeks. The vertical grey dashed line indicates the switch in presidents. The newspaper sample is described in Section 3.1.

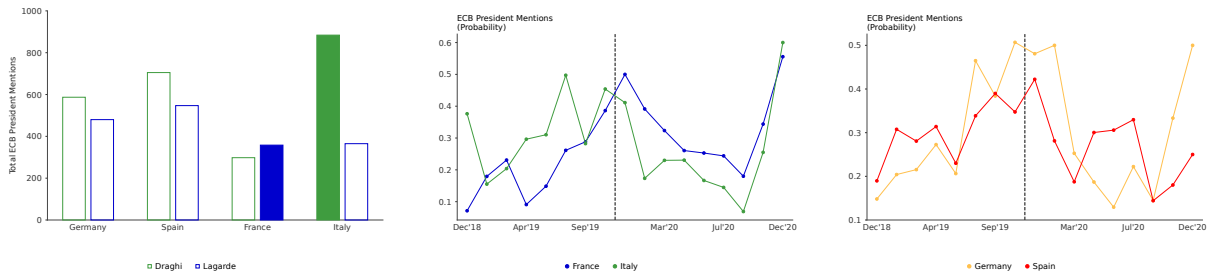


Figure A30: Salience of ECB President

Notes: The left panel shows total ECB president name mentions across all articles by country and presidency. Filled bars indicate the ingroup country for the incumbent president – France under Lagarde, Italy under Draghi. The middle (right) panel plots the share of French and Italian (German and Spanish) articles mentioning the ECB president at least once per press-conference cycle. The vertical dashed line marks the November 2019 transition from Draghi to Lagarde. Dots are placed at cycle-ending press conference dates; the first post-transition dot spans both presidencies. The first and last cycles are shorter because the data cover exactly one year per president. Data are from LexisNexis and cover the two years surrounding the presidential transition (November 2018 – October 2020).

B.3 External Validity: Social Media

Data Cleaning. I collected roughly 11 million tweets via the Academic API between November 2022 and March 2023. From the raw data, I remove ECB-unrelated tweets using language-specific filters (e.g., excluding the English Cricket Board), exclude “Draghi”/“Lagarde” mentions without ECB references, lowercase and de-noise (URLs/emojis/handles, except “lagarde”/“ecb”), translate all text to English using Google Translate’s Neural Machine Translation to standardize dictionary processing and eradicate any spelling mistakes, and fully spell out contracted words (e.g., “can’t” to “cannot” or “we’ll” to “we will”). The final dataset has 8,031,937 tweets. An audit of frequent words using word clouds confirms the cleanliness of the samples. Figure A31 shows these word clouds, indicating that most of the final sample is indeed comprised of tweets about central banking.

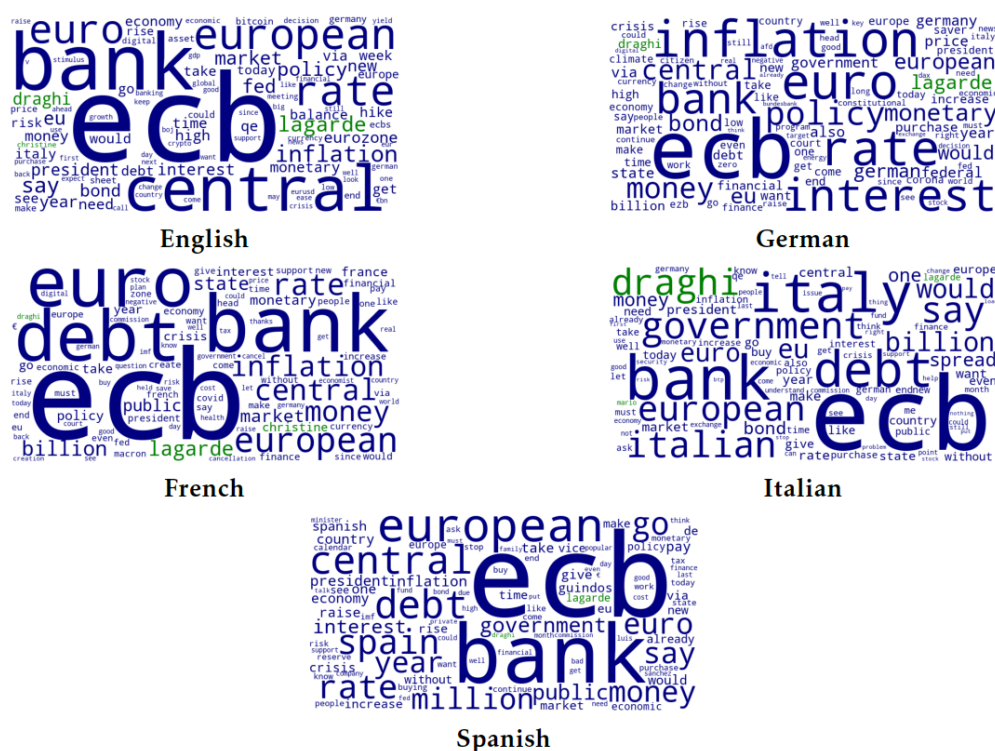


Figure A31: Word Clouds by Language

Notes: The 100 most frequent words between 2016-2022. Samples are split by original tweet language. Word size indicates word frequency. The following terms are highlighted in green for better visibility: “christine,” “lagarde,” “mario” and “draghi.” Word cloud is based on cleaned, translated and lemmatized tweets. Stopwords are removed. No ngrams are included.

Tweet Volume. English tweets make up almost half the sample (46%), reflecting English being the ECB’s official language and the lingua franca. Across languages, high-volume days

correspond to major ECB decisions or board appointments, confirming the success of the data cleaning. Specifically, volume peaks differ across languages and coincide with events of particular national importance (Table A11), supporting the use of tweet language as a proxy for nationality. The French peak in July 2019 reflects Lagarde’s nomination as incoming ECB president, while the Spanish peak in February 2018 followed Eurogroup support for de Guindos. The German peak in May 2020 coincides with constitutional judges ruling parts of the ECB’s asset purchase program unconstitutional. The Italian peak in March 2020 occurred a day after the PEPP announcement, which sharply lowered Italy’s borrowing costs amid the early Covid-19 outbreak. The English peak in July 2022 marks the ECB’s first rate hike in 11 years, suggesting English tweets focus more on Euro-area-wide monetary policy, while other languages emphasize country-specific news. This underscores the importance of non-English tweets for understanding the general dynamics of central bank communication, while making it impossible to infer any nationality from them.

Time trends. Tweet volumes and Twitter users rise over time i.e., between Draghi and Lagarde. Growth in tweets is largest for French and smallest for Italian (Table A11). However, these descriptives may reflect ingroup effects as well as other platform growth that is unrelated to ECB presidency, and should therefore not be interpreted causally.

Table A11: Tweet Volume and Peaks by Language

Language	Tweet Volume			Peak		
	Draghi	Lagarde	Growth	Volume	Date	Event
English	1,595,810	2,102,124	31.7%	25,624	21 Jul 2022	ECB raising rates for first time in 11 years
German	168,006	397,848	136.8%	7,098	05 May 2020	German constitutional court ruling
Spanish	623,046	1,106,234	77.6%	26,599	19 Feb 2018	Eurogroup backing of Luis de Guindos
French	238,455	585,257	145.4%	11,905	03 Jul 2019	Announcement of Lagarde as incoming ECB president
Italian	543,504	671,653	23.6%	22,050	19 Mar 2020	Day after PEPP announcement

Notes: Tweet volumes by language and presidency (“Draghi”: November 2016 – October 2019; “Lagarde”: November 2019 – October 2022). Growth is the percentage change in tweet volume between presidencies. Peak volume is the maximum daily tweet count (CET time zone) within each language sample.

Sentiment Measure. Sentiment of tweet content is measured by a dictionary-based method that provides a continuous estimate of how positive or negative a text snippet is. A sentiment score is computed for each tweet using the polarity measure of the Python package *TextBlob*, which is based on Princeton University’s WordNet lexicon (Loria, 2018). Sentiment scores are estimated on tweets that are cleaned and translated (and thus spell-checked). This ensures that the same (English) dictionary can be used to create the sentiment scores. I normalize this sentiment indicator to assign values between 0 (most negative) and 1 (most positive) to

each tweet, where neutral text gets a value of 0.5. The original polarity sentiment indicator returns a value between -1 and 1, where the most positive (negative) text obtains a value of 1 (-1) and neutral text gets a value of 0.

Sentiment Across Languages. To be able to make meaningful comparisons of beliefs across languages, it is important that no language shows idiosyncratic differences in the measure of tweet sentiment. Given the strong similarity in the distribution of tweet sentiment, this seems to be indeed the case. An overview of tweet sentiment across languages can be found in Table A12. The average and standard deviation of tweet sentiment is remarkably similar across languages.⁴⁶

Table A12: Sentiment of Tweets by Language

Language	N	Mean	Std. dev.	p10	p90
English	3,697,934	0.53	0.10	0.45	0.65
German	565,854	0.52	0.10	0.42	0.63
Spanish	1,729,280	0.53	0.10	0.44	0.65
French	823,712	0.53	0.10	0.44	0.64
Italian	1,215,157	0.52	0.10	0.42	0.65

Notes: Sentiment ranges from 0 to 1, with more positive (negative) tweets closer to 1 (0) and neutral tweets at 0.5. Means and standard deviations are computed over all available tweets by language during 2016–2022.

Alternative Measure of Information Use: Size of Belief Updates. To limit assumptions compared to the analysis in Section 4.1.2, define information use as the absolute sentiment change relative to the average update across all N_t users at each press conference: $RelativeUpdate_{i,t} = \Delta Sent_{i,t} - \frac{1}{N_t} \sum_{k=1}^{N_t} \Delta Sent_{k,t}$, where $\Delta Sent_{i,t} = |Posterior_{i,t} - Prior_{i,t}|$. Taking the absolute difference prevents positive and negative changes from canceling out, and subtracting the conference average controls for information content, isolating deviations from the common trend.

Beliefs shift more strongly following ECB press conferences when the president shares the audience’s nationality: Italians update relatively more under Draghi, and the French under Lagarde. By contrast, the Spanish and German samples serve as comparison groups, where updating should not vary across time (Figure A32).⁴⁷ Controlling for language and press conference fixed effects, ingroup users update their beliefs by 0.017 more in absolute terms

⁴⁶Average tweet sentiment is close to neutral between 0.52 and 0.53, and the standard deviation is 0.10 throughout. Even the 10th and the 90th percentiles of tweet sentiment do not vary considerably across countries, with values for the 10th percentile ranging between 0.42 and 0.45, and for the 90th between 0.63 and 0.65, respectively. It is noteworthy, however, that across languages most tweets have rather neutral sentiment around 0.5, and that tweets are slightly more positive than negative.

⁴⁷Germans show no difference in updating behavior across presidents. The stronger Spanish updating

Table A13: Information Use Around Press Conferences

	(1) All	(2) All	(3) All	(4) Experts	(5) Non-Experts
Prior	0.492** (0.239)	0.835*** (0.033)	0.811*** (0.039)	0.818*** (0.044)	0.851*** (0.034)
Ingroup $\times$ Prior	-0.455* (0.237)	-0.806*** (0.038)	-0.782*** (0.045)	-0.701*** (0.060)	-0.795*** (0.046)
Signal		0.166*** (0.044)	0.197*** (0.053)	0.188*** (0.052)	0.163*** (0.057)
Ingroup $\times$ Signal			-0.164*** (0.057)	-0.169*** (0.051)	-0.147*** (0.055)
Ingroup	0.255* (0.129)	0.440*** (0.022)	0.498*** (0.019)	0.457*** (0.022)	0.487*** (0.022)
Fixed Effects	PC	—	—	—	—
Observations	63,757	63,757	63,757	11,478	5,509
R-squared	0.954	0.946	0.946	0.957	0.944

Notes: OLS regressions without constants and with two-way clustered standard errors (by press conference and user) in parentheses. Column (1) includes press conference fixed effects; columns (2)–(5) replace them with the signal, measured as the EA-MPD OIS-2Y surprise ([Altavilla et al., 2019](#)). Columns (4) and (5) restrict the sample to non-experts and experts, respectively, following [Ehrmann and Wabitsch \(2022\)](#). Priors, posteriors, and the signal are normalized to $[0, 1]$. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

on the $[0, 1]$ sentiment scale ($p < 0.05$), which is sizable given that average belief updates across conferences lie within ± 0.020 (Table [A14](#)).

under Lagarde may reflect the contemporaneous presence of Spanish Vice President de Guindos, who also participates in ECB press conferences.

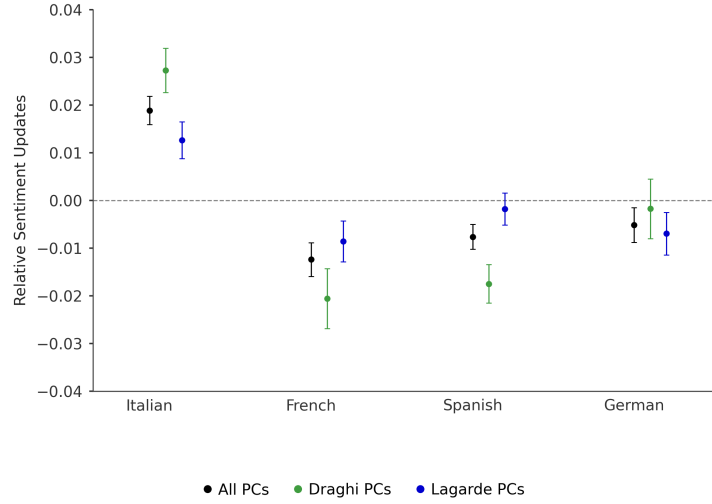


Figure A32: Ingroup Effects on Relative Belief Updating

Notes: Absolute belief updates relative to the conference-specific average, calculated as the absolute sentiment change in user tweets before and after press conferences. Green and blue indicate press conferences (PCs) in Draghi’s and Lagarde’s presidencies, respectively. Black indicates the entire sample.

Table A14: Relative Belief Updating Around Press Conferences

<i>Panel A: Regression results</i>			<i>Panel B: Descriptive statistics</i>	
	(1)	(2)		
Ingroup	0.014*** (0.002)	0.017** (0.008)	German	-0.005
Constant	0.184*** (0.001)	0.150*** (0.005)	Spanish	-0.008
			French	-0.012
			Italian	0.019
Language FE		✓		
Press Conference FE		✓		
Observations	63,757	63,757	Range	[-0.012, 0.019]
R-squared	0.001	0.019		

Notes: The dependent variable is the absolute change in sentiment between a user’s last tweet during the seven-day quiet period preceding a press conference and their first tweet within 24 hours after it. Panel A reports OLS regressions: column (1) gives the unconditional difference with robust standard errors; column (2) adds language and press conference fixed effects with standard errors two-way clustered by press conference and user. Standard errors in parentheses. Panel B reports mean updates by language, demeaned by press conference, and their range. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A15: Ingroup Effect on Information Availability on Twitter

	(1) All	(2) Experts	(3) Non-Experts
Ingroup	0.105*** (0.022)	0.024* (0.013)	0.078*** (0.022)
Germany	0.001 (0.016)	0.094*** (0.010)	-0.020 (0.020)
Spain	0.258*** (0.022)	0.224*** (0.011)	0.223*** (0.026)
Italy	0.090*** (0.022)	0.187*** (0.013)	0.071*** (0.022)
Constant (France)	0.136*** (0.018)	0.118*** (0.013)	0.162*** (0.030)
Press Conference Cycle FE	✓	✓	✓
Observations	200	200	200
R-squared	0.582	0.738	0.505

Notes: OLS regressions with robust standard errors in parentheses, following the specification in Section 3.2. All columns use six years of Twitter data at the level of six-week press conference cycles, where the cycle spanning the presidential transition is split by presidency. Column (1) uses all users; columns (2) and (3) restrict to experts and non-experts, following [Ehrmann and Wabitsch \(2022\)](#). The dependent variable is the share of tweets by language. *Ingroup* equals 1 for the language corresponding to the ECB president's country of origin during their term (Italian under Draghi, French under Lagarde). Baseline language: French. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

B.4 External Validity: Household Inflation Expectations

Table A16: Household Inflation Forecasts

	(1) Mean distance	(2) Median distance	(3) Uncertainty
Belgium	0.695** (0.265)	0.494** (0.197)	0.691*** (0.043)
Germany	0.007 (0.250)	0.253 (0.206)	-0.102* (0.054)
Spain	1.400*** (0.284)	0.567** (0.216)	0.578*** (0.045)
Italy	3.100*** (0.382)	1.807*** (0.288)	1.476*** (0.053)
Netherlands	0.240 (0.263)	0.393* (0.200)	0.767*** (0.049)
Constant (France)	1.679*** (0.237)	0.510*** (0.171)	1.440*** (0.035)
Wave FE	✓	✓	✓
Observations	89	89	270
R-squared	0.808	0.731	0.901

Notes: OLS coefficients are relative to France; wave fixed effects included; robust standard errors in parentheses. Columns (1)–(2) cover the 15 survey waves coinciding with ECB projection releases (April 2020–December 2023); column (3) covers all 45 monthly waves. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

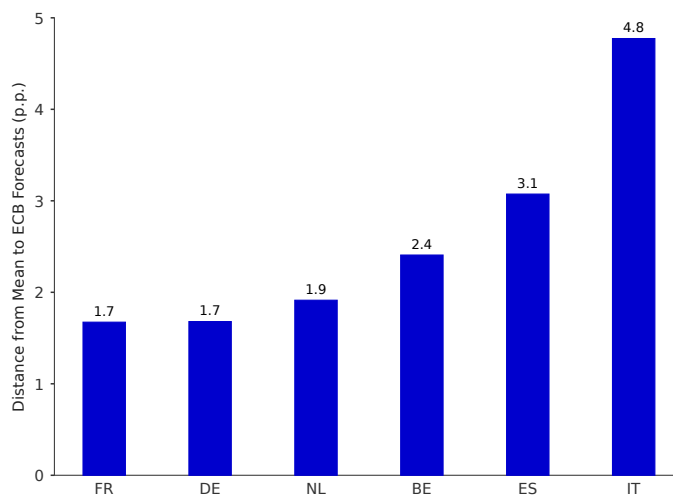


Figure A33: Comparing Mean Household Inflation Forecasts to ECB Projections

Notes: Average difference between mean household inflation forecasts over the next twelve months and the ECB's projection (lagged by one month); 15 survey waves coincide with ECB projection releases between April 2020 and December 2023, whereby data for Belgium are missing for the first wave (April 2020).

C Model Details, Proofs and Derivations

C.1 Equilibrium in Linear Strategies

To decide on their actions, agents account for the average action. Since they do not observe others' actions, they form expectations about $\bar{a}$ from beliefs about others' posteriors of the fundamental (i.e., $E_i[y_j|\Omega_i] = E_i[x|\Omega_i]$). Agents form posteriors based on their information sets (shown in Equation (9)). Given diffuse priors,

$$E_i[x|\Omega_i] = \begin{cases} \frac{\tau_y y_i + \tau_Y Y}{\tau_y + \tau_Y}, & \text{if } i \text{ is type } g, \\ \frac{\tau_y y_i + \rho \tau_Y Y}{\tau_y + \rho \tau_Y}, & \text{if } i \text{ is type } o^I \text{ where } |Y| \geq d_o, \\ y_i, & \text{if } i \text{ is type } o^U \text{ where } |Y| < d_o. \end{cases} \quad (21)$$

Recall that all agents follow a linear strategy shown in Equation (12): $a_i = \kappa_i y_i + (1 - \kappa_i)Y$.

Uninformed Outgroup Agents. An uninformed outgroup agent simply chooses $a_{o^U}(y_i) = y_i$; since Y is unobserved, $E_{o^U}[x|y_i] = y_i$.

Ingroup Agents. An ingroup agent g , who always receives the public signal, updates without bias and expects all agents in the economy to do the same. She assigns the same signal extraction weight κ_g to both ingroup agents and informed outgroup agents who receive Y , assuming they process the signal without bias, unaware of heterogeneity in information use. Knowing the fraction of outgroup agents A that receive the public signal and the overall share of outgroup agents $(1 - \alpha)$, her expected average action of the population is

$$\begin{aligned} E_g[\bar{a}|y_i, Y] &= [\alpha \kappa_g + (1 - \alpha)A \kappa_g + (1 - \alpha)(1 - A)] E_g[y_j|y_i, Y] \\ &\quad + [\alpha(1 - \kappa_g) + (1 - \alpha)A(1 - \kappa_g)] Y \\ &= Y \left[(1 - \kappa_g)(\alpha + (1 - \alpha)A) + \frac{\tau_Y}{\tau_y + \tau_Y} \left(\kappa_g(\alpha + (1 - \alpha)A) + (1 - \alpha)(1 - A) \right) \right] \\ &\quad + y_i \left[\frac{\tau_y}{\tau_y + \tau_Y} \left(\kappa_g(\alpha + (1 - \alpha)A) + (1 - \alpha)(1 - A) \right) \right]. \end{aligned} \quad (22)$$

Thus, the ingroup agent g 's optimal action is

$$\begin{aligned} a_g(y_i, Y) &= (1 - r)E_g[x|y_i, Y] + r E_g[\bar{a}|y_i, Y] \\ &= \left[\frac{\tau_y}{\tau_y + \tau_Y} \left(1 - r + r[\kappa_g(\alpha + (1 - \alpha)A) + (1 - \alpha)(1 - A)] \right) \right] y_i \\ &\quad + \left[1 + \frac{\tau_y}{\tau_y + \tau_Y} \left(r[-\kappa_g(\alpha + (1 - \alpha)A) + \alpha + (1 - \alpha)A] - 1 \right) \right] Y. \end{aligned} \quad (23)$$

Comparing coefficients with (12) gives

$$\kappa_g = \frac{\tau_y}{\tau_y + \tau_Y} [1 - r + r [\kappa_g(\alpha + (1 - \alpha)A) + (1 - \alpha)(1 - A)]] ,$$

from which

$$\kappa_g^* = \frac{\tau_y q}{\tau_Y + \tau_y q}, \quad \text{where} \quad q \equiv 1 - r[\alpha + (1 - \alpha)A].$$

Informed Outgroup Agents. The informed outgroup agent o^I – who receives both signals but updates with a bias – looks very similar to the ingroup agent but downweights the public signal by the resonance weight ρ . His expected average action of the population is

$$\begin{aligned} E_{o^I}[\bar{a}|y_i, Y] &= [\alpha\kappa_{o^I} + (1 - \alpha)A\kappa_{o^I} + (1 - \alpha)(1 - A)] E_{o^I}[y_j|y_i, Y] \\ &\quad + [\alpha(1 - \kappa_{o^I}) + (1 - \alpha)A(1 - \kappa_{o^I})] Y \\ &= Y \left[(1 - \kappa_{o^I})(\alpha + (1 - \alpha)A) + \frac{\rho\tau_Y}{\tau_y + \rho\tau_Y} (\kappa_{o^I}(\alpha + (1 - \alpha)A) + (1 - \alpha)(1 - A)) \right] \\ &\quad + y_i \left[\frac{\tau_y}{\tau_y + \rho\tau_Y} (\kappa_{o^I}(\alpha + (1 - \alpha)A) + (1 - \alpha)(1 - A)) \right]. \end{aligned} \tag{24}$$

Thus, the informed outgroup agent o^I 's optimal action is

$$\begin{aligned} a_{o^I}(y_i, Y) &= (1 - r)E_{o^I}[x|y_i, Y] + r E_{o^I}[\bar{a}|y_i, Y] \\ &= \left[\frac{\tau_y}{\tau_y + \rho\tau_Y} (1 - r + r[\kappa_{o^I}(\alpha + (1 - \alpha)A) + (1 - \alpha)(1 - A)]) \right] y_i \\ &\quad + \left[1 + \frac{\tau_y}{\tau_y + \rho\tau_Y} (r[-\kappa_{o^I}(\alpha + (1 - \alpha)A) + \alpha + (1 - \alpha)A] - 1) \right] Y. \end{aligned} \tag{25}$$

Comparing coefficients with (12) yields

$$\kappa_{o^I} = \frac{\tau_y}{\tau_y + \rho\tau_Y} [1 - r + r [\kappa_{o^I}(\alpha + (1 - \alpha)A) + (1 - \alpha)(1 - A)]] ,$$

hence

$$\kappa_{o^I}^* = \frac{\tau_y q}{\rho\tau_Y + \tau_y q}.$$

Remark. The optimal signal extraction weights for informed in- and outgroup agents are

$$\kappa_g^* = \frac{\tau_y q}{\tau_Y + \tau_y q} \quad \text{and} \quad \kappa_{o^I}^* = \frac{\tau_y q}{\rho\tau_Y + \tau_y q}.$$

Substituting these weights into the optimal actions specified above returns the equilibrium actions as in Proposition 1

$$a_g(y_i, Y) = \frac{\tau_Y Y + \tau_y q y_i}{\tau_Y + \tau_y q}, \quad a_{o^I}(y_i, Y) = \frac{\rho \tau_Y Y + \tau_y q y_i}{\rho \tau_Y + \tau_y q}, \quad \text{and} \quad a_{o^U}(y_i) = y_i.$$

Average Action. Let $a_i(x, Y) \equiv \mathbb{E}[a_i(y_i, Y)|x, Y]$ denote the representative (type-average) action for informed agents of types g and o^I , and let $a_{o^U}(x) \equiv \mathbb{E}[a_{o^U}(y_i)|x] = x$ for uninformed outgroup agents. By symmetry within types, the population average action equals a weighted average of representative type actions

$$\bar{a} = \alpha a_g + (1 - \alpha)A a_{o^I} + (1 - \alpha)(1 - A) a_{o^U}. \quad (26)$$

Taking expectations conditional on (x, Y) and using $\mathbb{E}[y_i|x] = x$ yields

$$\mathbb{E}[\bar{a}|x, Y] = \alpha a_g(x, Y) + (1 - \alpha)A a_{o^I}(x, Y) + (1 - \alpha)(1 - A)x. \quad (27)$$

C.1.1 Uniqueness of the Linear Equilibrium

Proposition 1 claims that the equilibrium strategies derived constitute the unique linear equilibrium of the model. To prove that the equilibrium strategy is indeed a linear combination of the available signals, and that the equilibrium is therefore unique, I extend the proof by [Morris and Shin \(2002\)](#) to accommodate heterogeneous agent types with biased information use. The best-response map in (κ_g, κ_{o^I}) is a contraction with modulus bounded by $r(\alpha + (1 - \alpha)A) < 1$, ensuring a unique fixed point. Alternatively, one can note that the recursive higher-order beliefs representation implied by

$$a_i(y_i, Y) = q \sum_{s=0}^{\infty} (r(\alpha + (1 - \alpha)A))^s E_i[\bar{E}_i^s(x|y_i, Y)]$$

remains linear in (y_i, Y) under ρ -downweighting; thus the linear solution constitutes the unique equilibrium.

C.2 Derivation for Proposition 2: Optimal Disclosure

To derive the condition under which disclosure improves welfare, I differentiate expected welfare with respect to τ_Y , holding r, α, ρ, τ_y fixed

$$\frac{\partial \mathbb{E}[W]}{\partial \tau_Y} = \alpha \frac{\tau_Y - \tau_y q(1 - 2q)}{(\tau_Y + \tau_y q)^3} + (1 - \alpha)A \rho \frac{\rho^2 \tau_Y - \tau_y q(\rho - 2q)}{(\rho \tau_Y + \tau_y q)^3}, \quad (28)$$

where $q \equiv 1 - r(\alpha + (1 - \alpha)A) = 1 - rs$. The sign of this derivative depends on the parameters $c = \tau_y/\tau_Y$, r , α , A , and ρ , leading to the thresholds stated in Proposition 2. Since $q \in (0, 1]$,

$\tau_Y, \tau_y > 0$, and $\rho \in (0, 1]$, the denominators in (28) are strictly positive; hence the sign is determined by the numerators. The numerators are non-negative whenever

$$\tau_Y - \tau_y q(1 - 2q) \geq 0 \iff c \leq \frac{1}{q(1 - 2q)} \quad \text{if } q < \frac{1}{2},$$

$$\rho^2 \tau_Y - \tau_y q(\rho - 2q) \geq 0 \iff c \leq \frac{\rho^2}{q(\rho - 2q)} \quad \text{if } q < \frac{\rho}{2}.$$

Therefore, a sufficient condition for $\frac{\partial \mathbb{E}[W]}{\partial \tau_Y} \geq 0$ is satisfied whenever one of the following holds

$$\begin{cases} q \geq \frac{1}{2}, \\ \frac{\rho}{2} \leq q < \frac{1}{2} \quad \text{and} \quad c \leq \frac{1}{q(1 - 2q)}, \\ 0 < q < \frac{\rho}{2} \quad \text{and} \quad c \leq \min \left\{ \frac{1}{q(1 - 2q)}, \frac{\rho^2}{q(\rho - 2q)} \right\}. \end{cases}$$

A simple corollary is that $c \leq 8$ is a uniform sufficient bound, since $q(1 - 2q) \leq \frac{1}{8}$ for $q \in [0, \frac{1}{2}]$ (attained at $q = \frac{1}{4}$) and $q(\rho - 2q)/\rho^2 \leq \frac{1}{8}$ for $q \in [0, \frac{\rho}{2}]$ (attained at $q = \rho/4$).

For all $c \geq 0$ and $s > 0$, if the following coordination parameter condition holds

$$q \geq \max \left\{ \frac{1}{2}, \frac{\rho}{2} \right\} \iff r \leq \min \left\{ \frac{1}{2s}, \frac{1 - \rho/2}{s} \right\},$$

then each numerator in (28) is non-negative regardless of c , so $\partial \mathbb{E}[W]/\partial \tau_Y \geq 0$ for any $c \geq 0$.

This completes the derivation underlying Proposition 2.

C.3 Raising Private Signal Precision May Harm Welfare

Proposition 4. *The effect of increasing the precision of private signals (τ_y) on expected welfare is ambiguous and depends on the coordination parameter r and the presence of outgroup agents ($\frac{\partial \mathbb{E}[W]}{\partial \tau_y} \gtrless 0$).*

Taking the derivative of $\mathbb{E}[W]$ with respect to τ_y shows that $\frac{\partial \mathbb{E}[W]}{\partial \tau_y} \gtrless 0$ and depends on the particular parameter values.

$$\begin{aligned}
\frac{\partial \mathbb{E}[W]}{\partial \tau_y} = & -\alpha \left[\frac{q^2}{(\tau_Y + \tau_y q)^2} - \frac{2q(\tau_Y + \tau_y q^2)}{(\tau_Y + \tau_y q)^3} \right] \\
& - (1 - \alpha)A \left[\frac{q^2}{(\rho\tau_Y + \tau_y q)^2} - \frac{2q(\rho^2\tau_Y + \tau_y q^2)}{(\rho\tau_Y + \tau_y q)^3} \right] \\
& + \frac{(1 - \alpha)(1 - A)}{\tau_y^2}
\end{aligned} \tag{29}$$

At first glance, more precise private signals improve alignment with the fundamental and thus welfare. However, the sign depends on $(r, \alpha, A, \rho, \tau_Y/\tau_y)$: with strong coordination and a sizable, biased informed outgroup, increasing τ_y can lower welfare. This occurs because of projection: informed outgroup agents underweight public information and expect others to do the same; under high r , they tilt actions toward private signals to coordinate on what they believe others will do, increasing action dispersion. Negative effects require both sufficiently strong strategic complementarities and sufficiently large bias and informed outgroup. The resonance weight needed for negative effects is much lower than what I find empirically.⁴⁸

This contrasts with [Morris and Shin \(2002\)](#) and [Cornand and Heinemann \(2008\)](#), and aligns with the broader insights in [Hellwig \(2005\)](#) and [Angeletos and Pavan \(2007\)](#): the welfare effect of higher private precision depends on the trade-off between coordination on public signals and dispersion from idiosyncratic ones. With strong coordination motives and sizable processing biases, greater reliance on private signals raises dispersion and can reduce welfare; otherwise, higher private precision improves welfare.

Corollary 2. *When there are significant biases in outgroup agents' information use, increasing τ_y may reduce social welfare.*

C.4 Derivation for Proposition 3: Optimal Delegation

Non-linearities make a closed-form analytical solution for α^* infeasible. The Weierstrass Extreme Value Theorem guarantees the existence of a welfare-maximizing α^* for an expected welfare function $\mathbb{E}[W(a, x)|x]$ that is continuous on the compact set $[0, 1]$. For the interior characterization, assume $\mathbb{E}[W(a, x)|x]$ is twice continuously differentiable on $(0, 1)$. Then α^* is either a boundary point ($\alpha^* \in \{0, 1\}$) or, if interior, satisfies $\frac{\partial \mathbb{E}[W]}{\partial \alpha} = 0$ and $\frac{\partial^2 \mathbb{E}[W]}{\partial \alpha^2} \leq 0$.

⁴⁸Specifically, $\rho \lesssim 0.3$ is needed for negative effects, given $r \gtrsim 0.5$, roughly equal public and private precision, and at least half the population being outgroup and informed. For milder biases or smaller informed outgroups, the effect is non-negative.

$$\begin{aligned} \frac{\partial \mathbb{E}[W]}{\partial \alpha} = & \frac{\alpha 2r\tau_y(1-A)q}{(\tau_Y + \tau_y q)^2} - \frac{\alpha 2r\tau_y(1-A)(\tau_Y + \tau_y q^2)}{(\tau_Y + \tau_y q)^3} + \frac{(1-\alpha)2A(1-A)r\tau_y q}{(\rho\tau_Y + \tau_y q)^2} \\ & - \frac{(1-\alpha)2A(1-A)r\tau_y(\tau_y q^2 + \rho^2\tau_Y)}{(\rho\tau_Y + \tau_y q)^3} + \frac{A(\tau_y q^2 + \rho^2\tau_Y)}{(\rho\tau_Y + \tau_y q)^2} - \frac{\tau_Y + \tau_y q^2}{(\tau_Y + \tau_y q)^2} + \frac{1-A}{\tau_y} = 0 \end{aligned} \quad (30)$$

Since an analytical solution is infeasible, Figure A34 reports α^* numerically across coordination r and public signal precision τ_Y , holding $\tau_y = 1$. Panels vary (i) signal size via availability A (top: large signal, $A = 0.8$; bottom: small signal, $A = 0.2$) and (ii) updating bias (left: $\rho = 1$; right: $\rho = 0.5$). Four forces determine optimal delegation:

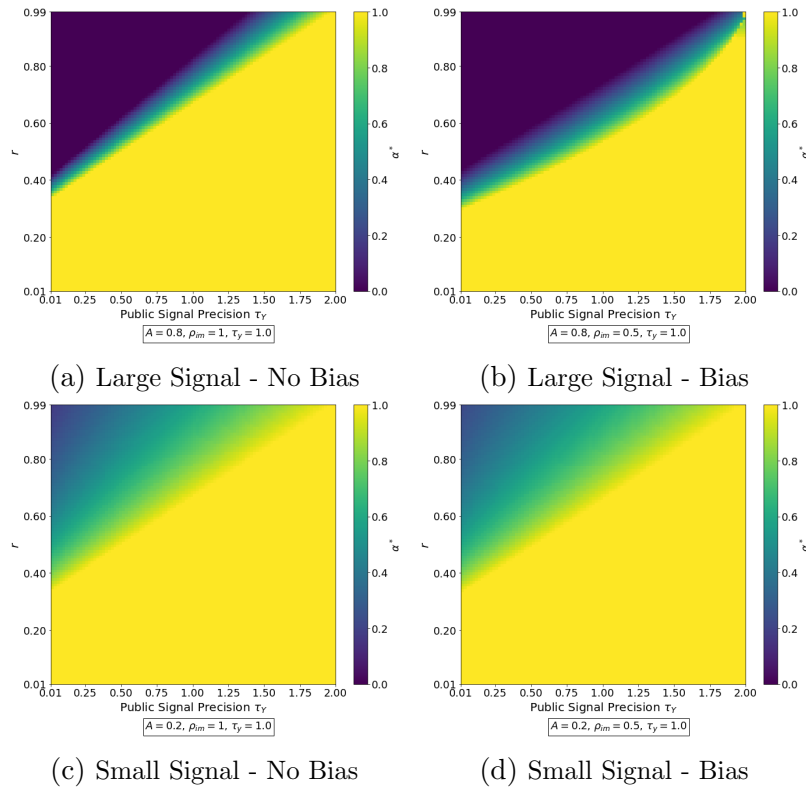


Figure A34: Optimal Share of Ingroup Agents

Notes: Welfare-maximizing α^* as a function of (r, τ_Y) with $\tau_y = 1$. Top: large signal (holding $A = 0.8$ fixed); bottom: small signal ($A = 0.2$). Left: no updating bias ($\rho = 1$); right: outgroup strongly underweights the public signal ($\rho = 0.5$).

(i) α^* is (weakly) increasing in relative *public signal precision* τ_Y : when the public signal is precise, delegation is desirable ($\alpha^* \rightarrow 1$); when it is noisy, α^* falls: when $\tau_Y < \tau_y$ in high coordination environments, raising α can reduce welfare unless public precision is sufficiently high.⁴⁹

⁴⁹The sufficient level of strategic complementarity depends on updating bias and availability. It is $r > 1/3$

(ii) α^* is decreasing in the *coordination motive* r whenever the public signal is insufficiently precise: stronger complementarities raise the cost of coordinating on a noisy public signal, making it optimal to retain some outgroup ($\alpha^* < 1$).

(iii) α^* is shifted down with *bias in updating* ($\rho < 1$): relative to the unbiased case, the region with $\alpha^* < 1$ expands, especially when τ_Y is large (see Figures A34a vs. A34b). This is because biased outgroup agents underweight the public signal, providing a buffer against over-coordination on noisy public signals, and increasing ingroup agents removes this buffer.

(iv) α^* decreases with *signal size* $|Y|$ (i.e., *availability* A): large signals make the outgroup buffer operative: more outgroup agents actually receive Y , so their bias reduces α^* more strongly (see Figures A34b vs. A34d). With small signals (low A), many outgroup agents miss Y , so bias has little additional effect and the surface resembles the no-bias case (see Figures A34c vs. A34d).

C.5 Delegation With Expertise Loss

Proposition 5. *If delegation to increase α lowers public signal precision, $\tau_Y = \tau_Y(\alpha)$ with $\tau_Y'(\alpha) < 0$, then delegation is locally welfare-improving iff*

$$\frac{d\mathbb{E}[W]}{d\alpha} > 0 \iff \frac{\partial\mathbb{E}[W]}{\partial\alpha} > k(\alpha) \frac{\partial\mathbb{E}[W]}{\partial\tau_Y}, \quad (31)$$

where $k(\alpha) \equiv -\tau_Y'(\alpha) > 0$ is the marginal expertise loss (precision drop) per unit increase in α .

Proof sketch. Delegation changes α directly and τ_Y indirectly. By the chain rule,

$$\frac{d\mathbb{E}[W]}{d\alpha} = \frac{\partial\mathbb{E}[W]}{\partial\alpha} + \frac{\partial\mathbb{E}[W]}{\partial\tau_Y} \tau_Y'(\alpha) = \frac{\partial\mathbb{E}[W]}{\partial\alpha} - k(\alpha) \frac{\partial\mathbb{E}[W]}{\partial\tau_Y}. \quad (32)$$

Local welfare improves iff $\frac{d\mathbb{E}[W]}{d\alpha} > 0$, which yields Equation (31). For exposition, let $\tau_Y(\alpha) = \tau_Y^0 - k\alpha$ with $k > 0$. Then $\tau_Y'(\alpha) = -k$ and (32) becomes

$$\frac{d\mathbb{E}[W]}{d\alpha} = \frac{\partial\mathbb{E}[W]}{\partial\alpha} - k \frac{\partial\mathbb{E}[W]}{\partial\tau_Y}. \quad (33)$$

The delegation–expertise trade-off is exactly Equation (31) with $k(\alpha) \equiv k$. Therefore, delegation raises welfare only when the marginal benefit of raising α exceeds the marginal cost from the induced loss in public signal precision. Figure A35 visualizes this trade-off.

when outgroup agents update without bias (regardless of availability), and lower when updating is biased and signals are widely available. For example, with large signals ($A \rightarrow 1$) and a small updating bias $\rho = 0.95$, $r > 0.156$ suffices. Larger signals lower this threshold; stronger biases raise it.

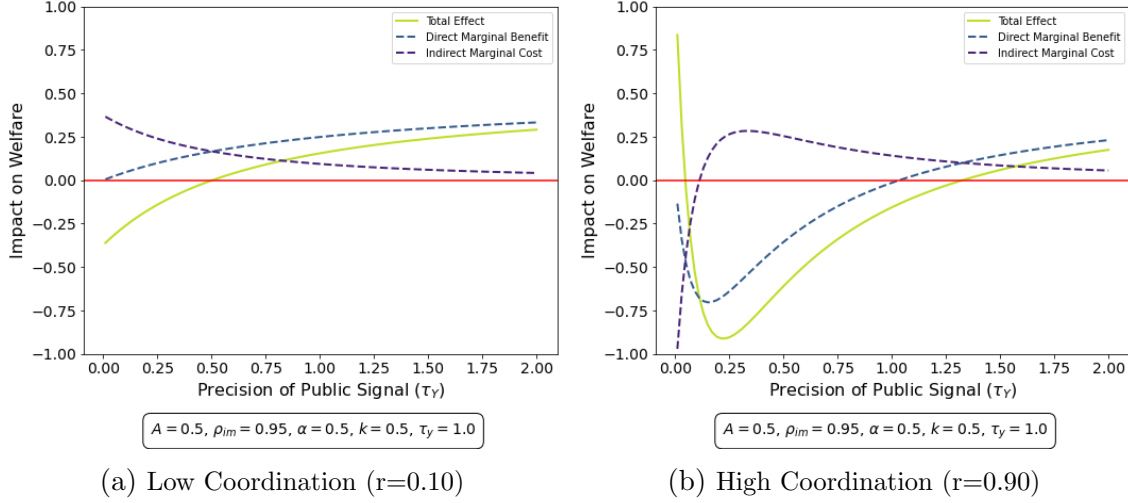


Figure A35: Social Welfare Impact of Delegation With Expertise Loss

Notes: Welfare effects of increasing α via delegation, accounting for reduced public signal precision from lower messenger expertise. Green: total effect; blue: marginal benefit; purple dashed: marginal cost. Parameters: $\tau_y = 1, A = k = \alpha_0 = 0.5$. Arrows indicate welfare ascent. Panel (a): low strategic complementarities ($r = 0.10$). Panel (b): high complementarities ($r = 0.90$).

Notably, when $\frac{\partial \mathbb{E}[W]}{\partial \tau_Y} > 0$ (disclosure is beneficial), expertise loss imposes a positive marginal cost $k(\alpha) \frac{\partial \mathbb{E}[W]}{\partial \tau_Y}$ that delegation must overcome; when $\frac{\partial \mathbb{E}[W]}{\partial \tau_Y} < 0$ (disclosure is harmful), expertise loss could help at the margin – yet so would centralization as shown in Section 5.5.